



MAX-M10S

Standard precision GNSS module

Professional grade

Integration manual



Abstract

This document describes the features and application of the u-blox MAX-M10S module. The MAX-M10S module provides an ultra-low-power standard precision GNSS receiver for high-performance asset-tracking applications.

Document information

Title	MAX-M10S	
Subtitle	Standard precision GNSS module	
Document type	Integration manual	
Document number	UBX-20053088	
Revision and date	R05	28-Apr-2026
Disclosure restriction	C1-Public	

This document applies to the following products:

Product name	Type number	FW version	IN/PCN reference	RN reference
MAX-M10S	MAX-M10S-00B-01	ROM SPG 5.10	UBX-22012689	UBX-22001426

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1 System description

This section gives an overview of the MAX-M10S receiver, and outlines the basics of operation with the receiver.

1.1 Overview

MAX-M10S module features the u-blox M10 standard precision GNSS platform and provides exceptional sensitivity and acquisition time for all L1 GNSS signals.

The M10 platform supports concurrent reception of four GNSSs (GPS, GLONASS, Galileo, and BeiDou). The high number of visible satellites enables the receiver to select the best signals. This maximizes the position availability, in particular under challenging conditions such as in deep urban canyons.

u-blox Super-Signal (Super-S) technology offers great RF sensitivity and can improve the dynamic position accuracy with small antennas or in non-line-of-sight scenarios.

The extremely low power consumption of 25 mW in continuous tracking mode allows great power autonomy for all battery-operated devices, such as asset trackers, without compromising on GNSS performance.

For maximum sensitivity in passive antenna designs, MAX-M10S integrates an LNA followed by a SAW filter in the RF path.

MAX-M10S offers backwards pin-to-pin compatibility with products from the previous u-blox generations, which saves the designer's effort and reduces costs when upgrading designs to the advanced low-power u-blox M10 GNSS technology.

1.2 Architecture

The MAX-M10S receiver provides all the necessary RF and baseband processing to enable multi-constellation operation. The block diagram below shows the key functionality.

1.2.1 Block diagram

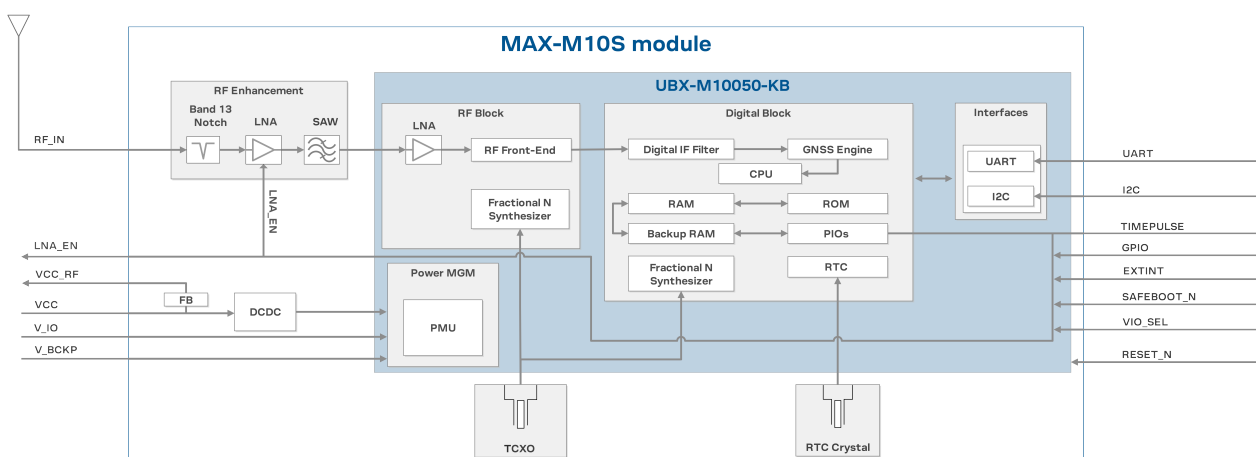


Figure 1: MAX-M10S block diagram

1.3 Pin assignment

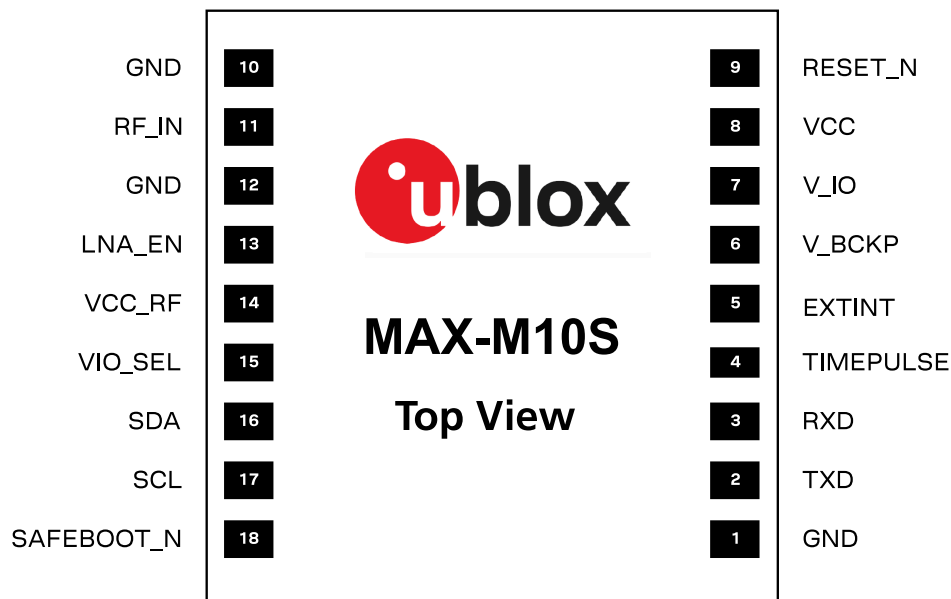


Figure 2: MAX-M10S pin assignment

Pin no.	Name	PIO no.	I/O	Description	Remarks
1	GND	-	-	-	Connect to GND
2	TXD	1	O	UART TX	If not used, leave open. Alternative functions ¹ .
3	RXD	0	I	UART RX	If not used, leave open. Alternative functions ¹ .
4	TIMEPULSE	4	O	Time pulse signal	See section TIMEPULSE for more information. Alternative functions ¹ .
5	EXTINT	5	I	External interrupt	See EXTINT for more information. Alternative functions ¹ .
6	V_BCKP	-	I	Backup voltage supply.	Leave open if no external backup supply. See V_BCKP for more information.
7	V_IO	-	I	IO voltage supply	See V_IO for more information.
8	VCC	-	I	Main voltage supply	See VCC for more information.
9	RESET_N	-	I	System reset (active low)	It has to be low for at least 1 ms to trigger a reset. Leave open if not used. See RESET_N section for more information.
10	GND	-	-	-	Connect to GND
11	RF_IN	-	I	GNSS signal input	The RF signal line is DC blocked internally. The line must match the 50 Ω impedance. See sections RF front-end and Layout for more information about the RF signal considerations.
12	GND	-	-	-	Connect to GND
13	LNA_EN	-	O	On/Off external LNA or active antenna	This pin cannot be used for another purpose as it also controls the internal LNA. See LNA_EN for more information.

¹ Alternatively, this pin can be used for [ANT_DETECT](#), [ANT_SHORT_N](#), [TX_READY](#), and [Data batching](#). Care must be taken when the assigned function sets the pin as an output.

Pin no.	Name	PIO no.	I/O	Description	Remarks
14	VCC_RF	-	O	Output voltage RF section	This pin supplies a filtered voltage that can be used for optional external active antenna or LNA. This pin is internally connected to VCC through a ferrite bead.
15	VIO_SEL	-	I	Voltage selector for V _{IO} supply	Connect to GND for 1.8 V supply, or leave open for 3.3 V supply
16	SDA	2	I/O	I2C data	If not used, leave open. Alternative functions ¹ .
17	SCL	3	I	I2C clock	If not used, leave open. Alternative functions ¹ .
18	SAFEBOOT_N	-	I	Safeboot mode	To enter safeboot mode, set this pin to low at receiver's startup. Otherwise, leave it open. The SAFEBOOT_N pin is internally connected to TIMEPULSE pin through a 1 kΩ series resistor.

Table 1: MAX-M10S pin assignment

2 Receiver configuration

The configuration determines all aspects of the GNSS receiver operation and therefore, information in this section is essential for the successful integration of MAX-M10S.

MAX-M10S is configured using UBX configuration interface keys. The configuration database in the receiver's RAM holds the current configuration, which is used by the receiver at runtime. It is constructed at the receiver startup from several sources of configuration. For more information on the receiver configuration, see the Interface description [3].

The configuration can be stored in the RAM and the battery-backed RAM (BBR) memory.

The permanence of the stored configuration and the actions to clear it in each memory are listed in Table 2.

Memory	Permanence of storage	Clearing actions
RAM	Settings remain effective until power-down	- Activating the RESET_N pin - A UBX-CFG-RST message excluding GNSS stop (resetMode 0x08) and GNSS start (resetMode 0x09) - Entering software standby mode - Using external control in power save mode (PSM) to enter the inactive state - Entering the off state of the PSM on/off operation (PSMOO) - Entering the inactive state of the PSM cyclic tracking operation (PSMCT)
BBR	The receiver retains the settings stored as long as the backup power supply remains	- Activating the RESET_N pin - A UBX-CFG-RST message with reset mode set to a hardware reset (resetMode 0x00 and 0x04)

Table 2: Permanence of storage and clearing actions for each memory

For more information about the UBX-CFG-RST message, refer to [Forcing receiver reset](#).

 **CAUTION** The configuration interface has changed from earlier u-blox positioning receivers. Users must adopt the configuration interface described in this document.

The configuration interface settings are stored in a database consisting of separate configuration items. An item is made up of a pair consisting of a key ID and a value. Related items are grouped together and identified under a common group name: CFG-GROUP-*; a convention used in u-center 2 and within this document. Within u-center 2, a configuration group is identified as "Group name" and the configuration item is identified as the "item name" in the "Device configuration" window.

The UBX messages available to change or poll the configurations are the UBX-CFG-VALSET, UBX-CFG-VALGET, and UBX-CFG-VALDEL messages. For more information about these messages and the configuration keys, see the configuration interface section in the Interface description [3].

2.1 Basic receiver configuration

This section summarizes the most commonly used, basic receiver configurations.

2.1.1 Basic hardware configuration

The MAX-M10S receiver is preconfigured in module production and is fully operational after connecting a proper power supply, the communication interfaces with the host application device, and a suitable antenna signal.

2.1.2 Internal LNA mode configuration

u-blox 10 receivers feature an internal low-noise amplifier (LNA) with three operational modes: normal gain, low gain and bypass mode. The MAX-M10S default is the low gain mode. With a high-

gain external active antenna, use the bypass mode to save power. The normal gain mode is not recommended for MAX-M10S.

The internal LNA mode can be configured at run time in the BBR and RAM memory using the configuration item CFG-HW-RF_LNA_MODE and applying a suitable software reset by sending a UBX-CFG-RST message. The suitable software reset type depends on the configured memory layer. For more information, refer to [Forcing receiver reset](#).

The internal LNA mode can also be permanently configured in the receiver's one-time programmable (OTP) memory.

The default gain mode is pre-configured in the receiver and does not require configuration in production. The configuration string for setting the internal LNA mode in the OTP memory is given in [Table 3](#).

 OTP configuration is permanent and cannot be reverted. Configuration at run time in BBR and RAM memory is still possible.

Internal LNA mode	Configuration string
Low gain	Default
Bypass	B5 62 06 41 10 00 03 00 05 1F 79 B2 0A E5 28 EF 12 05 9F FF FF FF 62 FB

Table 3: Internal LNA mode configuration in OTP memory

Configuring the internal LNA mode in the OTP memory:

1. Power up the system.
2. Test the communication interface by polling the UBX-MON-VER message.
3. Send the desired configuration string in [Table 3](#).
4. Power cycle the receiver or apply a hardware reset by sending a UBX-CFG-RST message. The configured internal LNA setting is applied at startup.
5. Verify that the configuration item is correctly set by polling CFG-HW-RF_LNA_MODE at RAM layer using the UBX-CFG-VALGET message.

2.1.3 GNSS signal configuration

MAX-M10S supports concurrent reception of four major GNSS constellations using the GPS L1C/A, Galileo E1, BeiDou B1C, and GLONASS L1OF signals. The default configuration is concurrent reception of GPS, Galileo and BeiDou B1I with QZSS and SBAS enabled.

 BeiDou B1I signal cannot be used simultaneously with the BeiDou B1C or GLONASS L1OF signals.

GNSS constellations and signals can be configured using the CFG-SIGNAL-* configuration group. Each GNSS constellation can be enabled or disabled independently except for QZSS and SBAS, which are functional only with GPS. In addition to the configuration key for each constellation, there is a configuration key for each signal supported by the firmware.

For example, if CFG-SIGNAL-GPS_ENA is set to zero, all signals from the GPS constellation are disabled. Alternatively, if CFG-SIGNAL-GPS_L1CA_ENA is set to zero, only the GPS L1 C/A signal is disabled.

Unsupported combinations are rejected with a UBX-ACK-NAK message, and the warning "inv sig cfg" is sent via UBX-INF and NMEA-TXT messages (if enabled).

 Any change to the signal configuration items triggers a restart of the GNSS subsystem. During the restart, the host application should wait for message acknowledgement and a margin of 0.5 seconds prior to sending any further commands.



To mitigate possible cross-correlation issues between the signals, it is recommended to enable also the QZSS L1C/A when the GPS L1C/A signal is enabled.

For more information on the CFG-SIGNAL-* configuration group, refer to the Interface description [3].

2.1.3.1 BeiDou B1I and B1C signals

BeiDou B1I and B1C signals differ in terms of the center frequency, bandwidth, and modulation. Therefore, there are differences in performance and supported GNSS signals and features.

Benefits of using BeiDou B1I signal:

BeiDou B1I offers superior GNSS performance. High start-up sensitivity and fast time-to-first-fix (TTFF) enable acquisition of a large number of BeiDou satellites. The tracking and reacquisition sensitivity for acquired signals is approximately at the same level for BeiDou B1I and B1C.

- **Faster TTFF and higher start-up sensitivity.** BeiDou B1I signals are acquired significantly faster and at a lower signal level than BeiDou B1C signals.
- **Better availability.** Higher start-up sensitivity results in a larger number of BeiDou satellites tracked and used in navigation solution especially at low signal level.
- **AssistNow support.** Enhanced start-up sensitivity and TTFF.
- **Power save mode (PSM) support.** PSM cyclic tracking (PSMCT) and on/off (PSMOO) mode operation are supported.

Benefits of using BeiDou B1C signal:

BeiDou B1C signal has the same center frequency as GPS L1 C/A enabling concurrent use of 4 GNSS constellations. Multi-GNSS configurations with BeiDou B1C also have a lower power consumption compared to those with BeiDou B1I.

- **Concurrent reception of 4 GNSSs** with GPS L1 C/A, Galileo E1, BeiDou B1C, and GLONASS L1OF.
- **Lower power consumption.** No additional frequency band required for BeiDou B1C in multi-GNSS constellations resulting in a lower power consumption during acquisition and tracking phases.

2.1.4 Communication interface configuration

Several configuration groups allow configuring the operation mode of the communication interfaces. These include parameters for the data framing, transfer rate and enabled input/output protocols. See [Communication interfaces and PIOs](#) section for details. The configuration groups available for each interface are:

Interface	Configuration groups
UART	CFG-UART1-*
	CFG-UART1INPROT-*
	CFG-UART1OUTPROT-*
I2C	CFG-I2C-*
	CFG-I2CINPROT-*
	CFG-I2COUTPROT-*
	CFG-TXREADY-*

Table 4: Interface configuration



The UART baudrate in MAX-M10S is configured to 9600 baud which is different to the firmware default. This ensures backwards compatibility with previous generations of u-blox MAX modules.

2.1.5 Message output configuration

The receiver supports two protocols for output messages: industry-standard NMEA and u-blox UBX. Any message type can be enabled or disabled individually and the output rate is configurable.

The message output rate is related to the frequency of an event. For example, the output message UBX-NAV-PVT (position, velocity, and time solution) is related to the navigation event, which generates a navigation epoch. In this case, the rate for each navigation epoch is defined by the configuration keys CFG-RATE-MEAS and CFG-RATE-NAV. For configuration examples of CFG-RATE-MEAS and CFG-RATE-NAV, see [Table 5](#).

Set the navigation rate value higher than one when the raw measurement data output rate needs to be higher than the navigation data rate.

Configuration example	CFG-RATE-MEAS	CFG-RATE-NAV	Measurement interval	Navigation epoch interval	Description
Example 1	1000	1	1000 ms	1000 ms	Measurement every 1000 ms, navigation solution for each measurement.
Example 2	1000	2	1000 ms	2000 ms	Measurement every 1000 ms, navigation solution for every second measurement.
Example 3	2000	1	2000 ms	2000 ms	Measurement every 2000 ms, navigation solution for each measurement.
Example 4	500	4	500 ms	2000 ms	Measurement every 500 ms, navigation solution for every fourth measurement.

Table 5: Measurement rate vs navigation rate configuration examples

The output rate for each message is defined in the CFG-MSGOUT-* configuration group. If the output rate of the message is set to one (1) on the UART interface, CFG-MSGOUT-UBX_NAV_PVT_UART1 = 1, the message is output for every navigation epoch. If the rate is set to two (2), the message is output for every other navigation epoch. If the rate is zero (0), then corresponding message is not output. As seen in this example, the rates of the output messages are individually configurable per communication interface.

Some messages, such as UBX-MON-VER, are non-periodic and are only output as an answer to a poll request.

The UBX-INF-* and NMEA-Standard-TXT information messages are non-periodic output messages that do not have a message rate configuration. Instead they can be enabled for each communication interface via the CFG-INFMSG-* configuration group.



All message output is additionally subject to the protocol configuration of the communication interfaces. Messages of a given protocol are not output unless the protocol is enabled for output on the interface. See [Communication interface configuration](#) for details.



The output rate of the NMEA-GxGSV message on the UART interface is configured to 5 in MAX-M10S. This differs from the firmware default to avoid overloading the communication interface and buffers.

2.1.6 Antenna supervisor configuration

This section gives an overview of the antenna supervisor configuration keys. The implementation of the antenna supervisor and a detailed description can be found in [Antenna supervisor](#).

The antenna supervisor is used to control an active antenna. The configuration of the antenna supervisor allows the following:

- Control voltage supply to the antenna, which allows the antenna supervisor to cut power to the antenna in the event of a short circuit or optimize power to the antenna in power save modes.
- Detect a short circuit in the antenna and automatically recover the antenna supply after the short circuit is no longer present.
- Detect an open circuit, which can be used to indicate if the antenna has been disconnected.



Using some antenna supervisor features may require disabling the UART or I2C interface and reconfiguring the PIOs as antenna supervisor pins.

Table 6 describes the configuration items.

Configuration item	Description	Comments
CFG-HW-ANT_CFG_VOLTCTRL	Enable active antenna voltage control	
CFG-HW-ANT_CFG_SHORTDET	Enable short circuit detection	
CFG-HW-ANT_CFG_SHORTDET_POL	Short antenna detection polarity	Set to 1 if the required logic polarity is active-low (default).
CFG-HW-ANT_CFG_OPENDET	Enable open circuit detection	
CFG-HW-ANT_CFG_OPENDET_POL	Open antenna detection polarity	Set to 1 if the required logic polarity is active-low (default).
CFG-HW-ANT_CFG_PWRDOWN	Power down antenna supply if short circuit is detected	Requires CFG-HW-ANT_CFG_VOLTCTRL and CFG-HW-ANT_CFG_SHORTDET to be enabled.
CFG-HW-ANT_CFG_PWRDOWN_POL	Power down antenna logic polarity	Set to 1 if the required logic polarity is active-high (default).
CFG-HW-ANT_CFG_RECOVER	Enables auto-recovery in the event of a short circuit	To use this feature, enable short circuit detection and CFG-HW-ANT_CFG_PWRDOWN.
CFG-HW-ANT_SUP_SWITCH_PIN	PIO number of the pin used for switching antenna supply	PIO5 is recommended if available. This pin can be used as an LNA_EN signal to control an external LNA, especially if the software standby mode or the power save mode on/off (PSMOO) operation is used.
CFG-HW-ANT_SUP_SHORT_PIN	PIO number of the pin used for detecting a short circuit in the antenna supply	Unused UART or I2C pins can be reassigned for short circuit detection.
CFG-HW-ANT_SUP_OPEN_PIN	PIO number of the pin used for detecting open/disconnected antenna	Unused UART or I2C pins can be reassigned for open circuit detection.
CFG-HW-ANT_ON_SHORT_US	Time delay between the antenna supply being turned on and the short circuit detection being activated (in microseconds)	Increase the time delay to avoid a short circuit to be detected before the antenna supply voltage has stabilized. Recommended values: 500 us (default) to 5000 us.

Table 6: Antenna supervisor configuration

It is possible to obtain the status of the antenna supervisor from the UBX-MON-RF message. For information about the *antStatus* and *antPower* fields, refer to the Interface description [3]. In addition, any changes in the status of the antenna supervisor are reported to the host interface as ANTSTATUS in NMEA notice messages.

ANTSTATUS	Description
OFF	Antenna is off
ON	Antenna is on
DONTKNOW	Antenna power status is not known

Table 7: Antenna power status

2.1.7 High performance navigation update rate configuration

The navigation update rate is a GNSS configuration item which determines how many position fixes the receiver calculates and outputs per second. The maximum achievable navigation update rate depends on the number of GNSS signals the receiver is tracking. Depending on the geographical location and GNSS constellations enabled, the number of tracking channels in use may vary. In particular, in Asia there is a large number of BeiDou satellites available potentially limiting the maximum achievable navigation update rate. In regions with a lower number of available BeiDou satellites, a higher navigation rate is achieved.

The maximum navigation update rate given in the datasheet is provided for a minimum of 98% position fix rate, i.e., up to 2% position fixes can get lost in case of high availability of signals. The navigation update rate can be increased beyond the maximum value stated in the datasheet. However, this may result in a reduced fix rate if a very large number of satellites is tracked.

u-blox M10 devices are optimized for low power consumption and come with the default CPU clock rate that supports the default navigation update rate stated in the product datasheet. However, it is possible to achieve a higher navigation update rate by configuring the device for a higher clock rate. This supports the high performance navigation update rate with minor increase in power consumption.



For the high navigation update rates, increase the communication speed and reduce the number of enabled messages.

The high performance navigation update rate can be configured in the device's one-time programmable (OTP) memory. The OTP configuration is only done once, and is subsequently applied automatically at every startup. The configuration string for setting the high CPU clock rate in the OTP memory is given in Table 8. This occupies 18 bytes of OTP memory space.



Changes made in the OTP configuration are permanent and cannot be reverted.

CPU clock	Configuration string
Default CPU clock	Default
High CPU clock	B5 62 06 41 10 00 03 00 04 1F 54 5E 79 BF 28 EF 12 05 FD FF FF FF 8F 0D B5 62 06 41 1C 00 04 01 A4 10 BD 34 F9 12 28 EF 12 05 05 00 A4 40 00 B0 71 0B 0A 00 A4 40 00 D8 B8 05 DE AE

Table 8: High performance navigation update rate configuration in OTP memory

To configure the high performance navigation update rate in OTP memory:

1. Power up the device.
2. Test the communication interface by polling the UBX-MON-VER message.
3. Send the configuration string provided in Table 8. The device returns two UBX-ACK-ACK messages with sequence of bytes B5 62 05 01 02 00 06 41 4F 78.

4. Set the device reset to a hardware reset. Power the device off and on or send the UBX-CFG-RST message to the device. The higher clock setting is applied at startup.
5. To verify that the configuration item is correctly set,
 - send the following sequence to the receiver: B5 62 06 8B 14 00 00 04 00 00 01 00 A4 40 03 00 A4 40 05 00 A4 40 0A 00 A4 40 4C 15
 - receive a UBX-CFG-VALGET message with the following sequence: B5 62 06 8B 24 00 01 04 00 00 01 00 A4 40 00 B0 71 0B 03 00 A4 40 00 B0 71 0B 05 00 A4 40 00 B0 71 0B 0A 00 A4 40 00 D8 B8 05 76 81
 - receive a UBX-ACK-ACK message with the following sequence: B5 62 05 01 02 00 06 8B 99 C2
6. The OTP memory configuration is completed and verified.

2.2 Navigation configuration

This section presents various configuration options related to the navigation engine. These options can be configured through CFG-NAVSPG-* configuration keys.

2.2.1 Dynamic platform

The dynamic platform model can be configured through the CFG-NAVSPG-DYNMODEL configuration item. For the supported dynamic platform models and their details, see [Table 9](#) and [Table 10](#).

Platform	Description
Portable	Applications with low acceleration, e.g. portable devices. Suitable for most situations.
Stationary	Used in timing applications (antenna must be stationary) or other stationary applications. Velocity restricted to 0 m/s. Zero dynamics assumed.
Pedestrian	Applications with low acceleration and speed, e.g. how a pedestrian would move. Low acceleration assumed.
Automotive	Used for applications with equivalent dynamics to those of a passenger car. Low vertical acceleration assumed.
At sea	Recommended for applications at sea, with zero vertical velocity. Zero vertical velocity assumed. Sea level assumed.
Airborne <1g	Used for applications with a higher dynamic range and greater vertical acceleration than a passenger car. No 2D position fixes supported.
Airborne <2g	Recommended for typical airborne environments. No 2D position fixes supported.
Airborne <4g	Only recommended for extremely dynamic environments. No 2D position fixes supported.
Wrist	Only recommended for wrist-worn applications. Receiver will filter out arm motion.

Table 9: Dynamic platform models

Platform	Max altitude [m]	Max horizontal velocity [m/s]	Max vertical velocity [m/s]	Sanity check type	Max position deviation
Portable	12000	310	50	Altitude and velocity	Medium
Stationary	9000	10	6	Altitude and velocity	Small
Pedestrian	9000	30	20	Altitude and velocity	Small
Automotive	6000	100	15	Altitude and velocity	Medium
At sea	500	25	5	Altitude and velocity	Medium
Airborne <1g	80000	100	6400	Altitude	Large
Airborne <2g	80000	250	10000	Altitude	Large
Airborne <4g	80000	500	20000	Altitude	Large

Platform	Max altitude [m]	Max horizontal velocity [m/s]	Max vertical velocity [m/s]	Sanity check type	Max position deviation
Wrist	9000	30	20	Altitude and velocity	Medium

Table 10: Dynamic platform model details

Applying dynamic platform models designed for high acceleration systems (e.g. airborne <2g) can result in a higher standard deviation in the reported position.

If a sanity check against the limit of the dynamic platform model fails, the position solution becomes invalid. [Table 10](#) shows the types of sanity checks which are applied for a particular dynamic platform model.

2.2.2 Navigation input filters

The navigation input filters in the CFG-NAVSPG-* configuration group control how the navigation engine handles the input data that comes from the satellite signal.

Configuration item	Description
CFG-NAVSPG-FIXMODE	By default, the receiver calculates a 3D position fix if possible but it reverts to 2D position if necessary (auto 2D/3D). The receiver can be configured to only calculate 2D (2D only) or 3D (3D only) positions.
CFG-NAVSPG-CONSTR_ALT, CFG-NAVSPG-CONSTR_ALTVAR	The fixed altitude is used if fixMode is set to 2D only. A variance greater than zero must also be supplied.
CFG-NAVSPG-INFIL_MINELEV	Minimum elevation of a satellite above the horizon to be used in the navigation solution. Low-elevation satellites may provide degraded accuracy, due to the long signal path through the atmosphere.
CFG-NAVSPG-INFIL_MINSVS, CFG-NAVSPG-INFIL_MAXSVS	Minimum and maximum number of satellites to use in the navigation solution. There is an absolute maximum limit of 32 satellites that can be used for navigation.
CFG-NAVSPG-INFIL_NCNOTHRS, CFG-NAVSPG-INFIL_CNOTHRS	A navigation solution will only be attempted if there is at least the given number of satellites with signals at least as strong as the given threshold.

Table 11: Navigation input filter parameters

If the receiver has only three satellites for calculating a position, the navigation algorithm uses a constant altitude to compensate for the missing fourth satellite. This is called a 2D fix. The constant altitude value is taken from the last successful 3D fix using a minimum of four available satellites.



u-blox receivers do not calculate any navigation solution with fewer than three satellites.

2.2.3 Navigation output filters

The result of a navigation solution is initially classified by the fix type (as detailed in the `fixType` field of the UBX-NAV-PVT message). This distinguishes between failures to obtain a fix ("No Fix") and cases where a fix has been achieved, which are further subdivided into specific types of fixes (for example, 2D, 3D).

Where a fix has been achieved, the fix is checked to determine whether it is valid or not. A fix is only valid if it passes the navigation output filters as defined in CFG-NAVSPG-OUTFIL. In particular, both PDOP and accuracy values must be below the respective limits.



Important: Users are recommended to check the `gnssFixOK` flag in the UBX-NAV-PVT or the NMEA valid flag. Fixes not marked as valid should not be used.

UBX-NAV-STATUS message also reports whether a fix is valid in the `gpsFixOK` flag. These messages have only been retained for backwards compatibility and it is recommended to use the UBX-NAV-PVT message.

2.2.4 Odometer filters

2.2.4.1 Speed (3D) low-pass filter

The CFG-ODO-OUTLPVEL configuration item activates a speed (3D) low-pass filter. The output of the speed low-pass filter is available in the UBX-NAV-VELNED message (`speed` field). The filtering level can be set via the CFG-ODO-VELLPGAIN configuration item and must be between 0 (heavy low-pass filtering) and 255 (weak low-pass filtering).



The internal filter gain is computed as a function of speed. Therefore, the level defines the nominal filtering level for speeds below 5 m/s, as defined in the CFG-ODO-VELLPGAIN configuration item.

2.2.4.2 Course over ground low-pass filter

The CFG-ODO-OUTLPCOG configuration item activates a course over ground low-pass filter when the speed is below 8 m/s. The output of the course over ground (also named heading of motion 2D) low-pass filter is available in the UBX-NAV-PVT message (`headMot` field), UBX-NAV-VELNED message (`heading` field), NMEA-RMC message (`cog` field), and NMEA-VTG message (`cogt` field). The filtering level can be set via the CFG-ODO-COGLPGAIN configuration item and must be between 0 (heavy low-pass filtering) and 255 (weak low-pass filtering).



The filtering level defines the filter gain for speeds below 8 m/s, as defined in the CFG-ODO-COGLPGAIN configuration item. If the speed is 8 m/s or higher, no course over ground low-pass filtering is performed.

2.2.4.3 Low-speed course over ground filter

The CFG-ODO-USE_COG configuration item activates this feature and the CFG-ODO-COGMAXSPEED, CFG-ODO-COGMAXPOSACC configuration items are used to configure a low-speed course over ground filter (also named heading of motion 2D). This filter derives the course over ground from position at very low speed. The output of the low-speed course over ground filter is available in the UBX-NAV-PVT message (`headMot` field), UBX-NAV-VELNED message (`heading` field), NMEA-RMC message (`cog` field) and NMEA-VTG message (`cogt` field). If the low-speed course over ground filter is not configured, then the course over ground is computed as described in section [Freezing the course over ground](#).

2.2.5 Static hold

The static hold mode allows the navigation algorithms to decrease the noise in the position output when the velocity is below a predefined "Static Hold Threshold" level. This reduces the position wander caused by environmental factors such as multi-path and improves position accuracy especially in stationary applications.

By default, the static hold mode is disabled.

The CFG-MOT-GNSSSPEED_THRS configuration item defines the static hold speed threshold. If the speed drops below the defined "Static Hold Threshold", static hold mode is activated. Once static hold mode is active, the position output is kept static and the velocity is set to zero until there is evidence of movement again. Such evidence can be velocity, acceleration, changes of the valid flag (for example, position accuracy estimate exceeding the position accuracy mask, see also section [Navigation output filters](#)), position displacement, etc.

The CFG-MOT-GNSSDIST_THRS configuration item defines the static hold distance threshold. If the distance between the estimated position and the static hold position exceeds the defined threshold, the static hold mode is suspended or deactivated until there is evidence of no movement.

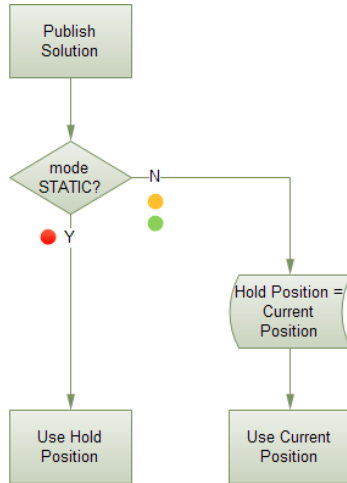


Figure 3: Position output in static hold mode



If the low-speed course over ground filter is deactivated or inactive (see section [Low-speed course over ground filter](#)), the receiver derives the course over ground from the GNSS velocity information. If the velocity cannot be calculated with sufficient accuracy (for example, with bad signals) or if the absolute speed value is very low (under 0.1 m/s), the course over ground value becomes inaccurate too. In this case the course over ground value is frozen, that is, the previous value is kept and its accuracy degrades over time. These frozen values will not be output in the NMEA messages NMEA-RMC and NMEA-VTG unless the NMEA protocol is explicitly configured to do so (see NMEA protocol configuration in the applicable Interface description [3]).

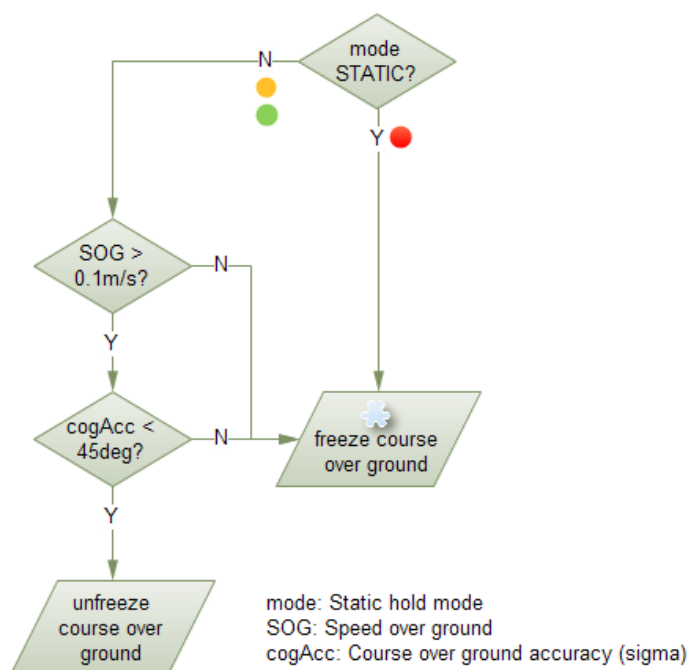


Figure 5: Flowchart of course over ground freezing

2.2.7 Super-Signal (Super-S) technology

This feature improves receiver performance when GNSS signals are weakened, for example, due to low-gain antennas or poor antenna placement.

Under normal conditions, the receiver down-weights or discards weak signals to reduce errors, such as those caused by multipath reflections. The Super-S feature modifies this behavior by adjusting the weighting of weak signals. By compensating for reduced signal strength, the receiver maintains reliable positioning in challenging conditions.

The weak signal compensation can be configured to three different modes with the CFG-NAVSPG-SIGATTCOMP configuration key as follows:

- Disabled: no weak signal compensation
- Automatic: the receiver automatically estimates and compensates for the weak signal
- Configured: the receiver compensates for the weak signal based on a configured value

When using the `configured` mode, set the maximum expected C/N0 observed in a clear-sky environment, excluding any outliers or unusually high values.

When enabling this feature, due to multi-path contamination or interference rather than poor signal reception, the receiver may overtrust distorted signals. This can reduce the positioning accuracy.

The Super-S feature (enabled by default) is not compatible with the [Protection level](#) feature. Disable Super-S before using the Protection level feature.

2.3 OTP memory configuration

MAX-M10S contains a one-time programmable (OTP) memory. This is a non-volatile memory for storing configuration settings and ROM patches permanently in the device. The stored data cannot be modified after it has been initially programmed. The device applies the settings and ROM patches on the device startup.

As the space in the OTP memory is limited, only essential system configuration settings should be stored. The total space used for device configuration and ROM patches in the OTP memory must not exceed 64 bytes. Other settings can be stored in the BBR or sent from the host to the device on each device startup.



Ensure that the final configuration stored and optional ROM patches do not require more than 64 bytes of OTP memory space.

3 Receiver functionality

This chapter describes MAX-M10S operational features and their configuration.

3.1 Augmentation systems

3.1.1 SBAS

MAX-M10S is capable of receiving multiple Satellite Based Augmentation System (SBAS) signals concurrently, even from different SBAS systems (WAAS, EGNOS, etc.). SBAS signals are recommended to be used only for correction data. These signals can also be used for navigation, but with their low weighting, they only have a minor impact on the navigation solution.

For receiving correction data, MAX-M10S automatically chooses the best SBAS satellite as its primary source. It selects only one satellite since the information received from other SBAS satellites is redundant and could be inconsistent. The selection strategy is determined by the proximity of the satellites, the services offered by the satellite, the configuration of the receiver (test mode allowed/disallowed, integrity enabled/disabled) and the signal link quality to the satellite.

If corrections are available from the chosen SBAS satellite and used in the navigation solution, the differential correction status is indicated in several output messages such as UBX-NAV-PVT, UBX-NAV-STATUS, UBX-NAV-SAT, UBX-NAV-SIG, NMEA-GGA, NMEA-GLL, NMEA-RMC, and NMEA-GNS. The UBX-NAV-SBAS message provides detailed information about the corrections available and applied. Refer to the Interface description [3] for a detailed description of the messages.

The most important SBAS feature for accuracy improvement is the ionosphere correction parameters. The measured data from regional Ranging and Integrity Monitoring Stations (RIMS) are combined to make a Total Electron Content (TEC) map. This map is transferred to the receiver via SBAS satellites to allow a correction of the ionosphere delays on each received signal.

Message type	Message content	Source
0(0/2)	Test mode	All
1	PRN mask assignment	Primary
2, 3, 4, 5	Fast corrections	Primary
6	Integrity	Primary
7	Fast correction degradation	Primary
9	Satellite navigation (ephemeris)	All
10	Degradation	Primary
12	Time offset	Primary
17	Satellite almanac	All
18	Ionosphere grid point assignment	Primary
24	Mixed fast / long-term corrections	Primary
25	Long-term corrections	Primary
26	Ionosphere delays	Primary

Table 12: Supported SBAS messages

Each satellite serves a specific region and its correction signal is only useful within that region. Planning is crucial to determine the best possible configuration, especially in areas where signals from different SBAS systems can be received:

- **Example 1 - SBAS receiver in North America:** In eastern parts of North America, make sure that EGNOS satellites do not take preference over WAAS satellites. The satellite signals from the EGNOS system should be disallowed by using the PRN scan mask (configuration key CFG-SBAS-PRNSCANMASK).
- **Example 2 - SBAS receiver in Europe:** Some WAAS satellite signals can be received in some parts of western Europe and GAGAN SBAS satellites in other parts of Europe. Therefore, it is recommended that satellites from all but the EGNOS system are disallowed using the PRN scan mask.



Although u-blox receivers try to select the best available SBAS correction data, it is recommended to configure them to exclude the unwanted SBAS satellites.

To configure the SBAS functionality, use the CFG-SBAS-* configuration group.

Parameter	Description
CFG-SIGNAL-SBAS_ENA	Enabled/disabled status of the SBAS subsystem
CFG-SBAS-USE_TESTMODE	Allow/disallow SBAS usage from satellites in test mode (enable when BDSBAS is used)
CFG-SBAS-USE_RANGING	Use the SBAS satellites for navigation (ranging)
CFG-SBAS-USE_DIFFCORR	Combined enable/disable switch for fast, long-term, and ionosphere corrections
CFG-SBAS-USE_INTEGRITY	Apply integrity information data
CFG-SBAS-PRNSCANMASK	Allows selectively enabling/disabling SBAS satellites (BDSBAS disabled by default)

Table 13: SBAS configuration parameters



When SBAS integrity data is applied, the navigation engine stops using all signals for which no integrity data is available (including all non-GPS signals). It is not recommended to enable SBAS integrity on borders of SBAS service regions in order not to inadvertently restrict the number of available signals.



SBAS integrity information is required for at least five GPS satellites. If this condition is not met, SBAS integrity data will not be applied.



When the receiver switches from a solution using correction data to a standard position solution, the reference frame of the output position switches as well. For an SBAS solution, the reference frame is aligned within a few centimeters of WGS84 (and modern ITRF realizations).



Other SBAS systems (such as KASS) can be enabled by adding the corresponding PRNs to the CFG-SBAS-PRNSCANMASK list.

3.1.2 QZSS SLAS

QZSS SLAS (Sub-meter Level Augmentation Service) is an augmentation technology, which provides correction data for pseudoranges of GPS, QZSS, and other major GNSS satellites. The correction stream is transmitted on the L1S signal at the L1 frequency (1575.42 MHz).

For more information on QZSS SLAS, visit qzss.go.jp/en/.

Multiple QZSS SLAS signals can be received simultaneously. When receiving QZSS SLAS correction data, MAX-M10S autonomously selects the best QZSS satellite. The selection strategy is determined by the quality of the QZSS L1S signals, the receiver configuration (test mode allowed or not), and the location of the receiver with respect to the QZSS SLAS coverage area. When outside of this coverage area, the receiver likely falls back to using SBAS corrections.

If QZSS SLAS corrections are used in the navigation solution, the differential status is indicated in several output messages such as UBX-NAV-PVT, UBX-NAV-STATUS, UBX-NAV-SAT, NMEA-GGA, NMEA-GLL, NMEA-RMC, and NMEA-GNS. The UBX-NAV-SLAS message provides detailed

information about which corrections are available and applied. Refer to the Interface description [3] for a detailed description of the messages.

Message type	Message content
0	Test mode
47	Monitoring station information
48	PRN mask
49	Data issue number
50	DGPS correction
51	Satellite health

Table 14: Supported QZSS L1S SLAS messages for navigation enhancement

Use the configuration key CFG-SIGNAL-QZSS_L1S_ENA to enable QZSS L1S signal. For further QZSS SLAS functionality, use the CFG-QZSS-USE_SLAS* configuration keys.

Parameter	Description
CFG-QZSS-USE_SLAS_DGNSS	Apply QZSS SLAS corrections
CFG-QZSS-USE_SLAS_TESTMODE	Allow the correction provided by QZSS satellites that are in test mode
CFG-QZSS-USE_SLAS_RAIM_UNCORR	If this configuration is set, the receiver will try to estimate the position by using only corrected measurements; if all corrected measurements are not available, it will not use any corrections. If this configuration is not set, the receiver will mix corrected and uncorrected measurements for the navigation solution.
CFG-QZSS-SLAS_MAX_BASELINE	Maximum distance from the closest ground monitoring station (GMS) to apply the SLAS corrections. Note that due to the nature of the service, the usefulness of corrections degrades with distance. When far away from GMS, SBAS may be a better correction source.

Table 15: QZSS SLAS configuration parameters



If the RAIM option is set, QZSS is the only GNSS time system that measurements can observe.

3.2 Communication interfaces and PIOs

MAX-M10S supports communication over UART and I2C interfaces with a host CPU. UBX and NMEA protocols can be enabled simultaneously with individual interface settings, e.g. for baud rate, message rates, and so on.

3.2.1 UART

MAX-M10S supports a Universal Asynchronous Receiver/Transmitter (UART) port consisting of an RX and a TX line. The UART can be used as a host interface which supports a configurable baud rate and protocol selection.



The UART interface does not support handshaking signals or hardware flow control signals.

The UART baud rate can be configured for selected speeds. Different rates than these speeds are not supported for transmission and reception.



The UART RX interface is disabled when more than 100 frame errors are detected during a one-second period. This can happen if the wrong baud rate is used or the UART RX pin is grounded. An error message appears when the UART RX interface is re-enabled at the end of the one-second period.

Baud rate	Data bits	Parity	Stop bits
4800	8	none	1
9600	8	none	1

Baud rate	Data bits	Parity	Stop bits
19200	8	none	1
38400	8	none	1
57600	8	none	1
115200	8	none	1
230400	8	none	1
460800	8	none	1
921600	8	none	1

Table 16: Possible UART interface configurations

Allow a short time delay of typically 100 ms between sending a baud rate change message and providing input data at the new rate. Otherwise some input characters may be ignored or the port could be disabled until the interface is able to process the new baud rate.



The default baud rate is 9600 baud. Using a lower baud rate may cause buffering problems.

If there is too much data for the interface's bandwidth, the output buffer will fill up. Once the buffer space is exceeded, new messages to be sent will be dropped. To prevent message loss, the baud rate and the number of enabled messages should be selected carefully.

3.2.2 I2C

An I2C interface is available for communication with an external host CPU in the I2C Fast-mode. Backwards compatibility with the Standard-mode I2C bus operation is not supported. The interface can be operated only in the peripheral mode with the maximum bit rate of 400 kbit/s. The interface can make use of clock stretching by holding the SCL line LOW to pause a transaction. In this case, the bit transfer rate is reduced. The maximum clock stretching time is 20 ms.

The SCL and SDA pins have internal pull-up resistors which should be sufficient for most applications. However, depending on the clock speed of the host and the capacitive load on the I2C lines, additional external pull-up resistors may be necessary. The higher the speed and the capacitance load, the lower the pull-up resistor needs to be.

To poll or set the I2C address, use the CFG-I2C-ADDRESS configuration item. Refer to Interface description [3] for details. The CFG-I2C-ADDRESS configuration item is an 8-bit value containing the I2C address in the 7 most significant bits plus a 0 as the least significant bit. Thus, the default address becomes 0x84(1000 0100).



In designs where the host uses the same I2C bus to communicate with more than one u-blox receiver, each receiver's I2C address must be configured with a different value.

3.2.2.1 I2C register layout

As shown in Figure 6, there are 256 registers. The data registers 0 to 252, at addresses 0x00 to 0xFC, contain reserved information and must not be used. Hence, only the last three registers are left for communication. The registers 0xFD and 0xFE contain the currently available number of bytes to be read, while the register 0xFF buffers the message stream. The 0xFF address delivers a 0xFF byte value if there is no data awaiting for transmission, or all the bytes have been read.

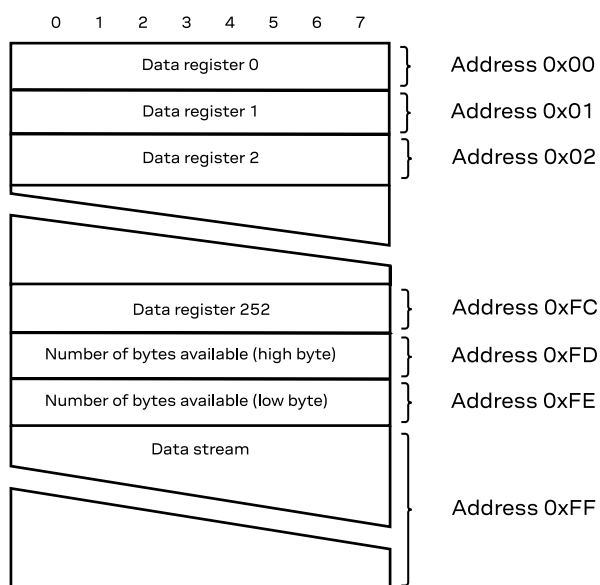


Figure 6: I2C register layout

3.2.2.2 Read access types

The host can choose one of the following two modes:

- Random read access: the controller first reads the number of available bytes at the 0xFD and 0xFE before accessing the data at 0xFF.
- Current address read access: the controller directly reads the data at the register 0xFF, without knowing first if there is any data waiting. If there is no data, the read result is a 0xFF byte value. This mode basically skips the first step of the "random read access", as it does not address to any particular register.

Figure 7 shows the format of the "random access" form of the request.

Following the start condition from the controller, the 7-bit device address and the R/W bit (which is a logic low for write access) are clocked onto the bus by the controller transmitter. The receiver answers with an acknowledge (logic low) to indicate that it recognizes the address.

Next, the 8-bit address of the register to be read must be written to the bus (0xFD for u-blox receivers). Following the receiver's acknowledgment, the controller again triggers a start condition and writes the device address, but this time the R/W bit is a logic high to initiate the read access. Now, the controller can read 1 to N bytes from the receiver. The receiver will first deliver the byte value at 0xFD, followed by the value at 0xFE. At this point the controller knows the number of bytes waiting at the 0xFF register, and by acknowledging again, the data stream follows. The data transfer will stop once the controller emits a not-acknowledge response or a stop condition is triggered after the last byte has been read.

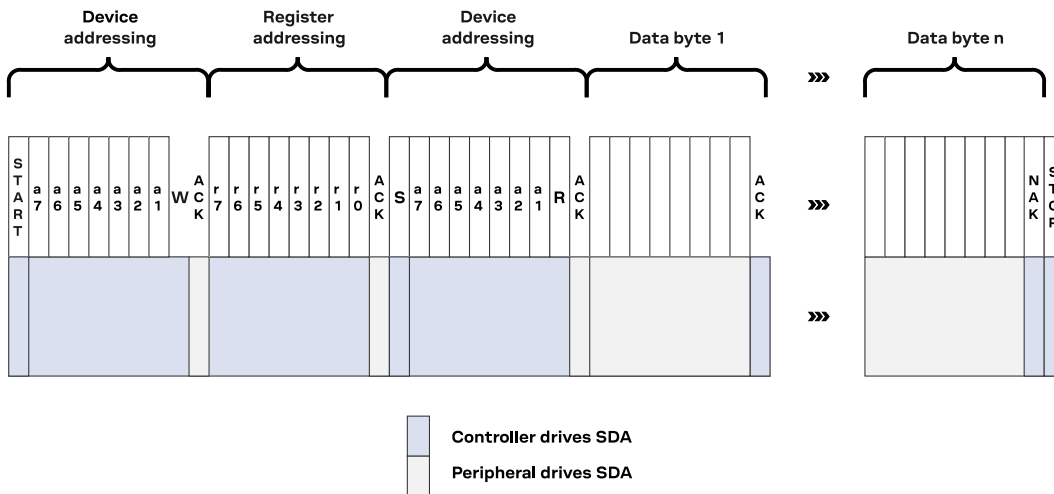


Figure 7: I2C random read access

If "current address" is used, an address pointer in the receiver is used to determine which register to read. This address pointer will increment after each read operation unless it is already pointing at register 0xFF, the highest addressable register, in which case it remains unaltered.

The initial value of this address pointer at startup is 0xFF, so by default all current address read operations will repeatedly read register 0xFF and receive the next byte of message data (or 0xFF value if no message data is waiting).

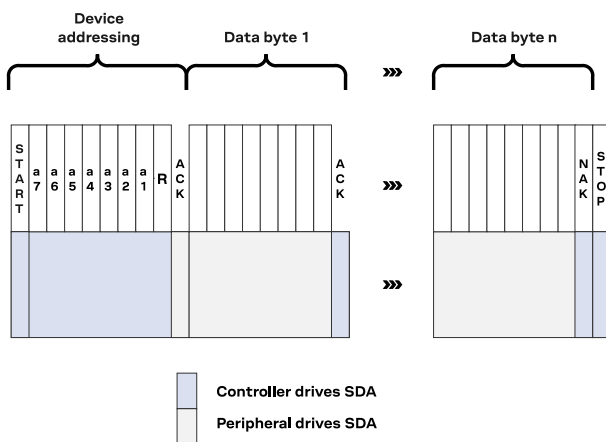


Figure 8: I2C current address read access

Only after addressing the peripheral, the receiver starts the data stream. If the controller does not read data from the receiver for a certain timeout, the receiver assumes that the communication is broken and stops the data stream, preventing an overflow of the output buffer. This timeout is 1.5 seconds by default. However, it can be extended by setting the CFG-I2C-EXTENDEDTIMEOUT configuration item to true. Refer to the Interface description [3] for details. By disabling the timeout, the receiver will only interrupt the data stream when the buffer is full. The buffer can store up to 4 kB and the time for an overflow event depends on the number of messages enabled.

3.2.2.3 Write access

The receiver does not provide any write access except for writing UBX and NMEA messages to the receiver, such as configuration or aiding data. Therefore, the register set mentioned in section [I2C register layout](#) is not writeable.

Following the start condition from the controller, the 7-bit device address and the R/W bit (which is a logic low for write access) are clocked onto the bus by the controller. The receiver answers with an acknowledge (logic low) response to indicate that it is responsible for the given address.

The controller can write 2 to N bytes to the receiver, generating a stop condition after the last byte being written. To properly distinguish from the write access to set the address counter in random read accesses, the number of data bytes must be at least 2.

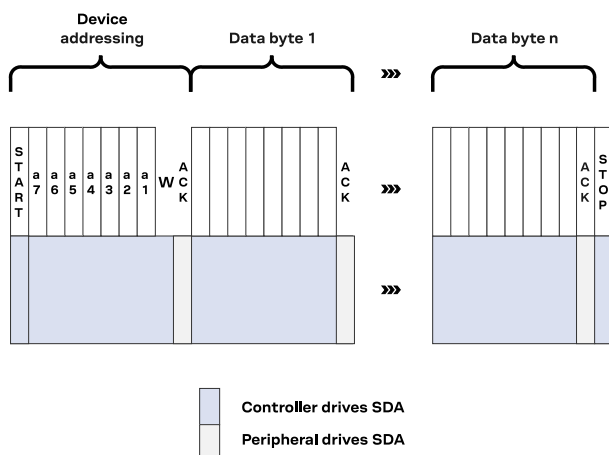


Figure 9: I2C write access

3.2.3 PIOs

This section describes the PIOs supported by MAX-M10S. All PIO active voltage levels are related to the V_{IO} supply voltage. All the inputs have internal pull-up resistors in normal operation and can be left open if unused.

- When assigning a different function to a PIO, ensure that the default function is disabled where applicable. For example, disable the I2C interface with the CFG-I2C-ENABLED configuration key if I2C pins are used for antenna supervisor functions.

3.2.3.1 RESET_N

MAX-M10S provides a RESET_N pin to reset the receiver. The RESET_N pin is input-only with an internal pull-up resistor to V_{IO} and should be left open for normal operation. Driving RESET_N low for at least 1 ms triggers a receiver reset. The RESET_N complies with the V_{IO} level and can be actively driven high.

- Use RESET_N only in critical situations to recover the receiver. RESET_N resets the receiver and clears the FW image in the code RAM and the BBR content including receiver configuration, real-time clock (RTC), and GNSS orbit data, triggering a cold start.
- No capacitor should be placed at RESET_N to GND, otherwise it could trigger a reset on every startup.

3.2.3.2 SAFEBOOT_N

The SAFEBOOT_N pin is for future service, updates and reconfiguration.

- The SAFEBOOT_N pin is internally connected to the TIMEPULSE pin through a 1 kΩ series resistor.

3.2.3.3 TIMEPULSE

MAX-M10S features one time pulse output at the TIMEPULSE pin. This can only be configured in PIO4. The details about this feature are explained in the section [Time pulse](#).



The TIMEPULSE and SAFEBOOT_N functions share the same internal IC function. If this pin is low at receiver startup, the receiver will enter safeboot mode. However, in normal operation the pin outputs the time pulse signal. Make sure this pin has no load that could pull it low at startup.

3.2.3.4 LNA_EN

The LNA_EN signal can be used to turn on and off an optional external LNA and an active antenna supply to optimize the power consumption in the backup modes and the power save mode on/off operation (PSMOO). The LNA and the active antenna supply are turned on when the LNA_EN signal is "high".

The LNA_EN signal is also used internally in MAX-M10S to control the integrated LNA. The polarity cannot be changed.

The LNA_EN signal can also be used as a part of an antenna supervisor circuit to control an active antenna power supply.

3.2.3.5 EXTINT

MAX-M10S supports external interrupts at the EXTINT pin. The EXTINT pin has a fixed input voltage threshold with respect to V_{IO}. It can be used for functions such as wake-up source from Software backup mode, accurate external [Frequency assistance](#), [Time assistance](#), and [Time mark](#) reporting.

The EXTINT pin enables [External control](#) for host-controlled on/off operation of the receiver, and as a wake-up source for power save mode on/off operation (PSMOO). If configured for host-controlled on/off operation, the internal pull-up is disabled. Make sure the EXTINT input is always driven within the defined voltage level by the host.

The EXTINT pin can also be configured for another functionality.



EXTINT functionality is only available at the EXTINT pin.

3.2.3.6 TX_READY

The receiver includes an internal message buffer that stores bytes to be sent to the host application. The TX_READY feature changes the polarity of the TX_READY pin to indicate when the buffer contains bytes ready for transmission. This allows the host application to wait for the signal instead of continuously polling the interface. The signal stays active until all of the bytes in the buffer have been transferred.

Each interface of the receiver includes a small additional transmit buffer. After the TX-ready signal deactivates, up to 16 bytes may still remain to be transferred to the host.

The TX-ready signal is enabled and configurable with the CFG-TXREADY-* configuration group.

The TX-ready signal can be assigned to any pin. Any previous configuration on the pin must be disabled before the assignment. The verification can be done with the UBX-MON-HW3 message.

The CFG-TXREADY-THRESHOLD parameter is specified in increments of 8 bytes. To configure the threshold, divide the desired byte count by 8. For instance, setting the parameter to 128 corresponds to a threshold of 1024 bytes (128 × 8 bytes). It is recommended not to exceed a threshold value of 256, equivalent to 2048 bytes, as higher values may result in delayed activation of the TX-ready signal and potential loss of messages.

3.2.3.6.1 Extended TX timeout

If the host does not communicate for more than 1.5 seconds, the receiver assumes that the host is no longer using this interface and no more packets are scheduled for this interface. This mechanism can be changed by enabling "extended TX timeouts" using the configuration key CFG-

I2C-EXTENDEDTIMEOUT. When enabled, the receiver delays idling the interface until the allocated and undelivered bytes for this interface reach 4 kB.

This feature is especially useful when using the TX-ready feature with a message output rate of less than once per second, and fetching data only when available, determined by the TX_READY pin becoming active.

3.3 Antenna

This section explains the antenna supervisor feature and the available implementation options.

3.3.1 Antenna supervisor

An active antenna supervisor provides the means to check the antenna for open and short circuits and to shut off the antenna supply if a short circuit is detected. Once enabled, the active antenna supervisor produces status messages that are reported in NMEA and/or UBX protocols. MAX-M10S supports two antenna supervisor variants: three-pin and two-pin implementations.

The three-pin antenna supervisor is able to detect short and open circuits and control the antenna supply. The two-pin antenna supervisor is a reduced version which is able to control the antenna supply and detect short circuits.

An overview of the two antenna supervisor variants is given in [Table 17](#). It is recommended to make use of the full capabilities of the antenna supervisor (detect open and short circuits, and control the antenna supply).

The antenna supervisor can be configured through the CFG-HW-ANT_* configuration items. This includes enabling and disabling as well as changing the polarity of each signal. The current configuration of the active antenna supervisor can also be checked by polling the related CFG-HW-ANT_* configuration items.

The active antenna status can be determined by polling the UBX-MON-RF message or checking the NMEA notice messages. If an antenna is connected, the initial state after power-up is reported in the UBX-MON-RF message in *antStatus* and *antPower* fields. For more information, refer to Interface description [\[3\]](#)

Features	Three-pin	Two-pin
Short detection	Yes	Yes
Open detection	Yes	No
External components	Discrete and IC	Discrete and IC
Number of PIOs needed	Three	Two

Table 17: Antenna supervisor overview

3.3.1.1 Three-pin antenna supervisor

An active antenna supervisor circuit uses the ANT_DETECT, ANT_OFF_N, and ANT_SHORT_N signals. The ANT_OFF_N signal is already enabled and assigned to the LNA_EN pin in MAX-M10S. The ANT_DETECT and ANT_SHORT_N signals can be assigned to any unused PIOs, which may require disabling the previous function of the PIOs. For example, the open circuit detection uses the ANT_DETECT signal, "high" = Antenna detected (antenna consumes current); "low" = Antenna not detected (no current drawn). To enable the three-pin antenna supervisor, the ANT_DETECT and ANT_SHORT_N signals must be enabled in the receiver configuration. The polarity of the ANT_DETECT and ANT_SHORT_N signals must also be defined in the receiver configuration based on the design use case.

Part	Description
R6	Defines the threshold of the comparator

Table 18: Components in antenna supervisor

The threshold voltage (V_{REF}) of the comparator is defined by R5 and R6. It can be calculated as:
 $V_{REF} = R6/(R6+R5)*V_{ANT}$.

 The open drain buffers shown in [Figure 10](#) are not needed if V_{ANT} has the same voltage level as V_{IO} .

3.3.1.2 Two-pin antenna supervisor

The reduced functionality antenna supervisor circuit is connected to two signals: antenna control (ANT_OFF_N) and antenna status detection (ANT_SHORT_N). The ANT_OFF_N signal is already enabled and assigned to the LNA_EN pin in MAX-M10S and the ANT_SHORT_N signal can be assigned to any unused PIO, which may require disabling the previous function of the PIO. To enable the reduced antenna supervisor, the ANT_SHORT_N signal must be enabled in the receiver configuration. The polarity of the ANT_SHORT_N signal must also be defined in the receiver configuration based on the design use case.

The antenna can be supplied by VCC_RF or an external supply. Note that the supply voltage must be clean, as any noise could directly couple into the RF part of the GNSS receiver affecting the overall GNSS performance.

Refer to [Reference designs](#) for antenna supervisor examples and the required configuration.

[Figure 11](#) presents the required two-pin antenna supervisor circuit and subsequent sections describe how to enable and monitor each feature.

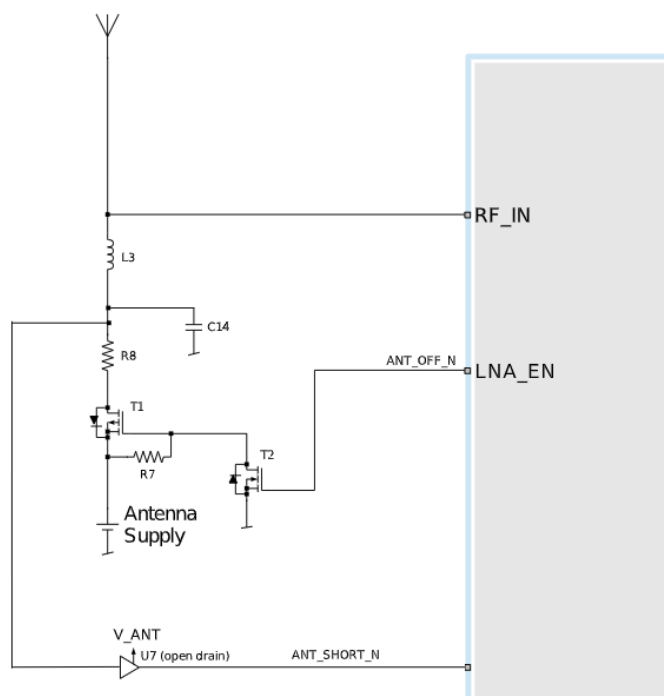

Figure 11: MAX-M10S two-pin antenna supervisor

Table 19 presents a list of the external components required for implementing the two-pin antenna supervisor design in Figure 11. Refer to External components for the recommended parts and specification.

Part	Description
C14	Filtering capacitor
L3	DC infeed inductor
T1, T2	p-channel, n-channel MOSFET acting as a switch to control the antenna supply
U7	Open drain buffers to shift voltage levels
R7	Passive pull-up to control T1
R8	Current limiter in the event of a short circuit

Table 19: Components in two-pin antenna supervisor

 The open drain buffer shown in Figure 11 is not needed if V_ANT is the same voltage level as V_IO.

3.3.1.3 Antenna voltage control - ANT_OFF_N

The antenna voltage control is enabled by default in MAX-M10S with the configuration item CFG-HW-ANT_CFG_VOLTCTRL set to true (1).

 The antenna status (as reported in UBX-MON-RF and UBX-INF-NOTICE messages) is not reported unless the antenna voltage control has been enabled.

Result:

- UBX-MON-RF: Antenna status = OK. Antenna power status = ON.
- ANT_OFF_N = active low. The pin is pulled high to enable an external antenna or LNA.

Startup message at power-up if the configuration is stored:

```
$GNTXT,01,01,02,ANTSUPERV=AC *00
```

```
$GNTXT,01,01,02,ANTSTATUS=INIT*3B
```

```
$GNTXT,01,01,02,ANTSTATUS=OK*25
```

ANTSUPERV=AC indicates that antenna control is activated.

3.3.1.4 Antenna short detection - ANT_SHORT_N

Enable the antenna short detection by setting the configuration item CFG-HW-ANT_CFG_SHORTDET to true (1).

Result:

- UBX-MON-RF: Antenna status = OK. Antenna power status = ON.
- ANT_OFF_N = active low to disable an external antenna. Therefore, the pin is pulled high to enable an external antenna.
- ANT_SHORT_N = active low to report a short circuit. The pin is default high (PIO pull-up enabled).

Startup message at power-up if the configuration is stored:

```
$GNTXT,01,01,02,ANTSUPERV=AC SD *37
```

```
$GNTXT,01,01,02,ANTSTATUS=INIT*3B
```

```
$GNTXT,01,01,02,ANTSTATUS=OK*25
```

ANTSUPERV=AC SD indicates that antenna control and short detection are activated.

If a short circuit is detected in the antenna (ANT_SHORT_N pulled low):

```
$GNTXT,01,01,02,ANTSTATUS=SHORT*73
```

- UBX-MON-RF: Antenna status = SHORT. Antenna power status = ON (automatic power-down is not enabled, CFG-HW-ANT_CFG_PWRDOWN or powerDown field in OTP memory file 0x37 is off by default).
- ANT_OFF_N = high (external antenna enabled).



If CFG-HW-ANT_CFG_PWRDOWN is already enabled (set to true), the polarity of the ANT_OFF_N signal changes to power down (disable) the antenna supply when a short is detected.



After a detected antenna short, the reported antenna status continues to be reported as a SHORT. If auto-recovery is enabled for the antenna short detection, the antenna status can recover after a timeout of 60 seconds. Recovering the antenna status immediately requires either a power cycle or switching the antenna short detection off and on again.

3.3.1.5 Antenna short detection auto-recovery

Enable the antenna short detection auto-recovery by setting the configuration item CFG-HW-ANT_CFG_RECOVER to true (1).



To use the auto-recovery feature, enable CFG-HW-ANT_CFG_PWRDOWN which requires CFG-HW-ANT_CFG_SHORTDET and CFG-HW-ANT_CFG_VOLTCTRL to be enabled.

Result:

- UBX-MON-RF: Antenna status = OK. Antenna power status = ON.
- ANT_OFF_N = active low. The pin is pulled high to enable an external antenna.
- ANT_SHORT_N = active low. The pin is default high (PIO pull-up enabled, to be pulled low if a SHORT is detected).

Startup message at power-up if the configuration is stored:

```
$GNTXT,01,01,02,ANTSUPERV=AC SD PDOS SR*3E
```

```
$GNTXT,01,01,02,ANTSTATUS=INIT*3B
```

```
$GNTXT,01,01,02,ANTSTATUS=OK*25
```

ANTSUPERV=AC SD PDOS SR (indicates short circuit recovery added - SR)

If short circuit is detected (ANT_SHORT_N pulled low):

```
$GNTXT,01,01,02,ANTSTATUS=SHORT*73
```

- UBX-MON-RF: Antenna status = SHORT. Antenna power status = OFF (automatic power-down is enabled).
- ANT_OFF_N = low (external antenna disabled).

After a timeout period of 60 seconds, the receiver retests the short circuit condition by enabling the antenna (i.e. pulling ANT_OFF_N high).

If a short is not present, the receiver reports antenna condition is OK:

```
$GNTXT,01,01,02,ANTSTATUS=OK*25
```

UBX-MON-RF: Antenna status = OK. Antenna power status = ON.

3.3.1.6 Antenna open circuit detection - ANT_DETECT

Enable the antenna open circuit detection by setting the configuration item CFG-HW-ANT_CFG_OPENDT to true (1).

Result:

- UBX-MON-RF: Antenna status = OK. Antenna power status = ON.
- ANT_OFF_N = active low. The pin is pulled high to enable an external antenna.
- ANT_SHORT_N = active low. The pin is default high (PIO pull-up enabled, to be pulled low if a SHORT is detected).
- ANT_DETECT = active high. The pin is default high (PIO pull-up enabled, to be pulled low if the antenna is not detected).

Startup message at power-up if the configuration is stored:

```
$GNTXT,01,01,02,ANTSUPERV=AC SD OD PDoS SR*15
```

```
$GNTXT,01,01,02,ANTSTATUS=INIT*3B
```

```
$GNTXT,01,01,02,ANTSTATUS=OK*25
```

ANTSUPERV=AC SD OD PDoS SR (indicates open circuit detection added - OD)

If ANT_DETECT is pulled low to indicate no antenna connected:

```
$GNTXT,01,01,02,ANTSTATUS=OPEN*35
```

If ANT_DETECT is left floating or pulled high to indicate antenna connected:

```
$GNTXT,01,01,02,ANTSTATUS=OK*25
```

3.3.1.7 Antenna status reporting

The antenna detection and antenna power status that is available in UBX-MON-RF and NMEA notice messages, is based on the antenna's physical state. The required antenna supervisor configuration keys depend on the selected antenna supervisor implementation (three-pin or two-pin).

Table 20 and Table 21 present a summary of the antenna status that is available in the *antStatus* and *antPower* fields of the UBX-MON-RF message. Refer to the Interface description [3] for more information.

Status	Description
DON'T KNOW	Antenna status is unknown.
INIT	Antenna power control feature is initialized (if CFG-HW-ANT_VOLTCTRL is enabled).
SHORT	A short is detected from the antenna input. That is, a lot of current is drawn from the active antenna.
OPEN	Antenna is not detected. That is, little or no current is drawn from the active antenna.
OK	Antenna is detected and no short is detected.

Table 20: Available antenna detection status

Status	Description
DON'T KNOW	CFG-HW-ANT_VOLTCTRL not configured.
OFF	GNSS OFF or a short is detected and CFG-HW-ANT_PWRDOWN is enabled. Note that this status also applies when the GNSS is restarted with the CFG-RST or implicitly with the CFG-GNSS messages, when the GNSS selection is reconfigured, or when the GNSS is stopped in the software standby mode and the off state of the power save mode is on/off (PSMOO).
ON	GNSS ON and no short/open is detected from the antenna input. Similarly, when there is a short and CFG-HW-ANT_PWRDOWN is not enabled or auto-recovery is enabled.

Table 21: Available antenna power status

Table 22 shows some possible combinations of the antenna supervisor configuration and the expected antenna status based on the physical state of the antenna. Note that the short detection takes priority over the open detection and "X" in Table 22 implies an unconfigured or undetected physical state. In addition, CFG-HW-ANT_PWRDOWN requires that CFG-HW-ANT_CFG_VOLTCTRL and CFG-HW-ANT_CFG_SHORTDET are enabled. Likewise, CFG-HW-ANT_RECOVER requires CFG-HW-ANT_PWRDOWN to be enabled.



The antenna supervisor is re-initialized after issuing a reset with RESET_N pin or changing any of the antenna supervisor configuration keys. Therefore, if the default configuration has changed, it is recommended to save the antenna supervisor configuration to BBR to ensure that the updated configuration is applied after a reset.

Configuration keys					Physical antenna state		Reported antenna status	
VOLTCTRL	SHORTDET	OPENDET	PWRDOWN	RECOVER	Short circuit	Open circuit	antPower	antStatus
TRUE	X	X	X	X	NO	NO	ON	OK
TRUE	FALSE	X	X	X	X	NO		
TRUE	X	FALSE	X	X	NO	X		
TRUE	FALSE	FALSE	X	X	X	X		
TRUE	FALSE	TRUE	X	X	X	YES	ON	OPEN
TRUE	X	TRUE	X	X	NO	YES		
TRUE	TRUE	X	FALSE	X	YES	X	ON	SHORT
TRUE	TRUE	X	TRUE	X	YES	X	OFF	SHORT
FALSE	TRUE	X	X	X	YES	X	UNKNOWN	SHORT
FALSE	FALSE	TRUE	X	X	X	YES	UNKNOWN	OPEN
FALSE	X	TRUE	X	X	NO	YES	UNKNOWN	OPEN

Table 22: Antenna supervisor configuration and antenna states

3.4 Forcing receiver reset

GNSS receivers typically make a distinction between cold, warm, and hot start based on the type of valid information the receiver has during the restart.

- **Cold start:** in the cold start mode, the receiver has no information from the last position (e.g. time, velocity, frequency etc.) at startup. Therefore, the receiver must search the full time and frequency space, and all possible satellite numbers. If a satellite signal is found, it is tracked to decode the ephemeris (18-36 seconds under strong signal conditions), while the other channels continue to search satellites. Once there is a sufficient number of satellites with valid ephemeris, the receiver can calculate position and velocity data. Other GNSS receiver manufacturers call this the Factory startup mode.
- **Warm start:** in the warm start mode, the receiver has approximate information for time, position, and coarse satellite position data (Almanac). In this mode, the receiver normally needs to download ephemeris after power-up before it can calculate position and velocity data. As the ephemeris data is usually outdated after 4 hours, the receiver typically starts with a warm start if it has been powered down for more than 4 hours. In this scenario, several augmentations are possible. See [Multiple GNSS assistance](#).
- **Hot start:** in the hot start mode, the receiver has been powered down only for a short time (4 hours or less), so that its ephemeris is still valid. Since the receiver does not need to download ephemeris again, this is the fastest startup method.

Using the UBX-CFG-RST message, you can force the receiver to reset and clear data, in order to see the effects of maintaining/losing such data between restarts. For this purpose, use the `navBbrMask` field in the UBX-CFG-RST message to initiate hot, warm, and cold starts, or a combination of startup modes.

The reset type can also be specified. This is not related to GNSS, but to the way the software restarts the system.

- **Hardware reset** uses the on-chip watchdog to electrically reset the chip. This is an immediate asynchronous reset. No stop events are generated.
- **Controlled software reset** terminates all running processes in an orderly manner. Once the system is idle, restarts the receiver operation, reloads its configuration and starts to acquire and track GNSS satellites.
- **Controlled software reset (GNSS only)** only restarts the GNSS tasks, without reinitializing the full system or reloading any stored configuration.
- **Hardware reset (after shutdown)** uses the on-chip watchdog to reset the receiver after shutdown.
- **Controlled GNSS stop** stops all GNSS tasks. The receiver is not restarted, but stops any GNSS-related processing.
- **Controlled GNSS start** starts all GNSS tasks.

Table 23 below contains an overview of the different reset types and the data that is cleared.

Reset type	Clears RAM	Clears BBR
0x00 - Hardware reset (immediately), 0x04 - Hardware reset (after shutdown)	Yes	Yes
0x01 - Controlled Software reset	Yes	No
0x02 - Controlled Software reset (GNSS only), 0x08 - Controlled GNSS stop, 0x09 - Controlled GNSS start	No	No
RESET_N pin	Yes	Yes

Table 23: Overview of the available reset types



After using any reset type that clears the BBR, the TTFF is similar to performing a cold start.

3.5 Security

The security concept of MAX-M10S covers:

- The integrity of the receiver
- Communication between the receiver and the GNSS satellites

Some security functions monitor and detect threats and report them to the host system. Other functions mitigate threats and allow the receiver to operate normally.

Table 24 gives an overview about possible threats and which functionality is available to detect and/or mitigate it.

Threat	u-blox solution
GNSS receiver integrity	Secure boot Receiver configuration lock
Over air signal integrity	Spoofing detection and monitoring Jamming interference detection and monitoring

Table 24: u-blox security options

3.5.1 GNSS receiver integrity

This section describes receiver security features implemented with MAX-M10S:

- Secure boot
- Receiver configuration lock

3.5.1.1 Secure boot

MAX-M10S boots only with firmware images that are signed by u-blox. This prevents the execution of non-genuine firmware images on the receiver.

3.5.1.2 Receiver configuration lock

The receiver configuration lock feature ensures that no configuration changes are possible once the feature is enabled. The configuration lock is enabled by setting the configuration item CFG-SEC-CFG_LOCK to "true".

The configuration lock can be applied to different configuration layers including the RAM and BBR. At startup, the receiver constructs the configuration database from different configuration layers and maintains it in the run-time RAM memory. When the configuration lock is set in the run-time RAM, the receiver configuration cannot be changed on any configuration layer.



For more information on the configuration layers including the order of priority they are applied in, see the applicable Interface description [3].

The configuration lock set on the RAM or BBR configuration layer is removed when the memory is cleared.

To test the lock functionality, set it on the RAM configuration layer. After a power cycle, the information on RAM layer is cleared and the lock is no longer set.



It is recommended to apply the configuration lock on the same layer the configuration is stored.

An example of use case is that the host application locks the receiver configuration. A user communicating with MAX-M10S through any of the available interfaces can poll, enable or send messages, but cannot change the configuration by sending UBX configuration messages.

3.5.2 Jamming and spoofing detection

3.5.2.1 Jamming and RF interference detection and monitoring

Intentional jamming signals and/or unintentional interference generated by nearby electronics can degrade the quality of GNSS signals and the receiver performance. The receiver has two independent mechanisms to detect and report the presence of RF interference or intentional jamming signals: jamming indicator and jamming and interference monitor (ITFM).

Jamming indicator

The jamming indicator detects narrow-band continuous wave (CW) signals over the configured frequency bands. The status is reported in the UBX-MON-RF message, `cwSuppression` flag. The value is always relative to the base level reported in an unjammed environment. A significant increase in the jamming indicator value indicates presence of a jamming signal. The jamming indicator is always enabled.

Jamming and interference monitor (ITFM)

Jamming and interference monitor detects any waveform over the configured frequency bands. The receiver monitors the background noise and looks for significant changes in the spectrum. The monitor status is reported in the UBX-MON-RF message, `jammingState` flag. The monitor is disabled by default.

The monitor is configured with the CFG-ITFM-* configuration group. The configuration keys are summarized in [Table 25](#).

Configuration key	Description
CFG-ITFM-ENABLE	Set to 1 to enable ITFM
CFG-ITFM-BBTHRESHOLD	The threshold level for broadband interference detection. The value is given in decibels (dB) above the level of the reference spectrum.
CFG-ITFM-CWTHRESHOLD	The threshold level for CW interference detection. The value is given in decibels (dB) above the level of the reference spectrum.
CFG-ITFM-ANTSETTING	The type of antenna (active or passive) used in the design.

Table 25: CFG-ITFM-* group configuration keys

The receiver measures the reference spectrum at the start-up after obtaining a good fix. Until then, the monitor reports "Unknown". Once the reference spectrum is available, the receiver measures the signal spectrum and compares it against the reference while applying the detection thresholds. The receiver also internally monitors other factors including changes in the average C/N0 level to determine the jamming state. The reported monitor states are summarized in [Table 26](#).

Value	Reported state	Description
0	Unknown	Monitor is not enabled, monitor is uninitialized, or the antenna is disconnected
1	OK	No RF interference is detected
2	Warning	Position OK but RF interference is visible (above the thresholds)
3	Critical	No reliable position fix and interference is visible (above the thresholds); jamming/RF interference is a probable reason for no position fix

Table 26: Jamming and interference monitor states


It is not recommended to restart the receiver when it is indicating jamming.

Evaluation

The detection of jamming or RF interference depends both on the type of jamming signal and the signal environment. It may not be always possible to detect jamming or RF interference signals. If the GNSS performance is degraded or the fix is completely lost, jamming or RF interference reported in `cwSuppression` and/or `jammingState` is a likely cause.

RF interference generated by the device itself or coupled from external sources is common and may be reported by the receiver. If jamming is reported but the C/N0 level and GNSS performance are not affected, the receiver may be able to mitigate the impact of jamming.

The jamming and RF interference detection feature can be evaluated by applying jamming signals relevant for the application and signal environment and observing the receiver behaviour.

3.5.2.2 Spoofing detection

Spoofing involves transmitting counterfeit GNSS signals with the intent of the target receiver misinterpreting them as genuine signals and producing an erroneous position fix and/or time solution.

The spoofing detector alerts the host when signals appear to be suspicious. The detection algorithms monitor multiple signal parameters for inconsistencies and suspicious changes to identify external manipulation. The detection algorithms rely on availability of signals from multiple GNSS constellations to improve the spoofing detection capabilities. The spoofing detector is always enabled.

The detection of spoofing requires a transition from initially genuine GNSS signals to the introduction of spoofed signals. Detection is therefore not possible if the spoofing signals are already present when the receiver starts up. Detection is most likely at the time when the spoofing signal is introduced, but it may take some time until spoofing is reported. The spoofing status

is reported in the UBX-NAV-STATUS message, `spoofDetState` flag. The reported states are summarized in [Table 27](#).



It is not recommended to restart the receiver when it is indicating spoofing.

Value	Reported state	Description
1	No spoofing indicated	No spoofing detectors indicate spoofing
2	Spoofing indicated	One of the spoofing detectors indicates spoofing
3	Multiple spoofing indications	Several types of spoofing detectors indicate spoofing

Table 27: Spoofing detection states

The detection of spoofing signals depends on the type of spoofing but also on the signal environment. It may not always be possible to detect spoofing attacks. However, for some spoofing scenarios the receiver may reject the inconsistent signals from the navigation solution, and in such case the receiver may not report detection of spoofing.

To evaluate the spoofing detection feature, apply spoofing signals relevant for the application and signal environment and observe the receiver behaviour.

3.6 Power management

u-blox receivers support different operating modes. These modes represent strategies of controlling the acquisition and tracking engines to achieve either the best possible performance or good performance with reduced power consumption.

3.6.1 Continuous mode

MAX-M10S uses dedicated signal processing engines optimized for signal acquisition and tracking. The acquisition engine actively searches for and acquires signals during cold starts or when insufficient signals are available during navigation. The tracking engine continuously tracks and downloads all the almanac data and acquires new signals as they become available during navigation. The tracking engine consumes less power than the acquisition engine.

The current consumption is lower when a valid position is obtained quickly after the start of the receiver navigation, the entire almanac has been downloaded, and the ephemeris for each satellite in view is valid. If these conditions are not met, the search for the available satellites takes more time and consumes more power.

3.6.2 Power save mode

Power save mode (PSM) allows a reduction in system power consumption by selectively switching parts of the receiver on and off. It is enabled with `CFG-PM-OPERATEMODE` and configured with items in the `CFG-PM` group.

Power save mode (PSM) has two modes of operation:

- **Power save mode cyclic tracking (PSMCT)** operation is used when position fixes are required in short periods of 0.5 s to 10 s.
- **Power save mode on/off (PSMOO)** operation is used for periods longer than 10 s, and can be in the order of minutes, hours, or days.

The mode of operation can be configured, and depending on the setting, the receiver demonstrates different behavior. In on/off operation the receiver switches between phases of startup/navigation

and phases with low or almost no system activity (backup/sleep). In cyclic tracking the receiver does not shut down completely between fixes, but uses low-power tracking instead.



In PSMOO mode, the RAM memory is cleared during the off periods. Consequently, store the configuration in the BBR memory to maintain the settings.



Likewise in PSMCT mode, the RAM memory is cleared when the receiver enters the "Inactive for search" state after signal loss and the [Acquisition timeout](#) is exceeded. Consequently, store the configuration in the BBR memory to maintain the settings.

GPS, GLONASS, BeiDou B1I, Galileo and QZSS signals are supported in power save mode. BeiDou B1C signal is not supported. The receiver is unable to download or process any SBAS data in power save mode and it is therefore recommended to disable SBAS.



BeiDou B1C is not supported in power save mode.

3.6.2.1 Operation

PSM is based on a state machine with five different states: Inactive for update, Inactive for search, Acquisition, Tracking and Power optimized tracking (POT) state.

- Inactive states: most parts of the receiver are switched off.
- Acquisition state: the receiver actively searches for and acquires signals. Maximum power consumption.
- Tracking state: the receiver continuously tracks and downloads data. Less power consumption than in the acquisition state.
- POT state: the receiver repeatedly loops through a sequence of tracking (Track), calculating the position fix (Fix), and entering an idle period (Idle). No new signal is acquired and no data is downloaded. The power consumption is much lower than in the tracking state.

The PSM state machine is described in [Figure 12](#).

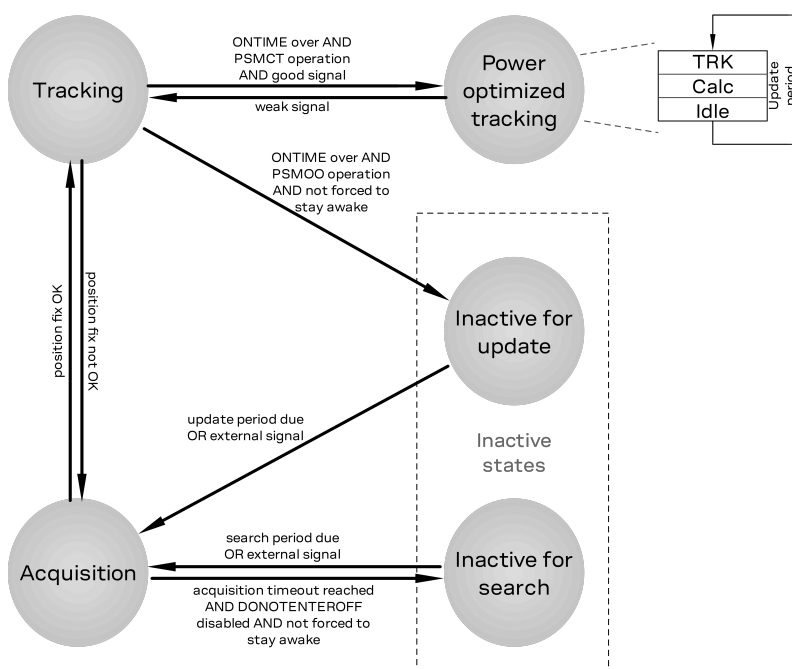


Figure 12: State machine

3.6.2.2 Acquisition timeout

The receiver has internal, external, and user-configurable mechanisms that determine the time to be spent in acquisition state. This logic is put in place to ensure good performance and low power consumption in different environments and scenarios. This collective logic is referred to as acquisition timeout.

The configuration items related to acquisition timeout are described in section [Configuration](#).

Internal mechanisms:

- The receiver transitions to the "Inactive for search" state after the timeout configured in MAXACQTIME or earlier, if the receiver is unable to acquire any signals or only acquires weak signals of insufficient quality to get a fix.

User-configurable mechanisms:

- MINACQTIME is the minimum time that the receiver will spend in the "Acquisition" state. MINACQTIME is applicable only when no or very poor GNSS signal is available.
- MAXACQTIME is the maximum time that the receiver will spend in the "Acquisition" state.
- DONOTENTEROFF forces the receiver to stay awake and in the "Acquisition" state even when a fix is not possible.

External mechanisms:

- The receiver is forced to stay awake if EXTINTWAKE is enabled and the EXTINT pin is set to "high". The receiver is forced to stay in the "Inactive for search/Fix" states if EXTINTBACKUP is enabled and the EXTINT pin is set to "low".
- The receiver is forced to stay awake if EXTINTINACTIVE is enabled and the EXTINT pin is toggled. If EXTINT pin state is not changed for a longer time than EXTINTINACTIVITY, the receiver enters the "Inactive for search/Fix" states.

3.6.2.3 Cyclic tracking

Power save mode cyclic tracking (PSMCT) operation is described in [Figure 13](#).



PSMCT supports 1 Hz and 2 Hz navigation update rates. In addition, longer update periods from 2 s to 10 s are supported at 1 s steps.

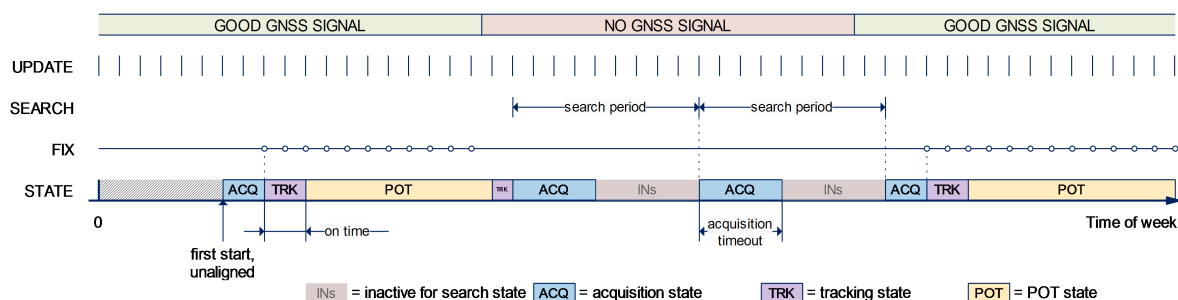


Figure 13: Cyclic tracking operation

- When the receiver is switched on, it first enters the "Acquisition" state. A larger number of signals tracked later helps the receiver to remain in the "POT" state if some signals get blocked and are lost. This may reduce the overall power consumption.
- If the receiver is able to acquire a valid position fix (one passing the navigation output filters) within the time given by the [Acquisition timeout](#), it switches to the "Tracking" state and the ONTIME starts. Otherwise it enters the "Inactive for search" state and restarts after the configured search period (minus a start-up margin).

- Once the ONTIME is over, the "POT" state is entered. Setting the ONTIME to zero causes the receiver to enter the "POT" state as soon as possible.
- In the "POT" state the receiver continues to output position fixes according to the CFG-RATE-*.
- If the signal becomes weak or is lost during the "POT" state, the "Tracking" state is entered.
- Once the signal is good again and the newly started ONTIME is over, the receiver will re-enter the "POT" state.
- If the receiver cannot get a position fix in the "Tracking" state, it enters the "Acquisition" state. Should the acquisition fail as well, the "Inactive for search" state is entered. If DONOTENTEROFF is enabled and no fix is possible, the receiver will remain in the "Acquisition" state until a fix is possible and it will never enter the "Inactive for search" state.

3.6.2.4 On/Off mode

Power save mode on-off (PSMOO) operation is described in Figure 14.



PSMOO requires an RTC to maintain time.

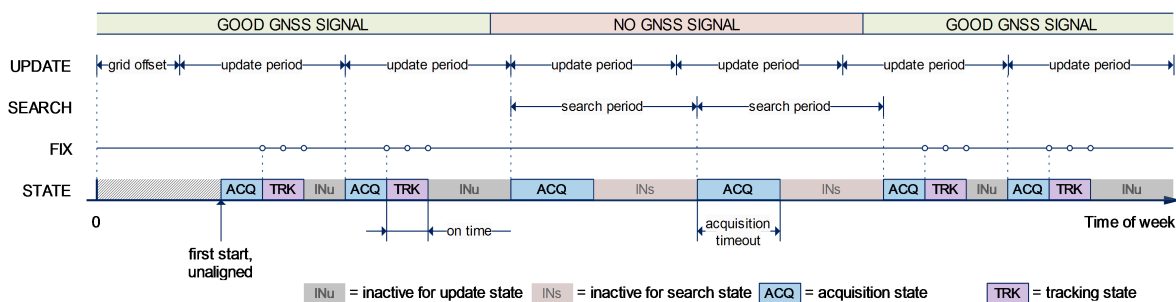


Figure 14: On/off mode operation

- When the receiver is switched on, it first enters the "Acquisition" state.
- If it is able to acquire a valid position fix (one passing the navigation output filters) within the time given by the [Acquisition timeout](#), it switches to the "Tracking" state and the ONTIME starts. Otherwise it enters the "Inactive for search" state and restarts after the configured search period (minus a startup margin).
- Once the ONTIME is over, the "Inactive for update" state is entered and the receiver restarts according to the configured update grid defined by GRIDOFFSET.
- If the signal is lost while in the "Tracking" state, the "Acquisition" state is entered. If the signal is not found within the acquisition timeout, the receiver enters the "Inactive for search" state. Otherwise the receiver will re-enter the "Tracking" state and stay there until the newly started ONTIME is over.



Entering the off state of the PSMOO operation clears the RAM memory including the receiver configuration. To maintain the configuration in PSMOO operation, store it on both RAM and battery-backed RAM (BBR) layers. Configuration in an optional flash memory is always maintained.

3.6.2.5 External control

The operation of power save mode can be controlled externally using the EXTINT pin. The external control allows the user to decide when to wake up the receiver to obtain a fix and when to force the receiver into the backup mode to save power. Operating the receiver externally through the EXTINT pin overrides internal functions that coincide with that specific operation.

Enabling EXTINTWAKE prevents the receiver from entering inactive states for as long as the EXTINT pin is held "high". In PSMOO, the receiver will therefore always be in the "Acquisition" or the "Tracking" state. In PSMCT, the receiver additionally be in the "POT" state. When EXTINT is set "low", the receiver continues with its configured behavior.

Enabling EXTINTBACKUP forces the receiver to enter inactive states for as long as the EXTINT pin is held "low" until the next wakeup event. Any wakeup event can wake up the receiver even if the EXTINT pin is held "low". In this case, the receiver only wakes up to read the configuration pins and then re-enters the inactive state.

If both EXTINTWAKE and EXTINTBACKUP are enabled at the same time, the receiver PSM operation is completely under external control. Setting EXTINT "high" wakes up the receiver to get a position fix and setting it "low" puts the receiver into the backup mode.

 EXTINT pin control can also be used in continuous mode.

In PSMOO, an external wakeup source can be set. Any wake-up event restarts the receiver to try to obtain a position fix. Wakeup signals have no effect if the receiver is already in the Acquisition, Tracking, or POT state.

Setting the update period POSUPDATEPERIOD to zero causes the receiver to wait in the Inactive for update state until the host wakes it up. Setting the search period ACQPERIOD to zero causes the receiver to indefinitely wait in the Inactive for search state after an unsuccessful startup.

 External wake-up source is required when setting update or search period to zero.

The wake-up sources are:

- Rising or falling edge on the UART RX pin
- Rising or falling edge on the EXTINT pin
- Rising or falling edge on the SPI CS pin
- Rising edge on RESET_N pin

[Backup modes](#) can also be used to control the receiver state externally.

3.6.2.6 Configuration

Power save mode (PSM) is enabled and disabled with CFG-PM-OPERATEMODE and configured with items in the CFG-PM group listed in [Table 28](#).

 When using power save mode on/off (PSMOO) operation, set the OPERATEMODE as the last PSM configuration key to prevent the receiver entering the off state before all intended PSM configuration keys are set.

Config key	Description
OPERATEMODE	Receiver mode of operation
POSUPDATEPERIOD	Time between two position fix attempts in on/off power save mode
ACQPERIOD	Time between two acquisition attempts if the receiver is unable to get a position fix
GRIDOFFSET	Time offset of update grid with respect to start of week
ONTIME	Time the receiver remains in the "Tracking" state and produces position fixes
MINACQTIME	Minimum time the receiver spends in the "Acquisition" state
MAXACQTIME	Maximum time in the "Acquisition" state
DONOTENTEROFF	Receiver does not enter the "Inactive for search" state if it cannot get a position fix but keeps indefinitely attempting a position fix instead
WAITTIMEFIX	Wait for time fix before entering the "Tracking" state
UPDATEEPH	Enables periodic ephemeris update
EXTINTWAKE	Enables EXTINT pin control to force receiver on
EXTINTBACKUP	Enables EXTINT pin control to force receiver in backup
EXTINTINACTIVE	Enter a backup state if EXTINT pin is inactive longer time than specified by EXTINTINACTIVITY

Config key	Description
EXTINTINACTIVITY	Specifies the inactivity period

Table 28: Power save mode configuration options in the CFG-PM group

OPERATEMODE The mode of operation to use mainly depends on the update period: For short update periods (in the range of a few seconds), cyclic tracking should be configured. For long update periods (in the range of minutes or longer), only use on/off operation. See section [On/Off mode](#) and [Cyclic tracking](#) for more information on the two modes of operation.

POSUPDATEPERIOD, ACQPERIOD The update period POSUPDATEPERIOD specifies the time between successive position fixes. If no position fix can be obtained within the acquisition timeout, the receiver will retry after the time specified by the search period ACQPERIOD. Update and search periods are fixed with respect to an absolute time grid based on reference time standard (i.e., GPS time or UTC). They do not refer to the time of the last valid position fix or last position fix attempt. Where multiple GNSS can operate simultaneously, UTC time is used as the reference time standard.

The update period setting is ignored if the receiver is set into cyclic tracking mode. It only has an impact if the receiver is set to on/off mode. New settings are ignored if the update period or the search period exceeds the maximum number of milliseconds in a week. In that case the previously stored values remain effective.

GRIDOFFSET Once the receiver has a valid time, the update grid is aligned to the start of the week of the reference time standard (midnight between Saturday and Sunday). Before having a valid time, the update grid is unaligned. A grid offset shifts the update grid with respect to the start of the week of the reference time standard. The grid offset is not used in cyclic tracking operation.

ONTIME This specifies how long the receiver stays in the "Tracking" state before switching to the "POT" state in PSMCT or the "Inactive for update" state in PSMOO.

MINACQTIME The receiver tries to obtain a position fix for at least the time given by MINACQTIME. If the receiver determines that it needs more time for the given starting conditions then it will automatically prolong this time. If MINACQTIME is set to zero, the receiver determines the time. Once the MINACQTIME has expired, the receiver will terminate the acquisition state if either a fix is achieved or if the receiver estimates that any signals received are insufficient (too weak or too few) for a fix to be possible.

MINACQTIME is applicable only when no or very poor GNSS signal is available.

MAXACQTIME This defines the maximum time that the receiver will spend in the "Acquisition" state. If the receiver is unable to acquire a valid position fix within this maximum time, it will transition to the "Inactive for search" state (if DONOTENTEROFF is disabled). Subsequently, the receiver will attempt to acquire another position fix according to the search period ACQPERIOD. If MAXACQTIME is set to zero, the receiver will autonomously determine the maximum time to spend in the "Acquisition" state. Note that shorter settings (below about 45 s) will degrade an unaided receiver's ability to collect new Ephemeris data at low signal levels.

DONOTENTEROFF If this option is enabled, then when the receiver cannot get a fix it keeps attempting to acquire a position fix instead of entering the "Inactive for search" state. In other words, the receiver will never be in the "Inactive for search" state and therefore the search period ACQPERIOD and the minimum acquisition time MINACQTIME will be ignored.

WAITTIMEFIX A time fix is a fix type in which the receiver will ensure that the time is accurate and confirmed to within the limits set in CFG-NAVSPG. Enabling the WAITTIMEFIX option will force the receiver to stay in the "Acquisition" state until the time is known to be within the configured limits,

then it will transition to the "Tracking" state. Take into account that enabling WAITTIMEFIX will delay the transition from the "Acquisition" state to the "Tracking" state by at least two extra seconds.

The quality of the position fixes can also be configured by setting the limits in the CFG-NAVSPG group. Setting harder limits in CFG-NAVSPG will typically prolong the time in the "Acquisition" state. When externally controlled, it is therefore necessary to ensure sufficient time for the receiver at startup. Refer to [Acquisition timeout](#) for more information. When internally controlled, the receiver can make good judgment on the time needed in the "Acquisition" state and no further adjustments will be needed.

UPDATEEPH To maintain the ability of a fast startup, the receiver needs to update its ephemeris data on a regular basis. This can be ensured by activating the update Ephemeris option UPDATEEPH. The ephemeris data is updated approximately every 30 minutes. Refer to [Satellite data download scheduler](#) for more information.

3.6.2.7 Satellite data download scheduler

The receiver is not able to download satellite data (e.g. the ephemeris) while it is working in on/off or cyclic tracking operation. Therefore, it has to temporarily switch to continuous operation for the time the satellites transmit the desired data. To save power, the receiver schedules the downloads according to an internal timetable and only switches to continuous operation when data of interest is being transmitted by the satellites.

Each satellite transmits its own ephemeris data. Ephemeris data download is feasible when the corresponding satellite has been tracked with a sufficient C/N0 over a certain period of time. The download is scheduled in a 30-minute grid or immediately when fewer than a certain number of visible satellites have valid ephemeris data.

Almanac, ionosphere, UTC correction, and satellite health data are transmitted by all satellites simultaneously. Therefore, these parameters can be downloaded when a single satellite is tracked with a sufficiently high C/N0.

Allowing more ephemerides to be downloaded before entering the POT or the "Inactive for update" state can improve the quality of the fixes and reduce the number of wake-ups needed to download ephemerides. However, this requires spending extra time in the acquisition state (only when an inadequate number of ephemerides are downloaded from tracked satellites).

3.6.3 Backup modes

A backup mode is an inactive state where the power consumption is reduced to a fraction of that in operating modes. The receiver maintains time information and navigation data to speed up the receiver restart after backup or standby mode.

MAX-M10S supports the following backup modes: hardware backup mode and software standby mode.

3.6.3.1 Hardware backup mode

The hardware backup mode allows entering a backup state and resuming operation by switching the main power supplies on and off while maintaining a V_BCKP supply via, e.g. a battery.

V_BCKP must be supplied to maintain the backup domain (BBR and RTC) to allow better TTFF, accuracy, availability and power consumption at the next startup compared with a cold start. As V_IO is not supplied, the PIOs cannot be driven by an external host processor. If driving of the PIOs cannot be avoided, buffers are required for isolating the PIOs.

3.6.3.2 Software standby mode

Software standby mode is entered using the UBX-RXM-PMREQ message. V_{IO} and VCC must be supplied, however VCC supply is internally disabled to save power. The V_{IO} supply maintains the battery-backed RAM (BBR), RTC, and PIOs.

Entering the software standby mode clears the RAM memory including the receiver configuration. To maintain the configuration, store it on BBR layers. For more information on permanence of the stored configuration, refer to [Receiver configuration](#).

The software standby mode can be set for a specific duration, or until the receiver is woken up by a signal at a wake-up source defined in UBX-RXM-PMREQ. The possible wake-up sources are UART RX and/or EXTINT pin. For more information on the UBX-RXM-PMREQ message, refer to the Interface description [3].

A system reset with the RESET_N signal also terminates the software standby mode, clears the BBR content and restarts the receiver.

As V_{IO} is supplied, the PIOs can be driven by an external host processor. No buffers are required for isolating the PIOs, which reduces cost.

-  The LNA_EN signal is set to the "LOW" state during the software standby mode.
-  The "force" flag must be set in UBX-RXM-PMREQ to enter software standby mode.
-  Leave V_BCKP open if it is not used.

3.7 Time

Maintaining receiver local time and keeping it synchronized with GNSS time is essential for proper timing and positioning functionality. This section explains how the receiver maintains local time and introduces the supported GNSS time bases.

3.7.1 Receiver local time

The receiver is dependent on a local oscillator for both the operation of its radio parts and also for timing within its signal processing. No matter what nominal frequency the local oscillator has, u-blox receivers subdivide the oscillator signal to provide a 1-kHz reference clock signal, which is used to drive many of the receiver's processes. In particular, the measurement of satellite signals is arranged to be synchronized with the "ticking" of this 1-kHz clock signal.

When the receiver first starts, it has no information about how these clock ticks relate to other time systems; it can only count time in 1 millisecond steps. However, as the receiver derives information from the satellites it is tracking or from aiding messages, it estimates the time that each 1-kHz clock tick takes in the time base of the chosen GNSS system. This estimate of GNSS time based on the local 1-kHz clock is called receiver local time.

As receiver local time is a mapping of the local 1-kHz reference onto a GNSS time base, it may experience occasional discontinuities, especially when the receiver first starts up and the information it has about the time base is changing. Indeed, after a cold start, the receiver local time initially indicates the length of time that the receiver has been running. However, when the receiver obtains some credible timing information from a satellite or an aiding message, it jumps to an estimate of GNSS time.

3.7.2 GNSS time bases

GNSS receivers must handle a variety of different time bases as each GNSS has its own reference system time. What is more, although each GNSS provides a model for converting their system time

into UTC, they all support a slightly different variant of UTC. So, for example, GPS supports a variant of UTC as defined by the US National Observatory, while BeiDou uses UTC from the National Time Service Center, China (NTSC). While the different UTC variants are normally closely aligned, they can differ by as much as a few hundreds of nanoseconds.

Although u-blox receivers can combine a variety of different GNSS times internally, the user must choose a single type of GNSS time and, separately, a single type of UTC for input (on EXTINT pins) and output (via the TIMEPULSE pin) and the parameters reported in corresponding messages.

The CFG-TP-TIMEGRID_TP* configuration item allows the user to choose between any of the supported GNSS (GPS, Galileo, BeiDou, etc.) time bases and UTC. Also, the CFG-NAVSPG-UTCSTANDARD configuration item allows the user to select which variant of UTC the receiver should use. This includes an "automatic" option which causes the receiver to select an appropriate UTC version itself, based on the GNSS constellation. The order of preference is:

- USNO if GPS is enabled
- SU if GLONASS is enabled
- NTSC if BeiDou is enabled
- NPLI if NAVIC is enabled
- NICT if QZSS is enabled
- European if Galileo is enabled

The receiver assumes that an input time pulse uses the same GNSS time base as specified for the time pulse output. So, if the user selects Galileo time for time pulse output, any time pulse input must also be aligned to the Galileo time (or to the separately chosen variant of UTC). When UTC is selected for the time pulse output, any GNSS time pulse input is assumed to be aligned with GPS time.



The receiver allows users to independently choose GNSS signals used in the receiver (using CFG-SIGNAL-*) and the input/output time base (using CFG-TP-*). For example, it is possible to instruct the receiver to use GPS and Galileo satellite signals to generate BeiDou time. This practice compromises time pulse accuracy if the receiver cannot measure the timing difference between the constellations directly and is therefore not recommended.



The information that allows GNSS times to be converted to the associated UTC times is only transmitted by the GNSS at relatively infrequent periods. For example, GPS transmits UTC(USNO) information only once every 12.5 minutes. Therefore, if a time pulse is configured to use a variant of UTC time, after a cold start, substantial delays can be expected before the receiver has sufficient information to start outputting the time pulse.

Each GNSS has its own time reference for which detailed and reliable information is provided in the messages listed in the table below.

Time reference	Message
GPS time	UBX-NAV-TIMEGPS
BeiDou time	UBX-NAV-TIMEBDS
GLONASS time	UBX-NAV-TIMEGLO
Galileo time	UBX-NAV-TIMEGAL
QZSS time	UBX-NAV-TIMEQZSS

Time reference	Message
UTC time	UBX-NAV-TIMEUTC

Table 29: GNSS time messages

3.7.3 Navigation epochs

Each navigation solution is triggered by the tick of the 1 kHz clock nearest to the desired navigation solution time. This tick is referred to as a navigation epoch. If the navigation solution attempt is successful, one of the results is an accurate measurement of time in the time base of the chosen GNSS system, called GNSS system time. The difference between the calculated GNSS system time and receiver local time is called the clock bias (and the clock drift is the rate at which this bias is changing).

In practice the receiver's local oscillator is not as stable as the atomic clocks to which GNSS systems are referenced and consequently clock bias tends to accumulate. However, when selecting the next navigation epoch, the receiver always tries to use the 1 kHz clock tick which it estimates to be closest to the desired fix period as measured in GNSS system time. Consequently, the number of 1 kHz clock ticks between fixes occasionally varies. This means that when producing one fix per second, there are normally 1000 clock ticks between fixes, but sometimes, to correct drift away from the GNSS system time, there are 999 or 1001 ticks.

The GNSS system time calculated in the navigation solution is always converted to a time in both the GPS and UTC time bases for output.

Clearly when the receiver has chosen to use the GPS time base for its GNSS system time, conversion to GPS time requires no work at all, but conversion to UTC requires knowledge of the number of leap seconds since GPS time started (and other minor correction terms). The relevant GPS-to-UTC conversion parameters are transmitted periodically (every 12.5 minutes) by GPS satellites, but can also be supplied to the receiver via the UBX-MGA-GPS-UTC aiding message. By contrast, when the receiver has chosen to use the GLONASS time base as its GNSS system time, conversion to GPS time is more difficult as it requires knowledge of the difference between the two time bases, but as GLONASS time is closely linked to UTC, conversion to UTC is easier.

When insufficient information is available for the receiver to perform any of these time base conversions precisely, predefined default offsets are used. Consequently, plausible times are nearly always generated, but they may be wrong by a few seconds (especially shortly after receiver start). Depending on the configuration of the receiver, such "invalid" times may well be output, but with flags indicating their state (e.g. the "valid" flags in UBX-NAV-PVT).



To support multiple GNSS systems concurrently, u-blox receivers employ multiple GNSS system times and/or receiver local times. For reporting GNSS system time or the receiver local time, users are recommended to use messages that report UTC time instead of using UBX messages. Other messages are retained only to support backwards compatibility.

3.7.4 iTow timestamps

The original designers of GPS chose to express time/date as an integer week number (starting with the first full week in January 1980) and a time of week (TOW) expressed in seconds. Manipulating time/date in this form is far easier for digital systems than the more conventional year/month/day, hour/minute/second representation. Therefore, most GNSS receivers use this time representation internally, and convert it to a more conventional form at external interfaces. In many UBX messages, the iTOW field provides an externally visible example of the internal time representation.

All the main UBX-NAV messages (and some other messages) contain an iTOW field to indicate the GPS time when the navigation epoch occurred. Messages with the same iTOW value can be assumed

to have come from the same navigation solution, and therefore, iTOW can be used to synchronize between these UBX messages. However, the iTOW values may not be valid (i.e., they may have been generated with insufficient conversion data). Therefore, it is not recommended to use the iTOW field for any other purpose.

If reliable absolute time information is required, use the UTC time related fields in the UBX-NAV-PVT message. Additionally, the UBX-NAV-PVT message contains information about the validity and the accuracy of the provided UTC time. See the section [Time validity](#) for further information.

 iTOW is always referenced to GPS time, and it should not be confused with the [UTC representation](#).

 The iTOW timestamps are not compensated for the [Leap seconds](#).

3.7.5 Time validity

Information about the validity of the time solution is given in the following form:

- **Time validity:** Information about time validity is provided in the `valid` flags (e.g. `validDate` and `validTime` flags in the UBX-NAV-PVT message). If these flags are set, the time is known and considered valid for use.
- **Time validity confirmation:** Information about confirmed validity is provided in the `confirmedDate` and `confirmedTime` flags in the UBX-NAV-PVT message. If these flags are set, the time validity can be confirmed by using an additional independent source, meaning that the probability of the time to be correct is very high. Note that information about time validity confirmation is only available if the `confirmedAvai` bit in the UBX-NAV-PVT message is set.

 `validDate` means that the receiver has knowledge of the current date. However, it must be noted that this date might be wrong for various reasons. Only when the `confirmedDate` flag is set, the probability of the incorrect date information drops significantly.

 `validTime` means that the receiver has knowledge of the current time. However, it must be noted that this time might be wrong for various reasons. Only when the `confirmedTime` flag is set, the probability of incorrect time information drops significantly.

 `fullyResolved` means that the UTC time is known without full seconds ambiguity. When deriving UTC time from GNSS time the number of leap seconds must be known, with the exception of GLONASS. It might take several minutes to obtain such information from the GNSS payload. When the one second ambiguity has not been resolved, the time accuracy is usually in the range of ~20s.

3.7.6 UTC representation

UTC time is used in many NMEA and UBX messages. In NMEA messages, time is always rounded to the nearest hundredth of a second and it is normally reported with two decimal places (e.g. 124923.52). Although compatibility mode (selected using CFG-NMEA-COMPAT) requires three decimal places, rounding to the nearest hundredth of a second remains, so the extra digit is always 0.

UTC time is also reported within some UBX messages, such as UBX-NAV-TIMEUTC and UBX-NAV-PVT. In these messages date and time are separated into seven distinct integer fields. Six of these (year, month, day, hour, min. and sec.) have fairly obvious meanings and are all guaranteed to match the corresponding values in NMEA messages generated by the same navigation epoch. This facilitates simple synchronization between associated UBX and NMEA messages.

The seventh field is called nano and it contains the number of nanoseconds by which the rest of the time and date fields need to be corrected to get the precise time. So, for example, the UTC time 12:49:23.521 would be reported as: hour: 12, min: 49, sec: 23, nano: 521000000.

It is however important to note that the first six fields are the result of rounding to the nearest hundredth of a second. Consequently the nano value can range from -5000000 (i.e. -5 ms) to +994999999 (i.e. nearly 995 ms).

When the nano field is negative, the number of seconds (and maybe minutes, hours, days, months or even years) have been rounded up. Therefore, some or all of them must be adjusted to get the correct time and date. Thus in an extreme example, the UTC time 23:59:59.9993 on 31st December 2011 would be reported as: year: 2012, month: 1, day: 1, hour: 0, min: 0, sec: 0, nano: -700000.

If a resolution of one hundredth of a second is adequate, negative nano values can simply be rounded up to 0 and effectively ignored.

The UBX-NAV-TIMEUTC message gives information about the UTC time reference clock.

The preferred variant of UTC time can be specified using the CFG-NAVSPG-UTCSTANDARD configuration item. The UTC time variant configured must correspond to a GNSS that is currently enabled. Otherwise the reported UTC time is inaccurate.

3.7.7 Leap seconds

Due to the slightly uneven spin rate of the Earth, UTC time gradually moves out of alignment with the mean solar time (that is, the sun no longer appears directly overhead at 0 longitude at midday). Occasionally, a "leap second" is announced to bring UTC back into close alignment with the mean solar time. Usually this means adding an extra second to the last minute of the year, but this can also happen on 30th June. When this happens, UTC clocks are expected to go from 23:59:59 to 23:59:60, and only then on to 00:00:00.

It is also possible to have a negative leap second, in which case there will only be 59 seconds in a minute and 23:59:58 will be followed by 00:00:00.

u-blox receivers are designed to handle leap seconds in their UTC output and consequently applications processing UTC times from either NMEA or UBX messages should be prepared to handle minutes that are either 59 or 61 seconds long.

Leap second information can be polled from the receiver with the message UBX-NAV-TIMEELS.

3.7.8 Date ambiguity

Each navigation satellite transmits the current date and time in the navigation message of the signal. The time of week (TOW) indicates the number of seconds elapsed since the start of the week (midnight between Saturday and Sunday). The week number (WN) indicates the number of weeks elapsed since the particular GNSS system was started. By combining these two values, the current date and time can be determined.

When a new week begins, the time of week numbering resets to zero, and the week number increments by one. Since the week number continuously increases, it eventually reaches the maximum value and must "roll over" back to zero. Modern satellite systems, such as GPS L1C, L2C, and L5, Galileo, BeiDou, and GLONASS transmit sufficient information to ensure the week number remains unambiguous for the foreseeable future. The first rollover will occur in 2137 for the modern GPS, 2078 for Galileo, and 2163 for BeiDou. Instead of week numbers, GLONASS transmits an unambiguous UTC date.

Unfortunately, when the original GPS L1C/A signal was designed, only 10 bits were allocated for the week number. As a result, the transmitted week number value "rolls over" back to zero every 1024 weeks (just under 20 years). Consequently, the GPS L1 signal cannot differentiate between years like 1980, 1999, 2019, or 2038. To calculate the correct week number, GPS L1 receivers must use additional methods. The receiver can obtain additional information from its permanent configuration or from the host to resolve the ambiguity. If the receiver initially gets an ambiguous date from the GPS L1 signal, it can later acquire a modern signal to resolve the ambiguity in the GPS date.



The receiver does not use date information for navigation, so rollover dates do not affect navigation calculations. The date is only used to present information in a convenient format for the host application.



GPS time starts on January 6, 1980; Galileo on August 22, 1999; BeiDou on January 1, 2006; and GLONASS on January 1, 1996. Do not test the receiver with simulated signals before these dates.

3.7.8.1 GPS-L1-only date resolution

If the receiver is configured to use only the GPS L1C/A signal, or if the reception of other signals is delayed during startup, only the week number information from the GPS L1C/A signal will be available. In this case, the receiver establishes the date by assuming that all week numbers must be at least as large as the configured reference rollover week number. The default value for the reference rollover week number is selected at the compile time of the receiver firmware and is typically set to a value a few weeks before the software is completed. This value can be overridden by the CFG-NAVSPG-WKNROLLOVER configuration item.

The following example illustrates how this works:

Assume that the reference rollover week number set in the firmware at compile time is 2148 (which corresponds to a week in calendar year 2021, but is transmitted by the satellites as 100). In this case, if the receiver sees transmissions containing week numbers in the range of 100 to 1023, they are interpreted as week numbers 2148 to 3071 (calendar years 2021 to 2038). Transmissions with week numbers from 0 to 99 are interpreted as week numbers 3072 to 3171 (calendar years 2038 to 2040). With this method, the receiver shows the correct date for at least 19 years after the firmware creation.



When supplying the receiver with simulated signals, it is important to set the reference rollover week number correctly, especially when the scenarios are in the past.

3.8 Time mark

The receiver can be used to provide an accurate measurement of the time at which a pulse was detected on the external interrupt pin. The reference time can be chosen by setting the time source parameter to UTC, GPS, GLONASS, BeiDou, Galileo, NAVIC or local time in the CFG-TP-* configuration group. The UTC standard can be set in the CFG-NAVSPG-* configuration group. The delay figures defined with CFG-TP-* are also applied to the results output in the UBX-TIM-TM2 message.

A UBX-TIM-TM2 message is output at the next epoch if

- The UBX-TIM-TM2 message is enabled, and
- a rising or falling edge was triggered since last epoch on the EXTINT pin.

The UBX-TIM-TM2 messages includes the time of the last time mark, new rising/falling edge indicator, time source, validity, number of marks and an accuracy estimate.



Only the last rising and falling edge detected between two epochs is reported since the output rate of the UBX-TIM-TM2 message corresponds to the measurement rate configured with CFG-RATE-MEAS (see [Figure 15](#) below).

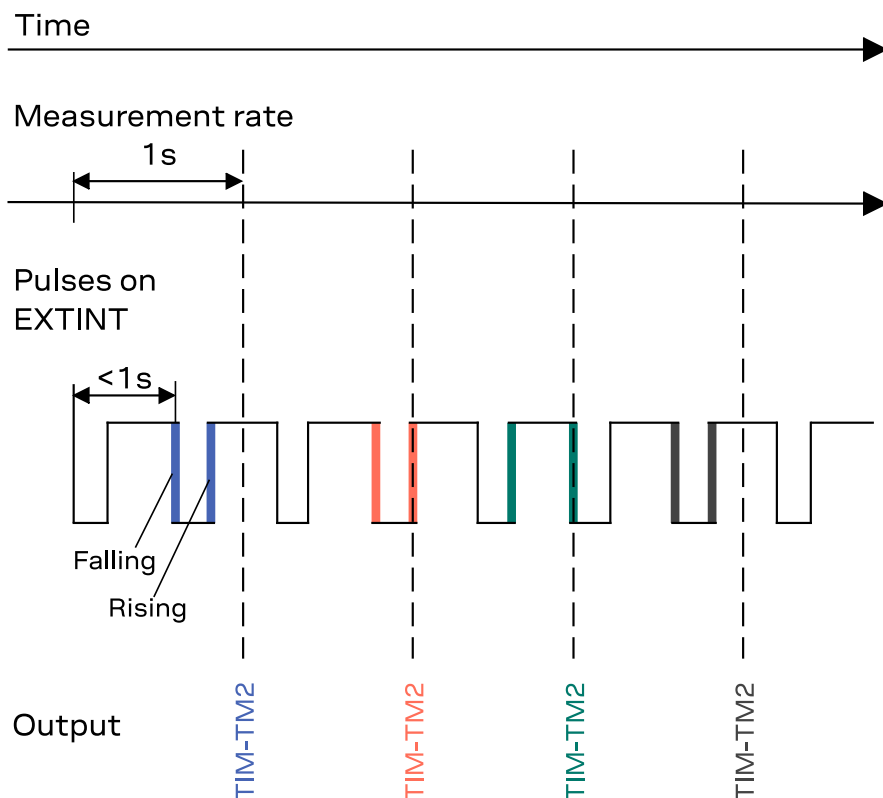


Figure 15: Time mark

3.9 Time pulse

The receiver includes a time pulse feature providing clock pulses with configurable duration and frequency. The time pulse function can be configured using the CFG-TP-* configuration group. The UBX-TIM-TP message provides time information for the next pulse and the time source.

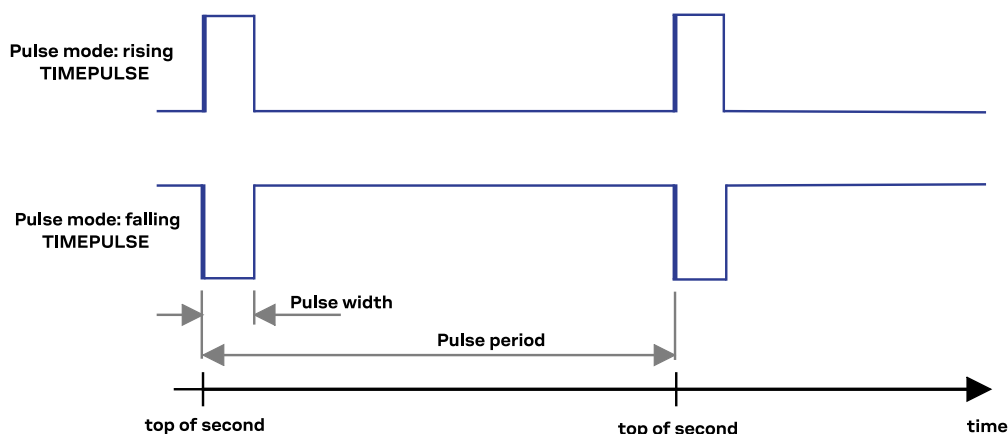


Figure 16: Time pulse

3.9.1 Recommendations

- The time pulse can be aligned to a wide variety of GNSS times or to variants of UTC derived from them. For further information, see [GNSS time bases](#). However, it is strongly recommended that the choice of time base is aligned with the available GNSS signals (for example, to produce GPS time or UTC(USNO), ensure GPS signals are available, and for Galileo time or UTC(EU) ensure the presence of Galileo signals, etc). This involves coordinating the setting of CFG-SIGNAL-* configuration group with the choice of time pulse time base.
- When using time pulse for precision timing applications it is recommended to calibrate the antenna cable delay against a reference timing source.
- To get the best timing accuracy with the antenna, a fixed and accurate position is needed.
- If relative time accuracy between multiple receivers is required, do not mix receivers of different product families. If this is required, the receivers must be calibrated accordingly, by setting cable delay and user delay.
- The recommended configuration when using the UBX-TIM-TP message is to set both the measurement rate (CFG-RATE-MEAS) and the time pulse frequency (CFG-TP-*) to 1 Hz.

The sequential order of the signal present at the TIMEPULSE pin and the respective output message for the simple case of 1 pulse per second (1PPS) is shown in the following figure.

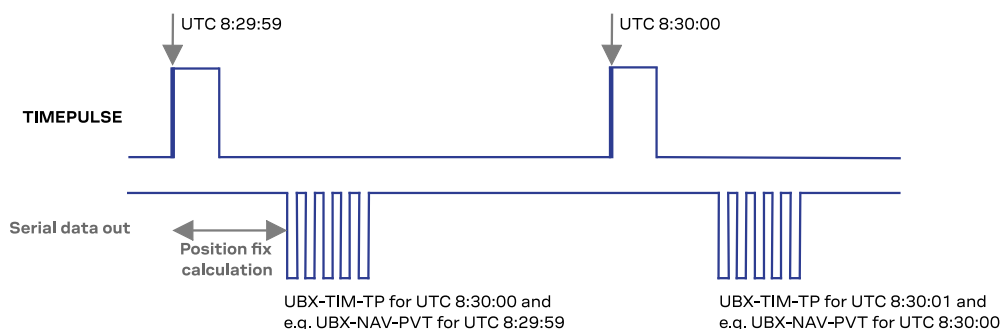


Figure 17: Time pulse and TIM-TP

3.9.2 Time pulse configuration

The time pulse (TIMEPULSE) signal has configurable pulse period, length and polarity (rising or falling edge).

It is possible to define different signal behavior (i.e. output frequency and pulse length) depending on whether or not the receiver is locked to reliable time source.

The configuration group CFG-TP-* can be used to change the time pulse settings, and includes the following parameters defining the pulse:

- **time pulse enable** - If this item is set, the time pulse is active.
- **frequency/period type** - Determines whether the time pulse is interpreted as frequency or period.
- **length/ratio type** - Determines whether the time pulse length is interpreted as length [us] or pulse ratio [%].
- **antenna cable delay** - Signal delay due to the cable between the antenna and the receiver.
- **pulse frequency/period** - Frequency or pulse time period when locked mode is not configured or not active.
- **pulse frequency/period lock** - Frequency or pulse time period for locked mode. In use as soon as the receiver has calculated a valid time from a received signal. Only used if the corresponding item is set to use another setting in locked mode.
- **pulse length/ratio** - Length or duty cycle of the generated pulse, specifies either time or ratio for the pulse to be on/off.
- **pulse length/ratio lock** - Length or duty cycle of the generated pulse for locked mode. In use as soon as the receiver has calculated a valid time from a received signal. Only used if the corresponding item is set to use another setting in locked mode.
- **user delay** - The cable delay from the receiver to the user device plus signal delay of any user application.
- **lock to GNSS freq** - If this item is set, uses the frequency gained from the GNSS signal information rather than the local oscillator's frequency.
- **locked other setting** - If this item is set, the alternative setting is used as soon as the receiver can calculate a valid time. This mode can be used, for example, to disable time pulse if the time is not locked, or to indicate a lock with different duty cycles.
- **align to TOW** - If this item is set, pulses are aligned to the top of a second.
- **polarity** - If set, the first edge of the pulse is a rising edge (pulse polarity: rising).
- **grid UTC/GNSS** - Selection between UTC (0), GPS (1), GLONASS (2), BeiDou (3), (4) Galileo and NAVIC (5) time grid. Also affects the time output by UBX-TIM-TP message.



The maximum pulse length cannot exceed the pulse period.



The high and the low period of the output cannot be less than 50 ns, otherwise pulses can be lost.

3.9.2.1 Example

The example below shows the 1PPS TIMEPULSE signal generated on the time pulse output according to the specific parameters of the CFG-TP-* configuration group:

- **CFG-TP-TP1_ENA** = 1
- **CFG-TP-PULSE_DEF** = 0 (PERIOD)
- **CFG-TP-PULSE_LENGTH_DEF** = 1 (LENGTH)
- **CFG-TP-PERIOD_TP1** = 1 000 000 μ s
- **CFG-TP-LEN_TP1** = 100 000 μ s
- **CFG-TP-TIMEGRID_TP1** = 1 (GPS)
- **CFG-TP-ALIGN_TO_TOW_TP1** = 1
- **CFG-TP-USE_LOCKED_TP1** = 1
- **CFG-TP-POL_TP1** = 1
- **CFG-TP-PERIOD_LOCK_TP1** = 1 000 000 μ s

- **CFG-TP-LEN_LOCK_TP1** = 100 000 μ s

The 1 Hz output is maintained whether or not the receiver is locked to GPS time. The alignment to TOW can only be maintained when GPS time is locked.

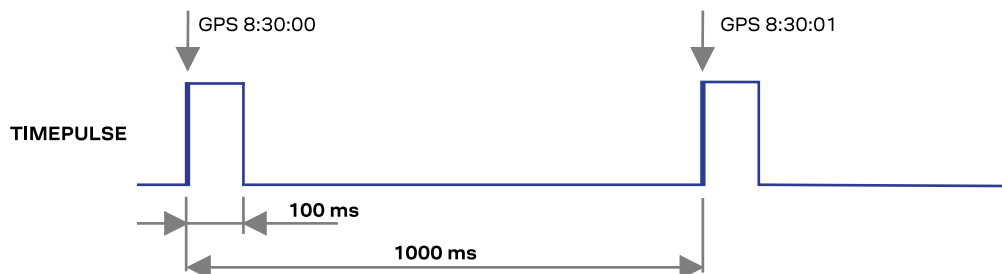


Figure 18: Time pulse signal with the example parameters

3.10 Time maintenance

Maintaining accurate time can improve the speed and performance of the receiver restart. Estimate of GNSS time can be maintained by a real-time clock, or it can be provided to the receiver by the host. Estimate of the clock drift of the receiver local oscillator or an external reference frequency can also be provided to improve the startup performance.

3.10.1 Real-time clock

The receiver contains a real-time clock (RTC). The RTC section is located in the backup domain and can keep time while the receiver is otherwise powered off. When the receiver powers up, it attempts to use the RTC to initialize receiver local time and in most cases this leads to considerably faster and more accurate first fixes.

3.10.2 Time assistance

The host can deliver time assistance to the receiver using UBX-MGA-INITIME-UTC or UBX-MGA-INITIME-GNSS for better startup performance.

The current GNSS time can be supplied to the receiver as a coarse value via the standard communication interfaces. This method suffers from communication latency and unpredictable delays so the accuracy of the supplied time is poor. The time aiding is also described in the [Transferring assistance data to the receiver](#).

Accuracy of the supplied time can be improved greatly if the host system has a very good sense of the current time and can deliver an exactly timed pulse to the EXTINT pin. This pulse informs the receiver when the supplied time assistance data is to be applied.

UTC time leap seconds and GPS-to-UTC conversion parameters are transmitted periodically by GPS satellites, but that happens only every 12.5 minutes. The receiver can normally calculate the correct leap seconds value from other GNSS systems immediately, but in some situations that is not possible. If the leap seconds information or the difference of time between GPS and UTC system is important for the host application, the information can be supplied to the receiver via the UBX-MGA-GPS-UTC aiding message.

3.10.3 Frequency assistance

Frequency assistance can improve the cold start speed in crystal-based designs. For TCXO-based designs, the frequency assistance has only minimal impact as the receiver is quick to acquire

accurate frequency from satellite transmissions. A stable external reference frequency can be used to speed up receiver testing in production test setup.

To supply hardware frequency assistance, connect a periodic rectangular signal with a frequency of up to 500 kHz to the EXTINT pin. The frequency can have an arbitrary duty cycle but the low/high phase duration must not be shorter than 50 ns. The applied frequency value must be submitted to the receiver using the UBX-MGA-INI-FREQ message.

3.10.4 Clock drift assistance

Estimate of the clock drift of the external local oscillator (TCXO or XTAL) can also be fetched from the receiver using the UBX-NAV-CLOCK message. This estimate can then be sent back to the receiver using the UBX-MGA-INI-CLKD message.

3.11 Protection level

3.11.1 Introduction

Critical applications need to know how much trust they can place in their GNSS receiver's output at any given moment. Computed by the GNSS receiver in real time, the protection level (PL) quantifies the reliability of the position information to allow systems to change their mode of operation and improve the efficiency and quality of the tasks being performed.

The GNSS receiver's protection level describes the maximum likely position error to a specified degree of confidence. For example, if a GNSS receiver determines its position with a 95% protection level of one meter, there is only a 5% chance that the reported position is more than one meter away from its true position. Like the accuracy estimate of the GNSS receiver, the protection level constantly fluctuates, influenced by all the common error sources that affect GNSS solutions. Unlike the accuracy estimate, the confidence level of the protection level is much higher and is validated against specific operating scenarios to ensure that the output bounds the true error.



The maximum navigation update rate for protection level is limited to 1 Hz.

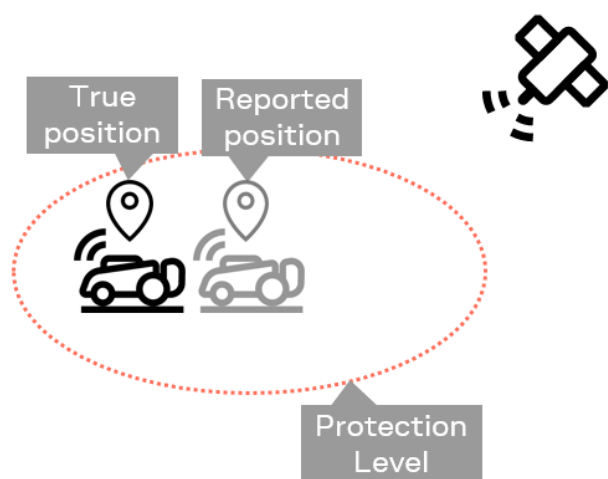


Figure 19: PL bounding true position error

3.11.2 Interface

The protection level bounds the true position error with a target misleading information risk (TMIR), for example 5% [MI/epoch] (read: 5% probability of having an MI per epoch). The target misleading information risk describes the probability per epoch of having misleading information (MI), meaning

that it is not possible to bound the true position error because it is larger than the protection level (see [Figure 20](#)).

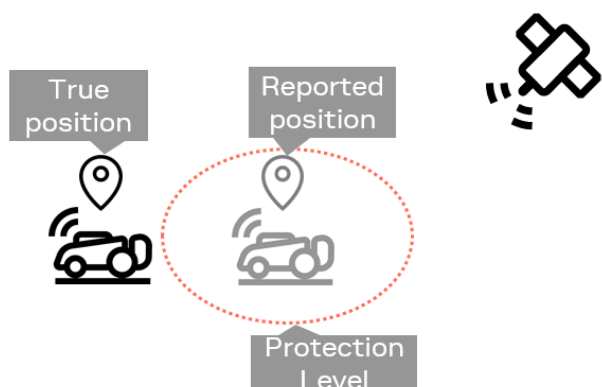


Figure 20: Misleading information

The output of the protection level is published through the UBX-NAV-PL message.

- TMIR is specified in one dimension for PL. It is not specified as a horizontal 2D or 3D value.
- The protection level values (UBX-NAV-PL.pIPos1/2/3) are confidence intervals around the reported position (for example, UBX-NAV-PVT or UBX-NAV-HPPOSLLH).
- The target misleading information risk is provided in exponential notation (UBX-NAV-PL.tmirCoeff and UBX-NAV-PL.tmirExp), for example UBX-NAV-PL.tmirCoeff = 5 and UBX-NAV-PL.tmirExp = 0 results in 5e0 (= 5).
- The true position error is generally unknown, unless a very accurate and reliable truth positioning system is reporting an estimate for the true position.

When the GNSS environment deviates significantly from the normal mode of operation as compared to scenarios where the PL has been validated, a validity flag is set to false to indicate these conditions. These conditions tend to be binary in nature, such as jamming has been detected, or the minimum number of satellites is being observed. UBX-NAV-PL reports a PL validity flag (see UBX-NAV-PL.pIPosValid), which indicates whether the PL is usable.

3.11.3 Validity requirements

The protection level performance depends on many external and internal factors. Some external factors such as a harsh GNSS environment may lead to degraded PL performance.

PL validity values	Description
UBX-NAV-PL.pIPosValid = 1	PL values are valid and can be used
UBX-NAV-PL.pIPosValid = 0	PL values are invalid and shall not be used

Table 30: PL validity

- The protection level validity flag and the misleading information are two separate, non-related parameters.

The PL feature is verified for the receiver configuration summarized in [Table 31](#).

Parameter	Details or required configuration key value
GPS system is enabled and used for navigation	CFG-SIGNAL-GPS_ENA = 1, CFG-SIGNAL-GPS_L1CA_ENA = 1

Parameter	Details or required configuration key value
Minimum 2 GNSS systems are enabled ²	Galileo, GLONASS, and/or BeiDou is enabled in addition to GPS CFG-SIGNAL-GAL_ENA = 1, CFG-SIGNAL-GAL_E1_ENA = 1, CFG-SIGNAL-GLO_ENA = 1, CFG-SIGNAL-GLO_L1_ENA = 1, and/or CFG-SIGNAL-BDS_ENA = 1, CFG-SIGNAL-BDS_B1C_ENA = 1 or CFG-SIGNAL-BDS_B1_ENA = 1
Automotive and portable dynamic models	CFG-NAVSPG-DYNMODEL = 0 or 4
Continuous mode	CFG-PM-OPERATEMODE = 0
Super-S signal feature is disabled	CFG-NAVSPG-SIGATTCOMP = 0
Static hold is disabled (optional)	CFG-MOT-GNSSSPEED_THRS = 0, CFG-MOT-GNSSSPEED_THRS = 0. Optional, disabling ensures that the static hold mode is not activated.
AssistNow Autonomous or AssistNow Offline are not used (optional)	CFG-ANA-USE_ANA = 0, AssistNow Offline data is not used. Optional, disabling ensures that the predicted orbits are not used for the navigation solution. This typically occurs during start-up before the ephemerides are decoded.

Table 31: Recommended configuration for using the PL feature


The Super-S feature is enabled by default.

The PL values are valid and can be used provided the conditions in [Table 32](#) are met.

Parameter	Condition
3D position fix	fixType = 3 in UBX-NAV-PVT message
Valid position fix flag	gnssfixOK = 1 UBX-NAV-PVT message
No jamming reported	jammingState flag in UBX-SEC-SIG
No spoofing reported	spoofingState flag in UBX-SEC-SIG
Valid and resolved time and date	validTime = 1, validDate = 1, and fullyResolved = 1 in UBX-NAV-PVT
Static hold mode is not activated	The static hold flag is not raised
Orbit prediction algorithm	AssistNow Autonomous or AssistNow Offline are not used for the navigation solution.

Table 32: Navigation solution requirements

3.1.1.4 Expected behavior

For each navigation epoch and for each coordinate axis, a PL value is provided. For example, if the coordinate frame reported is North/East/Down, then the UBX-NAV-PL contents can be interpreted as follows:

PL values	Description
UBX-NAV-PL.pIPos1	1 stands for the north axis
UBX-NAV-PL.pIPos2	2 stands for the east axis
UBX-NAV-PL.pIPos3	3 stands for the down axis

Table 33: Position PL values

If the PL coordinate frame is set to invalid (UBX-NAV-PL.pIPosFrame = 0), then the PL values shall not be used. If the PL validity flag is cleared (UBX-NAV-PL.pIValid = 0), the PL values shall not be used. Both of these cases must be checked.

Only if the PL is set to valid (UBX-NAV-PL.pIPosValid), the PL values (UBX-NAV-PL.pIPos1/2/3) can be used and are reliable with respect to the target misleading information risk.

² Refer to the data sheet [\[1\]](#) for the supported GNSS combinations.

3.12 AssistNow GNSS assistance

GNSS receivers require information on the exact time as well as satellite orbits for several satellites to obtain a position fix. At start-up, the receiver finds the satellite signals and decodes the information from the signals. This typically takes 20–30 s under good signal conditions. Under adverse signal conditions, finding signals and decoding the data may take several minutes or even completely fail.

Assistance data enables the receiver to find satellites faster and even compute the position without the need to download data from satellites. This significantly reduces the time-to-first-fix (TTFF), improves fix accuracy, and increases satellite availability, even under poor signal conditions.

The AssistNow service is a u-blox proprietary assisted GNSS (A-GNSS) service compatible with the u-blox GNSS receivers. The AssistNow Live Orbits and the AssistNow Predicted Orbits services are accessed over the HTTP or HTTPS protocols.

The AssistNow Autonomous feature runs locally on the receiver and generates mid- and long-term predicted orbits without an internet connection. The orbit prediction is based on broadcast ephemerides.

For more information on the AssistNow services, refer to the AssistNow service documentation [6].

Table 34 summarizes the AssistNow feature and services.

	AssistNow Live Orbits	AssistNow Predictive Orbits	AssistNow Autonomous
Type of feature	Service	Service	Stand-alone feature
Internet access	Always available	Sporadic	Not needed
Data types ³	EPH, ALM, TIME, AUX	ANO, ALM	ANO, ALM
Data validity	2–4 hours	1–14 days	3–6 days
Data transfer size	Medium	High	None (local)
Flash memory	Not required	Optional, receiver or host	Optional, receiver or host
TTFF performance	Best	Good	Good/fair

Table 34: AssistNow service overview

3.12.1 Legacy services AssistNow Online and AssistNow Offline

The legacy AssistNow services Online and Offline are replaced with the corresponding new services Live Orbits and Predictive Orbits. For more information on the availability of the legacy AssistNow services, refer to the AssistNow Product change note [5].

Use AssistNow Live Orbits and Predictive Orbits for new designs. For more information on migrating from the legacy AssistNow services, refer to the AssistNow service documentation [6].

3.12.2 AssistNow Live Orbits

AssistNow Live Orbits is used in GNSS receiver systems with direct internet access. At start-up, the host downloads the satellite ephemerides, time aiding, and other optional data from the service and sends it to the receiver. The assistance data can reduce the TTFF down to 1–2 s under good signal conditions.

For the supported GNSS constellations and signals, refer to the MAX-M10S Data sheet [1].

³ Data types: ephemeris (EPH), almanac (ALM), predictive orbits (ANO), auxiliary data (AUX), and time (TIME)

3.12.2.1 Live Orbits data types

Fast signal acquisition requires time information as well as satellite and orbital data. Current time is also needed to determine the validity and use other assistance data.

The time assistance message UBX-MGA-INI-TIME_UTC contains the current time and the accuracy for the time provided. Time accuracy is set by default to 10 s to consider the network latency and any processing delay on the host. The ephemerides contain precise short-term satellite and orbital data. The ephemerides are valid approximately 2–4 hours depending on the GNSS system.

Almanac contains a reduced-precision subset of the clock and ephemeris parameters. It is valid for a few weeks and enables the receiver to find the currently visible satellites faster. AssistNow Live Orbits provides also auxiliary information on satellite health status, ionospheric corrections to improve position accuracy, and time information on different GNSS systems.

The total AssistNow Live Orbits data size is approximately 2–4 kB per GNSS constellation. For detailed information on the supported GNSS constellations and satellites, data types, validity period, data size, and the full list of messages, refer to the AssistNow service documentation [6].

Table 35 summarizes the AssistNow Live Orbits assistance data and the related messages.

Type of data	Message(s)	Description
Time assistance (TIME)	UBX-MGA-INI-TIME_UTC	Current time (coarse)
Ephemeris (EPH) ⁴	UBX-MGA-XXX-EPH	Current satellite and orbital data for fix calculation
Almanac (ALM) ⁴	UBX-MGA-XXX-ALM	Coarse orbital information
Auxiliary data (AUX) ⁴	UBX-MGA-XXX-IONO	Simplified ionospheric corrections (Klobuchar)
	UBX-MGA-XXX-HEALTH	Satellite health information
	UBX-MGA-XXX-UTC	UTC time information
	UBX-MGA-XXX-TIMEOFFSET	Time offset relative to the GPS time

Table 35: AssistNow Live Orbits assistance data

3.12.2.2 Host operation

The host operation consists of the following steps:

- Start up the receiver
- Download the AssistNow Live Orbits assistance data from the service
- Transfer the Assistance data to the receiver immediately at start-up

The AssistNow C Client Toolkit [6] helps in implementing the AssistNow data download and transfer in the host application.

Downloading and storing assistance data

The host downloads the AssistNow Live Orbits assistance data from the service. The received data consists of UBX messages starting with time assistance followed by ephemerides, almanac, and auxiliary data.

The host can store the assistance data in the memory and use it for subsequent receiver start-ups until it expires. In addition, the receiver can maintain the assistance data and other information in its battery-backed RAM (BBR) memory.

Data download from the service is not required if valid assistance data is already available on the host system or on the receiver. However, it is recommended to replace ephemerides close to expiry with fresh data from the service.

⁴ The string XXX stands for the GNSS constellations GPS, GAL, BDS, GLO, or QZSS. Almanac and auxiliary data are not available for all GNSS constellations.

For more information on maintaining data on the receiver, refer to the section [Preserving AssistNow and operational data during power-off](#).

Transferring assistance data to the receiver

The assistance data must be sent to the receiver immediately at the receiver start-up. If downloading data from the service, send the messages directly after receiving them in the same order they were received: time assistance followed by ephemerides, almanac, and auxiliary data.

 Send the assistance data immediately at the receiver start-up

If valid assistance data stored on the host is used, the host must construct the UBX-MGA-INITIME_UTC message with current time and accuracy for it. The time accuracy provided must not be better than the actual accuracy of the host time.

 Do not provide too optimistic value for the time accuracy. This may degrade the start-up performance.

The internal time maintained in the receiver is generally significantly more accurate than the assisted time. If the receiver already has time information available, it therefore ignores the time assistance and continues to use its internal time. To force the receiver use the assisted time, clear the real-time-clock (RTC) time in the BBR memory before sending time assistance.

Apart from time assistance, sending new assistance data from the host clears the existing data in the receiver. Valid assistance data is stored in the receiver's BBR memory and used accordingly. However, expired assistance data is rejected resulting in a loss of assistance data.

 Do not send ephemerides close to expiring to the receiver. This may degrade the start-up performance.

3.12.3 AssistNow Predictive Orbits

AssistNow Predictive Orbits is used in GNSS receiver applications with infrequent internet access. The host downloads assistance data from the service and stores it on the host side to be available for use at every start-up. The data can also be stored on a flash memory connected to the receiver. The assistance data can reduce the TTFF to well below 10 s under good signal conditions.

For the supported GNSS constellations and signals, refer to the MAX-M10S Data sheet [1].

3.12.3.1 Predictive Orbits data types

AssistNow Predictive Orbits provides medium-term assistance data for fast signal acquisition. The data can be used for several receiver start-ups over a long period of time.

The time assistance message UBX-MGA-INITIME_UTC containing current time is only provided with Galileo or GLONASS almanac. The time accuracy is set by default to 10 s to consider the network latency and any processing delay on the host. Time assistance is only valid for an receiver start-up directly after the data download.

The UBX-MGA-ANO messages contain medium-term orbital data valid up to 14 days. There are one or more UBX-MGA-ANO messages for each satellite depending on the selected validity period. Each UBX-MGA-ANO message contains information on the GNSS constellation (gnssId), satellite ID (svId), and date (year, month, day). Time (hour) is always set to UTC 12:00. The data is most accurate at the time indicated.

Almanac contains a reduced-precision subset of the clock and ephemeris parameters. It is valid for a few weeks and enables the receiver to find the currently visible satellites faster. Refer to the Interface description for more information on the UBX-MGA-* messages.

The total AssistNow Predictive Orbits data size is of the order of 2 - 3 kB per one day and 35 kB per 14 days per GNSS constellation. For detailed information on the supported GNSS constellations and satellites, data types, validity period, data size, and the full list of messages, refer to the AssistNow service documentation [6].

Table 36 summarizes the AssistNow Predictive Orbits assistance data and the related messages.

Type of data	Message(s)	Description
Time assistance (TIME)	UBX-MGA-INI-TIME_UTC	Current time (coarse). Provided only with Galileo and GLONASS almanac.
Predictive Orbits (ANO)	UBX-MGA-ANO	Medium-term satellite and orbital data. Valid for 1 – 14 days.
Almanac (ALM) ⁵	UBX-MGA-XXX-ALM	Coarse orbital information

Table 36: AssistNow Predictive Orbits assistance data

3.12.3.2 Host operation

The host has two main tasks:

- Download the AssistNow Predictive Orbits assistance data from the service
- Transfer the assistance data to the receiver. There are two options available: the flash-based operation and the host-based operation.

The AssistNow C Client Toolkit [6] helps in implementing the AssistNow data download and transfer in the host application.

Downloading assistance data

The host downloads the AssistNow Predictive Orbits assistance data from the service. The received data consists of UBX messages starting with time assistance (optional) followed by almanac and Predictive Orbits data. The Predictive Orbits data is chronologically ordered starting with the earliest date.

The Predictive Orbits data is typically valid for several receiver start-ups. Data download from the service is not required if valid assistance data is already available on the host system or on the receiver.

Host-based operation

In host-based operation, the host downloads the assistance data from the service and stores it on the host system. At receiver start-up, the host selects valid assistance data and sends it to the receiver.

Typical steps in the host-based operation include:

- Download Predictive Orbits data from the service
- Select the UBX-MGA-ANO messages based on the GNSS constellation, satellite ID, and time that best matches the current time. There is only one UBX-MGA-XXX-ALM message for each satellite.
- At receiver start-up, construct and send the time aiding message UBX-MGA-INI-TIME_UTC (optional) followed by the Predictive Orbits and almanac data to the receiver.



Do not provide too optimistic value for the time accuracy. This may degrade the start-up performance.

The internal time maintained in the receiver is generally significantly more accurate than the assisted time. If the receiver already has time information available, it therefore ignores the time

⁵ The string XXX stands for the GNSS constellations GPS, GAL, BDS, GLO, or QZSS. Almanac is not available for all GNSS constellations.

assistance and continues to use its internal time. To force the receiver use the assisted time, clear the real-time-clock (RTC) time in the BBR memory before sending time assistance.

Apart from time assistance, sending new assistance data from the host clears the existing data in the receiver. Valid assistance data is stored in the receiver's BBR memory and used accordingly. However, expired assistance data is rejected resulting in a loss of assistance data.

For more information on maintaining data on the receiver, refer to the section [Preserving AssistNow and operational data during power-off](#).

3.12.4 Preserving AssistNow and operational data during power-off

The receiver stores the assistance data from the host in the navigation database. During operation, the receiver collects satellite broadcast data and updates the database when new data becomes available. Also time and position are continuously updated.

The receiver automatically copies the satellite data to the BBR to maintain it during the receiver power-off. Alternatively, the host can store satellite data on the host side or in the flash memory connected to the receiver.

The following mechanisms can be used instead or in parallel to the AssistNow feature. The maintained assistance data can be used for subsequent receiver start-ups until it expires.



The quality of the assistance data degrades over time resulting in longer TTFF and reduced initial position accuracy. Provide fresh assistance data for best start-up performance.

Maintain BBR and RTC time

The recommended way to preserve satellite and operational data during power-off is to maintain the BBR and RTC time. The satellite data, last position, user configuration, and calibration data are all stored in the BBR. The RTC must be present to maintain an estimate of time. The receiver automatically applies the data at the next startup.



The backup domain must be supplied to maintain BBR and RTC time.

Database dump

The database dump is used to store the receiver's navigation database on the host. Before power-off, the host reads the navigation database by polling the UBX-MGA-DBD message and stores the received UBX-MGA-DBD messages for later use. At the next startup, the host sends time assistance and the UBX-MGA-DBD messages back to the receiver. For more information on the UBX-MGA-DBD message, refer to the Interface description [3].

Save-on-shutdown (SOS)

The save-on-shutdown (SOS) feature is used to store the BBR contents to a flash memory connected to the receiver. Before power-off, the host instructs the receiver to store the BBR contents in the flash memory. The data is automatically restored from the flash at the next startup. For more information on the SOS feature, refer to the section [Save-on-shutdown](#).

3.12.5 AssistNow Autonomous

The *AssistNow Autonomous* feature provides a functionality similar to *AssistNow Predictive Orbits* without the need for a host and a connection. Based on a broadcast ephemeris downloaded from the satellite (or obtained by *AssistNow Live Orbits*), the receiver can autonomously (i.e. without any host interaction or online connection) generate an accurate satellite orbit representation ("AssistNow Autonomous data") that is usable for navigation much longer than the underlying broadcast

ephemeris was intended for. This makes downloading new ephemeris or aiding data for the first fix unnecessary for subsequent startups of the receiver.

3.12.5.1 Concept

The *AssistNow Autonomous orbit* is a feature designed to extend the usefulness of satellite orbit data beyond the limited validity of broadcast ephemerides. A broadcast ephemeris, which is downloaded directly from a satellite, provides a highly accurate snapshot of the satellite's orbit—typically valid for around four hours in the case of GPS. However, once this period expires, the data diverges significantly from the satellite's true trajectory and can no longer be used reliably for positioning.

To overcome this limitation, *AssistNow Autonomous* generates long-term orbit predictions by extending one or more broadcast ephemerides. Although these predictions are not perfectly precise, they are accurate enough to support reliable navigation over multiple satellite revolutions. The predictive orbit data is created automatically and stored in the receiver's battery-backed memory (BBR), and optionally in external flash memory or on the host device. The number of satellites for which data can be stored depends on the receiver's configuration and may vary during operation.

This entire process is transparent to the user and does not interfere with the normal functioning of the receiver. All calculations are performed in the background. If no broadcast ephemeris is available at the time of navigation, *AssistNow Autonomous* uses the stored data to generate the necessary orbit segments for positioning. The system also ensures that the data remains current, minimizing the time needed for orbit calculations when navigation is initiated.

To maintain accuracy, *AssistNow Autonomous* automatically invalidates orbit data that has become too old and could lead to significant positioning errors. This expiration threshold can be configured using the key CFG-ANA-ORBMAXERR. The quality of orbit predictions improves when a satellite has been observed multiple times, though this enhancement requires the presence of flash memory. Better prediction quality also extends the period during which the data remains usable.

AssistNow Autonomous supports satellites from GPS, GLONASS, Galileo, and BeiDou systems. However, it excludes satellites with high orbital eccentricity, specifically those with eccentricity greater than 0.05, such as Galileo E18.

The [Figure 21](#) illustrates the *AssistNow Autonomous* concept in a graphical way.

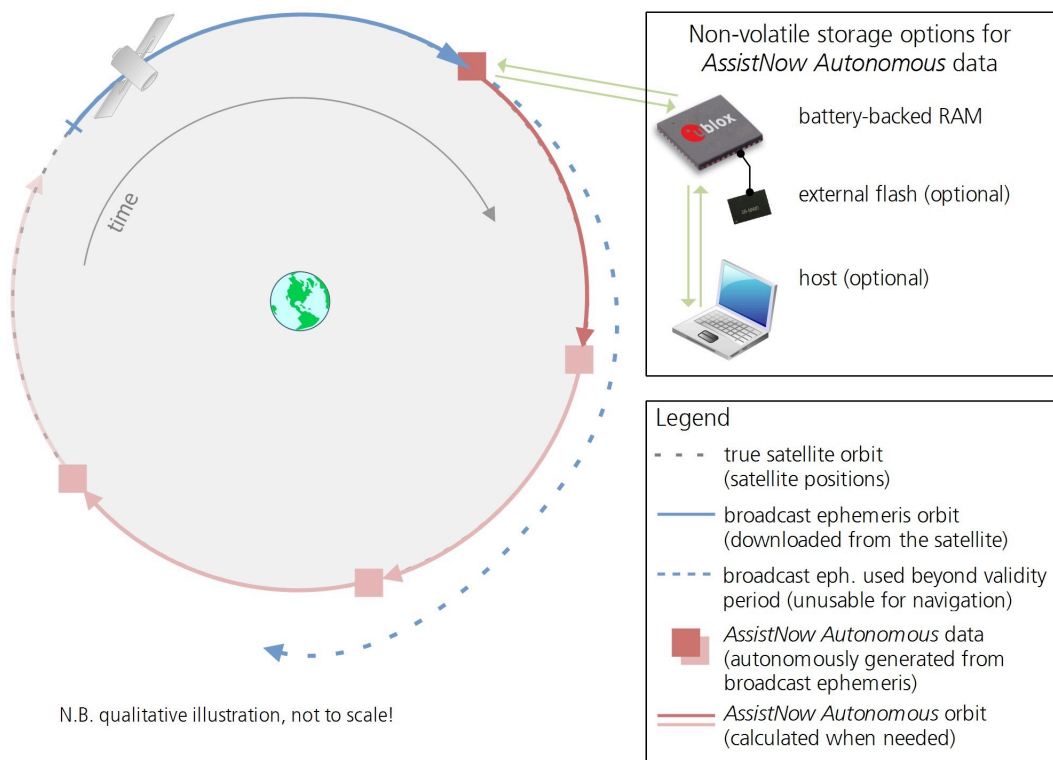


Figure 21: AssistNow Autonomous illustrative concept

3.12.5.2 Interface

The configuration keys and interface messages related to the *AssistNow Autonomous* are listed in Table 37.

Message name	Message Type	Description
CFG-ANA-USE_ANA	Configuration key	Used to enable or disable the <i>AssistNow Autonomous</i> feature. If the receiver uses flash memory, disabling the <i>AssistNow Autonomous</i> feature will delete all previously collected satellite observation data from the flash memory.
CFG-ANA-ORBMAXERR	Configuration key	Allows changing «maximum acceptable modeled orbit error» (in meters). Note that this number does not reflect the true orbit error introduced by extending the ephemeris. It is a statistical value that represents a certain expected upper limit based on a number of parameters. A rough approximation that relates the maximum extension time to this setting is: $maxError [m] = maxAge [d] * f$, where the factor f is 30 for data derived from satellites seen once and 16 for data derived for satellites seen multiple times during a long enough time period. It is recommended to use the firmware default value that corresponds to a default orbit data validity of approximately three days (for GPS satellites observed once) and up to six days (for satellites observed multiple times over a period of at least half a day).
UBX-NAV-AOPSTATUS	Output	Provides information on the current state of the <i>AssistNow Autonomous</i> subsystem. The status indicates whether the <i>AssistNow Autonomous</i> subsystem is currently idle (or not enabled) or busy generating data or orbits. Hosts should monitor this information and only power off the receiver when the subsystem is idle (that is, when the <code>status</code> field shows a steady zero).
UBX-NAV-SAT	Output	Indicates the use of <i>AssistNow Autonomous</i> orbits for individual satellites.
UBX-NAV-ORB	Output	Indicates the availability of <i>AssistNow Autonomous</i> orbits for individual satellites.

Message name	Message Type	Description
UBX-MGA-DBD	Input/Output	Provides a means to retrieve the <i>AssistNow Autonomous</i> data from the receiver to preserve the data in power-off mode where no battery backup is available. Note that the receiver requires the absolute time (i.e. full date and time) to calculate <i>AssistNow Autonomous</i> orbits. For the best performance, it is therefore recommended to supply this information to the receiver using the UBX-MGA-INITIME_UTC message in this scenario.

Table 37: AssistNow Autonomous related messages

3.12.5.3 Benefits and drawbacks

AssistNow Autonomous can provide quicker startup times by lowering the TTFF, provided that data is available for enough visible satellites. This is particularly true under weak signal conditions where it might not be possible to download broadcast ephemerides at all and therefore, no fix would be possible without *AssistNow Autonomous* (or A-GNSS). It is however required that the receiver roughly knows the absolute time, either from an RTC or from time-aiding and that it knows which satellites are visible, either from the almanac or from tracking the respective signals.

The *AssistNow Autonomous* orbit (satellite position) accuracy depends on various factors, such as the particular type of satellite, the accuracy of the underlying broadcast ephemeris, or the orbital phase of the satellite and Earth, and the age of the data (errors add up over time).

There is no direct relation between (true and statistical) orbit accuracy and positioning accuracy. The positioning accuracy depends on various factors, such as the satellite position accuracy, the number of visible satellites, and the geometry (DOP) of the visible satellites. Position fixes that include *AssistNow Autonomous* orbit information may be significantly worse than fixes using only broadcast ephemerides. Therefore, it might be necessary to adjust the limits of the navigation output filters (CFG-NAVSPG-OUTFIL_*).

Unknown future events form a fundamental deficiency of any system and can prevent precise satellite orbit predictions. Hence, the receiver will not be able to know about satellites that will have become unhealthy, have undergone a clock swap, or have had a maneuver. This means that the navigation engine might rarely mistake a wrong satellite position as the true satellite position. However, provided that there are enough other good satellites, the navigation algorithms will eventually eliminate a defective orbit from the navigation solution.

The repeatability of the satellite constellation is a potential pitfall for the use of the *AssistNow Autonomous* feature. For a given location on Earth, the (GPS) constellation (geometry of visible satellites) repeats every 24 hours. Hence, when the receiver «learned» about a number of satellites at some point in time, the same satellites will in most places *not* be visible 12 hours later, and the available *AssistNow Autonomous* data will not be of any help. However, after another 12 hours, usable data would be available because it was generated 24 hours ago.

The longer a receiver observes the sky, the more satellites it will see. At the equator, and with full sky view, approximately ten (GPS) satellites will show up in a one-hour window. After four hours of observation approximately 16 satellites (i.e. half the constellation), after 10 hours approximately 24 satellites (2/3rd of the constellation), and after approximately 16 hours the full constellation will have been observed (and *AssistNow Autonomous* data generated). Lower sky visibility reduces these figures (i.e. the number of satellites seen). Further away from the equator, the numbers improve because the satellites can be seen twice a day. For example, at 47 degrees north, the full constellation can be observed in approximately 12 hours with full sky view.

The calculations required for *AssistNow Autonomous* are carried out on the receiver. This requires energy and therefore, users may occasionally see increased power consumption during short periods (several seconds, rarely more than 60 seconds) when such calculations are running. Ongoing

calculations will automatically prevent the power save mode from entering the power-off state. The power-down will be delayed until all calculations are done.



AssistNow Autonomous should be enabled if the system has sporadic access to the AssistNow Predictive Orbits service. In this case, the receiver chooses intelligently the more reliable orbit predictions for each satellite. This way the autonomous prediction can provide performance improvements if the data becomes old or gets outdated.

3.13 Data batching

3.13.1 Introduction

The data batching feature allows position fixes to be stored in the RAM of the receiver to be retrieved later in one batch. Batching of position fixes happens independently of the host system, and can continue while the host is powered down.

[Table 38](#) lists all the batching-related messages:

Message	Description
UBX-MON-BATCH	Provides information about the buffer fill level and dropped data due to overrun
UBX-LOG-RETRIEVEBATCH	Starts the batch retrieval process
UBX-LOG-BATCH	A batch entry returned by the receiver

Table 38: Batching-related messages

3.13.2 Setting up the data batching

Data batching is disabled per default and it has to be configured before use via the CFG-BATCH-* configuration group.

The feature must be enabled and the buffer size must be set to greater than 0. It is possible to set up a PIO as a flag that indicates when the buffer is close to filling up. The fill level when this PIO is asserted can be set by the user separately from the buffer size. The notification fill level must not be larger than the buffer size.

If the host does not retrieve the batched fixes before the buffer fills up, the oldest fix will be dropped and replaced with the newest.

The RAM available in the chip limits the size of the buffer. To make the best use of the available space, users can select what data they want to batch. When batching is enabled, a basic set of data is stored and the configuration flags EXTRAPVT and EXTRAODO can be used to store more detailed information about the position fixes. However, enabling the EXTRAPVT and EXTRAODO flags reduces the number of fixes that can be batched.

The receiver will reject the configuration if it cannot allocate the required buffer memory. To ensure robust operation of the receiver the limits in [Table 39](#) are enforced:

EXTRAPVT	EXTRAODO	Maximum number of epochs
0	0	
0	1	
1	0	
1	1	

Table 39: Maximum number of batched epochs

-  It is recommended to disable all periodic output messages when using data batching. This improves system robustness and also helps ensure that the output of batched data is not delayed by other messages.
-  The buffer size is set up in terms of navigation epochs. This means that the time that can be covered with a certain buffer depends on the navigation rate. This rate can be set via the "CFG-RATE-" configuration group.

3.13.3 Retrieval

UBX-LOG-RETRIEVEBATCH message starts the process which allows the receiver to output batch entries. Batching must not be stopped for readout because all batched data is lost when the feature is disabled.

Batched fixes are always retrieved starting with the oldest fix in the buffer and progressing towards newer ones. There is no way to skip certain fixes during retrieval.

When a UBX-LOG-RETRIEVEBATCH message is sent the receiver transmits all batched fixes. It is recommended to send a retrieval request with `sendMonFirst` set. This way the receiver will send a UBX-MON-BATCH message first that contains the number of fixes in the batching buffer. This information can be used to detect when the u-blox receiver finishes sending data.

Once retrieval has started, the receiver will first send UBX-MON-BATCH message if `sendMonFirst` option was selected in the UBX-LOG-RETRIEVEBATCH message. After that, it will send UBX-LOG-BATCH messages with the batched fixes.

To maximize the speed of transfer, it is recommended that a high communication data rate is used.

-  The receiver will discard retrieval request while processing a previous UBX-LOG-RETRIEVEBATCH message.
-  The receiver does **not** acknowledge the reception of UBX-LOG-RETRIEVEBATCH message. The receiver responds with UBX-MON-BATCH (optional) and UBX-LOG-BATCH messages.

3.14 CloudLocate

In CloudLocate setup, the host processor of the customer application fetches a set of satellite signal measurements from the receiver and sends those to the u-blox CloudLocate service for position calculation. The CloudLocate service uses these measurements and current assistance data to calculate the receiver position. This data is then provided to the customer enterprise cloud for further use. Power saving up to 90% is possible compared to a cold start scenario.

The receiver starts to collect measurements as soon as it finds any satellite signals. It does not need to wait for a position fix for this. Collecting the measurements takes only a short time, so the application can quickly turn off the receiver or put it into a backup state.

3.14.1 CloudLocate measurements

The satellite signal measurements can be requested from the receiver either as a complete or compact raw measurement message.

The complete raw measurement message (UBX-RXM-MEASX) provides measurements for all visible satellites. The customer application can wait until the number of satellites in the message is sufficient and then send the message to the CloudLocate service. The amount of data in a MEASX message for five satellites is about 170 bytes. This increases by 24 bytes for each additional satellite.

Compact raw messages (UBX-RXM-MEAS50, UBX-RXM-MEAS20, UBX-RXM-MEAS12C, and UBX-RXM-MEAS12D) can be used when the amount of data to be sent to the cloud needs to be minimized. The data can be reduced to 50, 20, or even 12 Bytes, and the time for the receiver to stay on shall be limited to the minimum as well. These messages contain the measurement data in compressed format and use satellites only from a limited set of GNSS constellations. With the default settings, these messages contain measurement data only for a small number of satellites.

The raw measurement messages are enabled with the configuration keys in the CFG-MSGOUT configuration group. For example, setting the configuration key CFG-MSGOUT-UBX_RXM_MEAS50_UART1 to value 1 with UBX-CFG-VALSET message enables output of the UBX-RXM-MEAS50 message in the UART1 port for each navigation epoch.

The UBX-RXM-MEASX message can be sent with UBX header and checksum, and the message can be either in binary format or as encoded text. With the compact raw measurement messages, only the payload portion of the message is sent. Encoding the data would increase the size, so the compact messages should be sent in binary format.

In adverse conditions, satellite signal reception may take a long time or may not be possible. In such a situation, the payload of a compact raw measurement message is empty. The host application can wait until the payload contains data and only then switch off the receiver. In the case of using UBX-RXM-MEASX, the payload always has some data. The host application must check if the message contains enough information to calculate a position. For example, the application can wait for a confirmation from the customer enterprise cloud.

For more information on the raw measurement messages and on using the CloudLocate setup, see the Interface description [3] and the [u-blox website CloudLocate documentation](#).

3.15 Collecting debug logfiles: design-in guidance

The ability to collect logfiles with additional UBX protocol messages is essential throughout the MAX-M10S receiver's lifecycle: development, integration, validation, and field support. Enabling and collecting logfiles makes the integration and troubleshooting faster, more precise, and cost-effective, making it easier to resolve issues promptly, even if they occur in the field.

The additional messages offer deep insight into:

- Signal acquisition, tracking and navigation performance
- Protocol-level communication and interfaces
- Internal status and diagnostic data
- Unexpected behavior

To avoid support delays and ensure robust integration, incorporate logfile collection in both the hardware interface design and host software capabilities during the design-in phase.

3.15.1 Checklist for designing debug logfile collection

To ensure a successful and reliable integration of UBX message output within your system, follow the steps below. These cover both hardware and host configuration requirements.

Hardware interface

- Ensure at least one GNSS receiver interface is available for UBX message output.
- Provide access to an interface for external connection (for example to PC or data logger).
- Design the interface to handle the bandwidth required for UBX message output.
- Configure a baud rate that supports the expected debug message volume.

Host and configuration

- Enable the host to send UBX configuration strings to activate message output on the correct interface.
- Choose the message set based on the intended use case or guidance from u-blox technical support.
- Set up the host to store or stream the collected logfiles as needed.

Support and flexibility

- Request message definitions and estimated output payload sizes from u-blox technical support.
- Ensure updates or modifications to the message set are possible without firmware changes.

3.15.2 Hardware interface options for collecting debug logfiles

The receiver supports UBX message output over all available communication interfaces. When collecting logfiles, consider the scenarios in [Table 40](#).

Scenario	Hardware design guidance
Logs collected by host	Use a communication interface (e.g. UART) connected to the host processor. The host enables messages and stores the logfile locally.
Logs collected by external device	Expose a communication interface via a debug header or test points to allow connection to a PC or external logger.

Table 40: Logging interface options by scenario

Interfaces must be available and accessible at runtime. Messages are explicitly enabled per interface using configuration commands.

Enabling UBX output messages on a specific interface:

By default, the GNSS receiver does not output UBX messages used for in-depth analysis. Select the required messages based on the specific application or follow guidance from u-blox's technical support. Activate the messages on the appropriate interface using configuration messages.

- u-blox technical support provides a tailored set of binary CFG-MSGOUT configuration strings to enable the required messages on the selected interface.
- They also provide an estimated output payload size to help you assess the interface bandwidth requirements.
- These UBX configuration strings are sent to the MAX-M10S either by the host application or an external PC and the messages are output on the selected interface.

3.15.3 Host application and configuration requirements for collecting debug logfiles

The host must:

- Transmit the provided UBX configuration to the GNSS receiver
- Manage the interface on which the messages are enabled (host-connected or externally exposed)
- Log, buffer or stream the collected messages as required
- Provide a means to enter a debug mode and switch message sets based on a use case or support guidance
- Monitor interface health using MON-TXBUF to detect potential overflows

Message throughput depends on the message set and GNSS activity. Select appropriate baud rates and buffering strategies to avoid data loss.

3.15.4 Recommended message groups by use case

[Table 41](#) presents a preliminary set of a messages which can be enabled for logfile collection. These messages can assist technical support during the initial troubleshooting and help refine the list of debug messages to be provided for more detailed analysis.

Use case	Purpose	Typical UBX messages	Estimated interface load
General debug	OS-level issues, system exceptions	MON-HW, MON-COMMS, MON-SYS, MON-EXCEPT, INF-*	Low
RF / interference	Signal quality, jamming, spoofing analysis	SEC-SIG, SEC-SIGLOG, MON-RF, MON-SPAN, NAV-SIG	High
Tracking	Satellite tracking and signal acquisition	RXM-RAWX, RXM-MEASX, RXM-SFRBX	High
Navigation	Standard precision GNSS positioning solution	NAV-PVT, NAV-STATUS, NAV-SIG, NAV-SAT, NAV-CLOCK	Medium
Timing	GNSS time synchronization and pulse accuracy	TIM-TP, NAV-TIMEGPS, NAV-TIMEUTC, TIM-SVIN	Low

Table 41: Recommended message groups per use case



Only enable the message sets required for the specific use case. This keeps the system easier to manage and helps avoid unnecessary bandwidth usage. This is especially important in applications with limited buffer space or constrained network capacity. Selective message logging ensures optimal performance and cleaner diagnostic data.

4 Hardware integration

This chapter explains how the receiver can be integrated into an application design.

4.1 Power supply

MAX-M10S has the following power supply pins: **VCC**, **V_IO** and **V_BCKP**.

Power supply at VCC and V_IO must be present for normal operation. These two pins can either be connected together or supplied independently by the application. Power supply at V_BCKP is optional. If present, it enables the hardware backup mode when V_IO and VCC supplies are off.

The 3.3 V and 1.8 V operating voltages allow different combinations for the power supply design. These are listed in [Supply design examples](#).

Refer to the MAX-M10S Data sheet [1] for absolute maximum ratings, operating conditions, and power requirements.

4.1.1 VCC

VCC provides power to the core and RF domains and must be supplied during normal operation. For low power consumption, the VCC pin supplies power to the core via an internal DCDC converter. A filtered VCC supply is available on the VCC_RF pin. The VCC_RF output voltage is derived from the VCC supply and is available whenever VCC is supplied.



Do not add series resistance greater than 0.2 Ω on the supply line to avoid voltage ripple due to the dynamic current conditions.

4.1.2 V_IO

V_IO supplies all the digital IOs, clock, and the backup domain. The current drawn at V_IO depends on the activity and loading of the PIOs and the main oscillator.

A power interruption at V_IO will erase the battery-backed RAM (BBR) unless there is an external supply connected to V_BCKP.

V_IO allows two voltage ranges, 1.8 V or 3.3 V operation. For 1.8 V designs, the VIO_SEL pin must be connected to GND. For 3.3 V designs, it must be left open.



V_IO supply voltage must not be higher than VCC + 0.3 V.

4.1.3 V_BCKP

Power supply at V_BCKP is optional. If the power supply at V_IO is interrupted, but the V_BCKP pin is supplied, the receiver enters the hardware backup mode. In this mode, the RTC time and the GNSS orbit data in the BBR are maintained. Valid time and GNSS orbit data at startup improves positioning performance by enabling hot starts, warm starts, and AssistNow Autonomous. This ensures faster TTFF when V_IO is supplied again. To make these features available, connect an independent power supply to V_BCKP to ensure backup domain supply when V_IO is not supplied.

Designs using an external battery as a power source at the V_BCKP pin must consider the battery capacity. That is, the GNSS satellite ephemeris data is typically valid for up to 4 hours for hot starts. Furthermore, for products supporting AssistNow Offline and Autonomous, the assistance data is valid up to few days for warm starts.



Avoid high resistance on the V_BCKP line. During the switch to V_BCKP supply, a short current adjustment peak may cause a high voltage drop at the pin.

If the hardware backup mode is not used, leave the V_BCKP pin open.

4.1.4 Supply design examples

The two voltage ranges for V_IO allow several combinations when designing the receiver power supply. Depending on the chosen combination, there are certain requirements to be considered. These are summarized in [Table 42](#).

Option	Nominal supply (V)		Design case / Requirements
	V_IO	VCC	
1	3.3	3.3	3.3 V design where VCC and V_IO are connected together. See Figure 22 for designs using the hardware backup mode, and Figure 23 for designs without backup supply. - VIO_SEL pin left open. - Voltage at VCC_RF pin = VCC - 0.1 V.
2	1.8	1.8	1.8 V design with VCC and V_IO connected together. This design requires an accurate supply. See Figure 22 for designs using the hardware backup mode, and Figure 23 for designs without backup supply. - VIO_SEL is grounded. - Note that the voltage output at VCC_RF = VCC - 0.1 V. - Note that the maximum supply tolerance is 1.8 V $\pm$ 2 %. To enter the hardware backup mode, set the receiver to the software standby mode with the UBX-RXM-PMREQ message before switching off V_IO and VCC.
3	1.8	1.8 / 3.3	VCC and V_IO are supplied independently. V_IO is supplied with an accurate 1.8 V supply. VCC can be either supplied with 1.8 V or 3.3 V. For designs using the hardware backup mode, see Figure 24, and for designs without backup supply, see Figure 25. - VIO_SEL pin is grounded. - Note that the maximum supply tolerance is 1.8 V $\pm$ 2 %. - Voltage at VCC_RF pin = VCC - 0.1 V. To enter hardware backup mode, switch off V_IO 100 ms before VCC. Alternatively, the receiver can be set to software standby mode with the UBX-RXM-PMREQ message before switching off V_IO and VCC.

Table 42: Voltage supply options

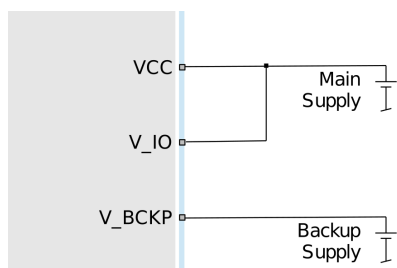


Figure 22: VCC and V_IO connected to the main supply, and external power supply at V_BCKP

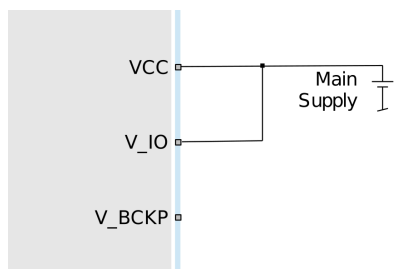


Figure 23: VCC and V_IO connected to the main supply. No external power supply at V_BCKP.

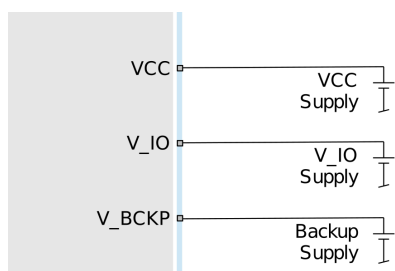


Figure 24: VCC and V_IO supplied by separate supplies, and external power supply at V_BCKP

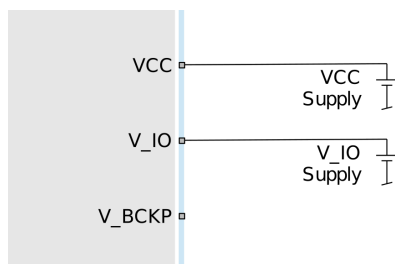


Figure 25: VCC and V_IO supplied by separate supplies. No external power supply at V_BCKP.

4.2 RF interference

The GNSS signal power received at the antenna is very low compared to other wireless communication signals. The received nominal -130 dBm GNSS signal strength makes the GNSS receiver susceptible to interference from any kind of nearby RF sources.

As an example, cellular applications emit signals with power levels of approximately $+30$ dBm, while the GNSS signal is less than -130 dBm when reaching the antenna. By simply comparing these numbers, it is obvious that interference issues must be seriously considered during the design phase.

4.2.1 In-band interference

Although the radio communications standards prevent intentional RF signal sources from interfering the GNSS frequencies, many devices emit RF power into the GNSS band at levels much higher than the GNSS signal itself.

One reason is that the frequency band above 1 GHz is not well regulated with regards to EMI, and even if permitted, signal levels are much higher than the GNSS signal power. In particular, all types of digital equipment, such as PCs, digital cameras, LCD screens, etc. tend to emit a broad frequency spectrum up to several GHz of frequency. Also wireless transmitters may generate spurious emissions that fall into the GNSS band.



The [Layout](#) section defines measures against in-band interference during the design phase of the application.

4.2.2 Out-of-band interference

Out-of-band interference is caused by signal frequencies that are different from the GNSS carrier frequency. The main sources are wireless communication systems such as LTE, GSM, CDMA, WCDMA, Wi-Fi, BT, etc. Typically, these systems may emit their specified maximum transmit power in close proximity to the GNSS receiving antenna, especially if such a system is integrated with the GNSS receiver. Even at reasonable antenna selectivity, destructive power levels may reach the RF input of the GNSS receiver. In addition, larger signal interferers may generate intermodulation

products inside the GNSS receiver front-end that fall into the GNSS band and contribute to in-band interference.

Measures against out-of-band interference include maintaining a good grounding concept in the design and adding a GNSS band-pass filter into the antenna input line to the receiver.



The sections [Out-of-band blocking immunity](#) and [Out-of-band rejection](#) provide more information about the RF immunity of the MAX-M10S module and mitigating out-of-band interference.

4.2.3 Spectrum analyzer

The UBX-MON-SPAN message can be enabled in u-center 2 to provide a low-resolution spectrum analyzer sufficient to identify noise or jammers in the reception band. Once enabled, u-center 2 includes a real-time chart that is updated once per second with the message data. See [Figure 26](#) for an example.

The design or device environment can generate interference at the in-band that can be analyzed from the spectrum in the UBX-MON-SPAN message. Hence, the shape of the spectrum as well as visible peaks help to identify in-band interference. Out-of-band interference can also cause peaks that appear in the in-band. However, there can be out-of-band interference that is not visible within the span of the spectrum. The presence of out-of-band interference may be seen as reduction in the PGA value.

The vertical axis compares the power level in dB for each frequency. A good spectrum shape is characterized by an even noise floor along with the GNSS band. For example, if any unwanted interference peak stands out, the vertical axis gives a rough approximation of the power level in dB compared to the noise floor.

Next to the chart, the center frequency, span, and resolution values set for the spectrum, and the PGA value are also displayed. The PGA value represents the internal gain set by the receiver, which depends on the external amplification of the GNSS input signal.

The vertical discontinuous lines in the chart area represent the offset to the center frequency in MHz. This helps to estimate the frequency of any spurious emission seen.

In addition, u-center 2 includes three functions commonly found in any spectrum analyzer. These features support the RF front-end design and help to spot out any jammer present during the application operation.

- **Hold:** if selected, the current spectrum shape freezes in a colored line. This allows for a comparison between the time the spectrum was frozen and the real-time spectrum. This is particularly helpful in assessing the impact of running other onboard components.
- **Average:** if selected, a colored line shows the averaged spectrum for each frequency. This supports the analysis over time and obtaining a less noisy shape.
- **Max hold:** if selected, a colored line shows the maximum amplitude measured at each frequency. This option helps to spot out any jammer over a period of time.

[Figure 26](#) shows the spectrum view in u-center 2 with the hold, average and max hold options selected. The green, yellow and red lines represent the frozen hold, average and max hold spectrums, while the blue line represents the current continuous spectrum.

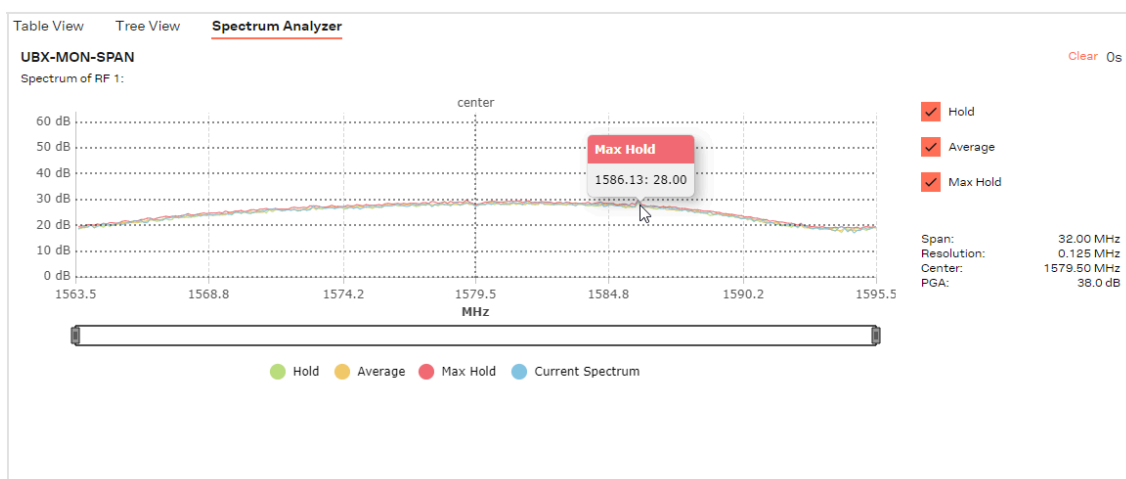


Figure 26: Spectrum analyzer view in u-center 2

-  By changing the enabled GNSS constellations, the span widens or narrows. This has a direct impact on the spectrum resolution, as the number of measured values is fixed to 256. For further details about this message and how to calculate each frequency, see the Interface description [3].
-  A peak may be visible around the center frequency. The signal comes internally from the receiver and it does not cause any degradation in the performance.

4.3 RF front-end

GNSS receivers operate with very low signal levels, ranging from -130 dBm to approximately -167 dBm. This alone is a challenge for the GNSS application design. Out-of-band sources of interference such as GSM, CDMA, WCDMA, LTE, Wi-Fi, or Bluetooth wireless systems with a much higher signal level require additional specific measures. The goal of the RF front-end design is to receive the in-band signal with minimum loss and added noise while suppressing the out-of-band interference.

The MAX-M10S RF front-end is designed for the highest sensitivity. The integrated RF circuit is matched to $50\ \Omega$ and it includes a built-in DC block, an LTE Band 13 notch filter, an LNA, and a SAW filter. For an overview of the RF front-end, refer to the [Block diagram](#). The MAX-M10S offers the best GNSS performance for designs with low or moderate RF interference levels.

For designs with other radio systems, an external SAW filter may be required to improve the immunity against RF interference. The external SAW filter converts the MAX-M10S RF front-end into an SAW-Band 13 notch-LNA-SAW circuit for the highest immunity, complete with built-in LTE Band 13 protection. The external SAW filter can be selected for an optimal trade-off between sensitivity and immunity.

Refer to the [Block diagram](#) for an overview of the RF front-end.

4.3.1 Internal LNA modes

In addition to the integrated LNA in the RF front-end circuit in MAX-M10S, there is also an internal LNA in the u-blox M10 receiver.

The receiver's internal LNA has three operating modes: normal gain, low gain, and bypass mode.

- By default, the internal LNA is configured for the low-gain mode for optimized sensitivity and immunity against RF interference.

- For RF front-end designs with 10 - 15 dB or higher total external gain, bypass mode is recommended to improve immunity. The power consumption is also slightly reduced in bypass mode.
- Normal-gain mode is not recommended for MAX-M10S.

The internal LNA mode can be configured at run time in BBR and RAM layers using the CFG-HW-RF_LNA_MODE configuration item and applying a reset, or set permanently in the one-time-programmable (OTP) memory in production. The configuration in the OTP memory is automatically applied at every startup. For more information, refer to [Internal LNA mode configuration](#).

For information on RF parameters, refer to the MAX-M10S Data sheet [1].

4.3.2 Out-of-band blocking immunity

Out-of-band RF interference may degrade the quality and availability of the navigation solution. Out-of-band immunity limit describes the maximum power allowed at the receiver RF input with no degradation in performance. Minor violation of the immunity limit may reduce C/N0 of the received signals but does not necessarily affect the overall receiver performance. However, a significant violation may reduce receiver sensitivity or cause a complete loss of signal reception. The severity of the interference depends on the repetition rate, frequency, signal level, modulation, and bandwidth of the signal.

[Figure 27](#) shows a typical out-of-band immunity level at the MAX-M10S RF input. The internal LNA is in the low-gain mode (default). The measurement has been done at room temperature using a test signal with 64QAM modulation and 10 MHz bandwidth similar to an LTE signal.

In general, the immunity is lower close to the receiver's in-band. At 500 MHz and 800 MHz ranges, the reduced immunity is due to harmonic multiples generated at the integrated LNA input falling at the receiver's in-band. Adding an external SAW filter in front of the RF input protects the LNA suppressing the harmonic generation. The SAW filter also further improves the overall immunity of the design.



If the out-of-band immunity limit is exceeded, it is recommended to verify that the receiver performance is not affected or is at an acceptable level in the presence of interference.

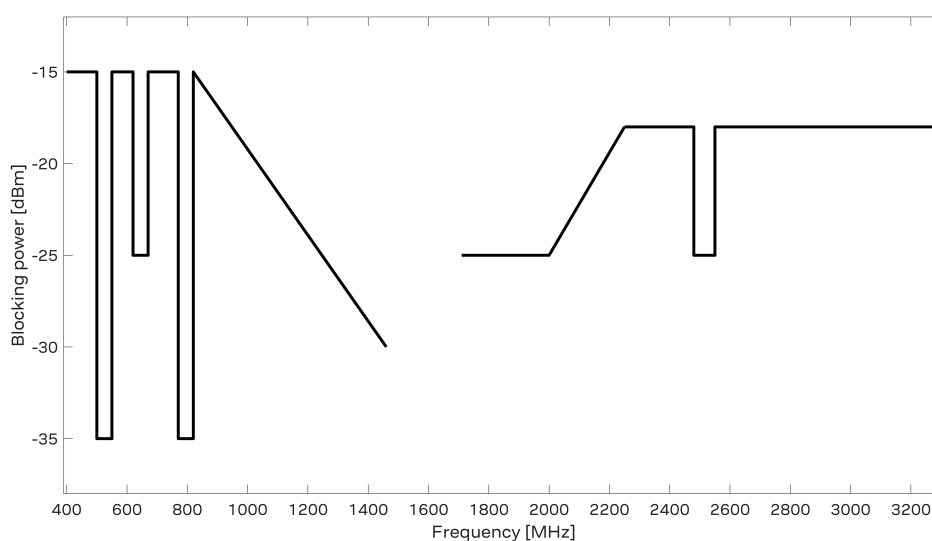


Figure 27: MAX-M10S out-of-band immunity level at 400–1460 MHz and 1710–3300 MHz for the low-gain mode (default).

Table 43 shows tabulated values for out-of-band immunity at selected cellular and Wi-Fi frequencies.

Parameter									
Frequency (MHz)	699	785	915	1710	1880	1980	2350	2440	2690
Immunity level (dBm)	-15	-35	-17	-25	-25	-25	-18	-18	-18

Table 43: MAX-M10S out-of-band immunity for the low-gain mode at selected frequencies.

4.3.3 Out-of-band rejection

RF interference is typically first coupled into the antenna and subsequently conducted into the receiver input. Typical out-of-band interference sources include transmitting antennas of other radio systems.

Estimation of the RF interference level coupled into the receiver antenna is a starting point for RF front-end design. For designs with other radio systems, the maximum power coupled into the antenna can be estimated from the maximum transmission power and the isolation between the antennas. Practical values for antenna isolation can range from 15 - 20 dB down to 6 - 10 dB for very small devices. RF interference may also couple from external sources such as nearby mobile devices or base stations.



A simplified test board can be used to estimate the isolation between two antennas. The size of the board and the placement of the antennas must match the final design. Connect the RF cables to the antenna inputs and measure S21 over the frequency band of interest with a vector network analyzer (VNA).

The required out-of-band rejection or isolation is the difference of the maximum power coupled into the antenna input terminal and the immunity level of the receiver RF input. The required isolation is realized with appropriate filtering, typically with one or two SAW filters. Amplification on the RF path reduces the out-of-band rejection and needs to be considered in filter selection. The type and number of filters are selected based on the estimated interference level and the immunity of the receiver.

RF interference from other parts of the design is more difficult to estimate. One option is to measure the interference level at the receiver input using a spectrum analyzer. Interference within the design is primarily a problem at the receiver in-band, where it cannot be addressed by filtering on the RF path. Outside the GNSS band, the required filtering is determined by the estimated interference level and the immunity of the receiver.

4.3.4 Antenna power supply

Figure 28 shows an active antenna supply network to connect the antenna supply to the RF signal line. The inductance L3 connects the antenna power supply to the RF signal line. The capacitance C14 filters out high-frequency interference from the power supply and the resistor R8 limits the short-circuit current.

The type and value of L3 is selected to have a resonance peak at GNSS frequencies. This provides a high series impedance above 500 Ω at GNSS L1 frequencies, creating an impedance mismatch with respect to the 50 Ω RF signal line. This minimizes the effect of the feed point on the RF signal line, and isolates the antenna supply from the RF signal line at GNSS frequencies. Both R8 and L3 must have sufficient current and power rating to withstand the short-circuit current. Example component values for the antenna supply network are given in [Standard resistors](#), [Standard capacitors](#), and [Inductors](#).

The VCC_RF pin can be used to supply an active antenna. VCC_RF is a RF filtered supply voltage derived from the VCC supply. Refer to the Data sheet[1] for the VCC_RF specification.

The [Antenna supervisor](#) can be used to detect open and short circuits on the antenna supply network and disconnect the antenna supply if a short circuit is detected.

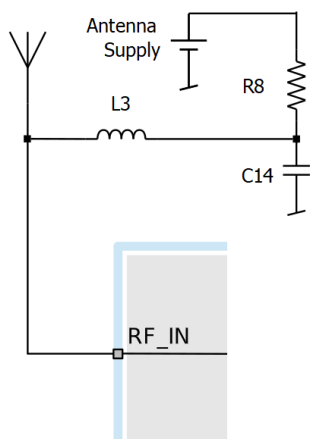


Figure 28: Antenna supply network

4.4 Layout

GNSS signals on the surface of the earth have a very low signal strength and are about 15 dB below the thermal noise floor. When integrating a GNSS receiver into a PCB, the placement of the components, as well as grounding, shielding, and interference from other digital devices are crucial issues that need to be considered very carefully.

An important factor in achieving high GNSS performance is the placement of the receiver with respect to other components on the PCB.

To minimize signal loss on the RF connection from the antenna to the receiver input and to avoid possible coupled interference, the connection to the antenna must be kept short while keeping some distance from the antenna to other electronic components.

The RF section should not be subject to noisy digital supply currents running through its GND plane. Make sure that critical RF circuits are clearly separated from any other digital circuits on the system board. To achieve this, position the receiver digital part towards the digital section of the system PCB and place the RF section and antenna as far away as possible from the other digital circuits on the board. Keep at least a 5 mm distance to any RF component and ensure proper grounding.



For applications using cellular antennas, increase the distance between both antennas as much as possible.

Another very important factor in GNSS applications is the grounding concept. Ensure good ground reference to the host ground by increasing the number of GND vias. The GND vias will improve the GND reference between all the layers, and the pads will serve as thermal relief.

Any stubs at the ground planes must be avoided or ended with a via to the reference ground. Otherwise, they could pick up and propagate interference.

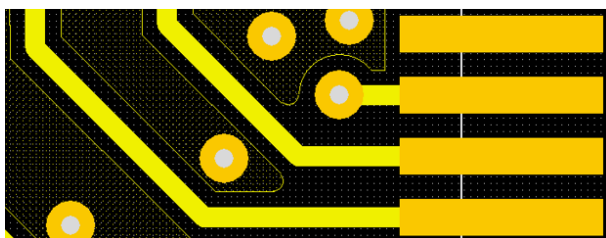


Figure 29: GND stub ended with a via

It is recommended to ground the area below the module, on the top and second layer. Avoid signal lines crossing below the module at these two layers.

For the RF signal line, it is best to use the co-planar waveguide with ground on the second layer. All the RF parts need a solid GND plane underneath in order to achieve the targeted impedance in the RF signal line.

The length and geometry in the RF signal line must be carefully analyzed. The impedance of the RF signal line must be $50\ \Omega$. Select the stack-up, copper, and dielectric properties of the PCB accordingly to fulfill this condition. The RF signal line should be as short as possible and the ground plane around should be filled with GND vias.

Avoid placing the receiver in the proximity of cooling fans or heat-emitting components, such as power devices. Temperature-sensitive components, such as the receiver oscillator, are sensitive to sudden changes in ambient temperature, which can adversely impact satellite signal tracking. Heating and cooling sources can include co-located power devices, cooling fans or thermal conduction via the PCB.

The GND planes can conduct heat to other elements, but they can act as heat dissipators as well. Increasing the number of GND vias helps to decrease sudden temperature changes.

 If needed, shield the receiver with temperature-sensitive components to reduce air convection and improve thermal stability.

4.4.1 Package footprint, copper and solder mask

The mechanical specification is available in the data sheet [1].

Figure 30 and Table 44 describe the footprint. Figure 31 and Table 45 provide recommendations for the paste mask.

 The copper and solder masks have the same size and position.

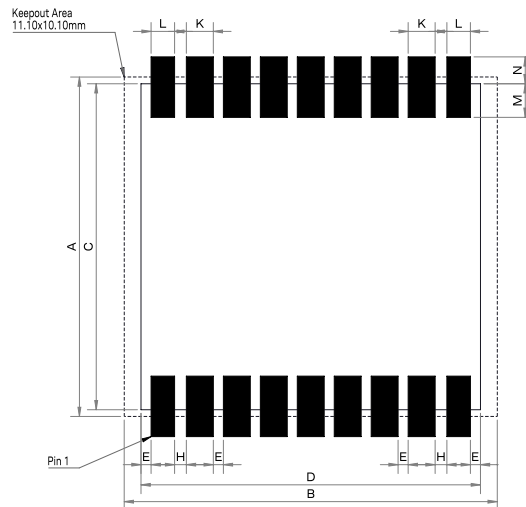


Figure 30: Recommended copper land and solder mask opening for MAX-M10S

Symbol	Dimension (mm)
A	10.1
B	11.1
C	9.7
D	10.1
E	0.3
H	0.35
K	0.8
L	0.7
M	1.0
N	0.8

Table 44: MAX-M10S footprint dimensions

To improve the wetting of the half vias, reduce the amount of solder paste under the module and increase it outside of the module by defining the dimensions of the paste mask to form a T-shape (or equivalent) extending beyond the copper mask.

Recommended stencil thickness is 150 µm.

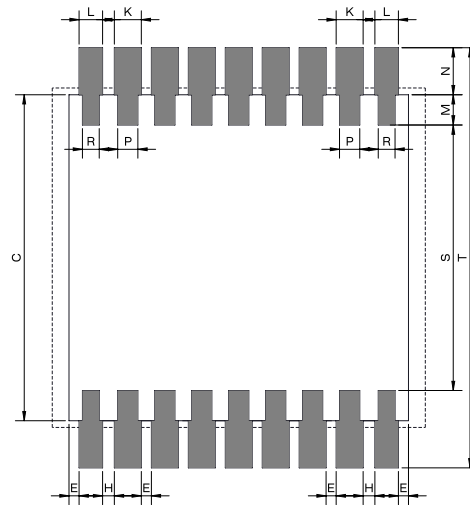


Figure 31: Recommended paste mask pattern for MAX-M10S

Symbol	Dimension (mm)
C	9.7
E	0.3
H	0.35
K	0.8
L	0.7
M	0.9
N	1.4
P	0.6
R	0.5
S	7.9
T	12.5

Table 45: MAX-M10S paste mask dimensions



These are only recommendations and not specifications. The exact geometry, distances, stencil thicknesses and solder paste volumes must be adapted to the customer's specific production processes (for example, soldering).

5 Product handling

5.1 Safety

5.1.1 ESD precautions

-  **CAUTION!** Risk of electrostatic discharge (ESD) damage. u-blox chips and modules are electrostatic sensitive devices containing highly sensitive electronic circuitry. A discharge of static electricity may damage the device or reduce the life expectancy of the device. To avoid ESD damage, adhere to the standard guidelines for handling ESD devices.

Consider the following:

Preventing electrostatic discharge

- Keep components in their original packages during transport.
- Open the package within an ESD-protected area (EPA), as in [Figure 32](#).
- At a workstation, store components in an EPA.
- Place ESD sensitive devices inside of shielding packaging or containers when transported outside of an EPA.
- Use protective clothing and proper personnel grounding at all necessary points when touching electrostatic sensitive device or assembly. For instance, wear ESD-safe clothing and shoes and wear an ESD wrist strap connected to a grounded workstation. Use heel straps when standing on conductive floors or dissipating floor mats.
- Hold the devices by the edges and avoid touching component contacts, pins, or circuitry

Product handling

- When handling RF transceivers and patch antennas, work in an EPA.
- When connecting test equipment or any other electronics to the module (as a standalone or PCB-mounted device), the first point of contact must always be between the local ground and the PCB ground.
- Before mounting a ceramic patch antenna, connect the device to ground.
- When handling the RF pin, do not touch any charged capacitors. Be especially careful when handling materials like patch antennas (~10 pF), coaxial cables (~50-80 pF/m), soldering irons, or any other materials that can develop charges.
- If there is any risk of touching an exposed antenna area in a non-ESD protected work area, implement proper ESD protection measures in the design.
- When soldering RF connectors and patch antennas to the receiver's RF pin, use an ESD-safe soldering iron (tip)

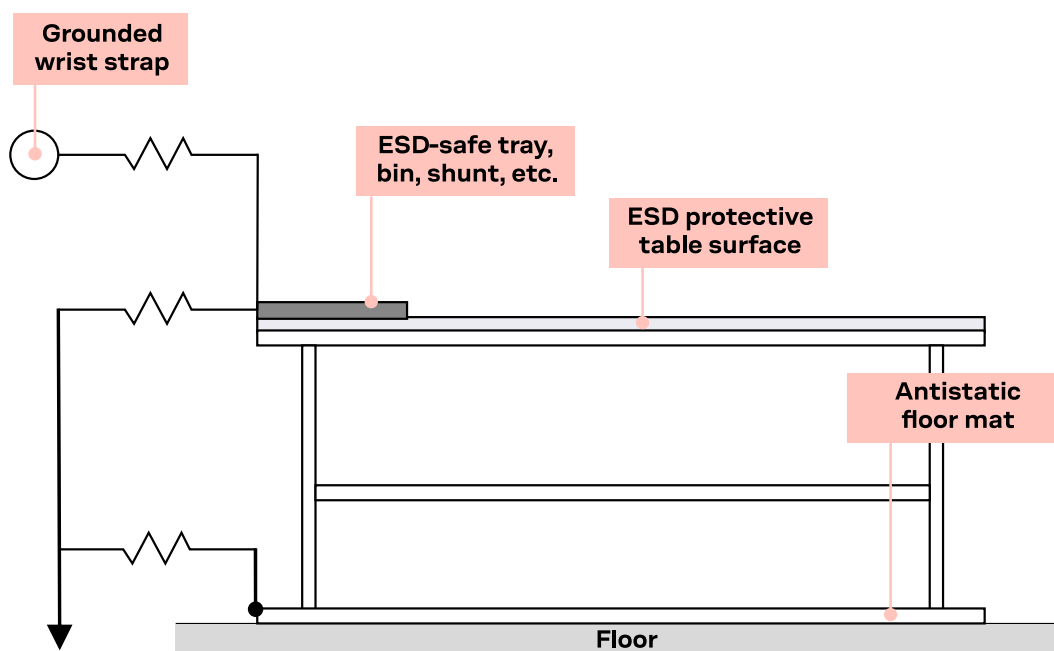


Figure 32: Standard workstation setup for safe handling of ESD-sensitive devices

5.1.2 Safety precautions

The MAX-M10S modules must be supplied by an external limited power source in compliance with the clause 2.5 of the standard IEC 60950-1. In addition to external limited power source, only Separated or Safety Extra-Low Voltage (SELV) circuits are to be connected to the module including interfaces and antennas.



For more information about SELV circuits see section 2.2 in Safety standard IEC 60950-1.

5.2 Soldering

Reflow soldering procedures are described in the IPC/JEDEC J-STD-020 standard [7].



When populating the modules, make sure that the pick and place machine is aligned to the copper pins of the module instead of the module edge.

Soldering paste

Use of “no clean” soldering paste is highly recommended, as it does not require cleaning after the soldering process. For instance, the following paste meets these criteria.

- Soldering paste: OM338 SAC405 / Nr.143714 (Cookson Electronics)
- Alloy specification: Sn 95.5/ Ag 4/ Cu 0.5 (95.5% tin/ 4% silver/ 0.5% copper)
- Melting temperature: 217 °C
- Stencil: The exact geometry, distances, stencil thicknesses and solder paste volumes must be adapted to the customer's specific production processes.

Reflow soldering



CAUTION. Risk of device damage. Exceeding the peak temperature of the recommended soldering profile may permanently damage the device.

The final soldering temperature chosen at the factory depends on additional external factors such as the choice of soldering paste, size, thickness and properties of the base board, etc.

As a reference, see “IPC-7530 Guidelines for temperature profiling for mass soldering (reflow and wave) processes”, published in 2001.

A convection-type soldering oven is highly recommended over the infrared-type radiation oven. Convection-heated ovens allow precise control of the temperature, and all parts will heat up evenly, regardless of material properties, thickness of components and surface color.

CAUTION. Risk of device damage. Modules must not be soldered with a damp heat process.

To avoid falling off, the modules should be placed on the topside of the board during soldering.

For the recommended soldering profile and conditions, see [Figure 33](#) and [Table 46](#)

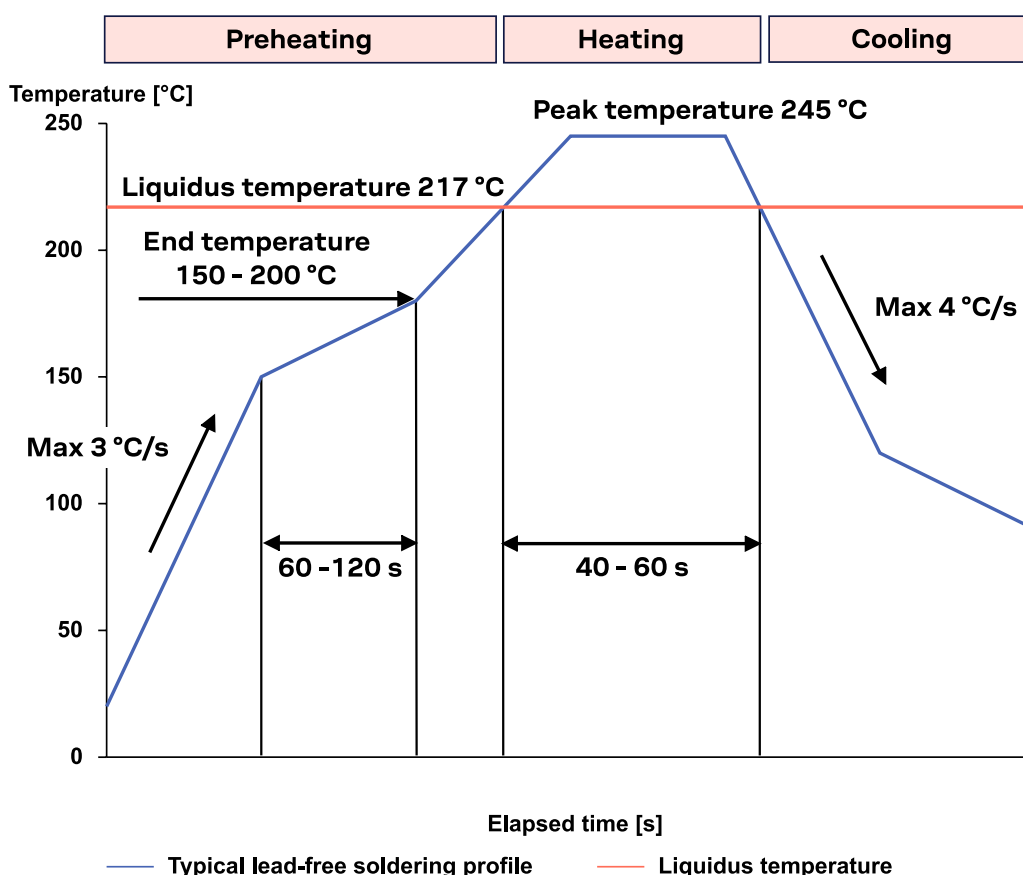


Figure 33: Recommended soldering profile

Phase	Value	Details
Preheating		
Temperature rise rate	Max 3 °C/s	If the temperature rise is too rapid in the preheat phase, excessive slumping may be caused.
Time	60 – 120 s	If the preheating is insufficient, rather large solder balls tend to be generated. Conversely, if performed excessively, fine balls and large balls will be generated in clusters.
End temperature	150 – 200 °C	If the temperature is too low, non-melting tends to be caused in areas containing large heat capacity.
Heating - reflow		

Phase	Value	Details
Time limit above 217 °C liquidus temperature	40 – 60 s	The temperature rises above the liquidus temperature of 217 °C. Avoid a sudden rise in temperature as the slump of the paste could become worse.
Peak reflow temperature	245 °C	
Cooling		
Temperature fall rate	Max 4 °C/s	A controlled cooling prevents negative metallurgical effects of the solder (solder becomes more brittle) and possible mechanical tensions in the products. Controlled cooling helps to achieve bright solder fillets with a good shape and low contact angle.

Table 46: Recommended conditions for reflow soldering

Optical inspection

After soldering the module, consider optical inspection.

Cleaning



Do not clean with water, solvent, or ultrasonic cleaner:

- Cleaning with water leads to capillary effects where water is absorbed into the gap between the baseboard and the module. The combination of residues of soldering flux and encapsulated water leads to short circuits or resistor-like interconnections between neighboring pins.
- Cleaning with alcohol or other organic solvents can result in soldering flux residues flowing underneath the module into areas that are not accessible for post-cleaning inspections. The solvent also damages the sticker and the printed text on the module.



CAUTION. Risk of device damage. Ultrasonic cleaning permanently damages the module, in particular the quartz oscillators.

The best approach is to use a “no clean” soldering paste to eliminate the cleaning step after the soldering.

Repeated reflow soldering



Repeated reflow soldering processes or soldering the module upside down are not recommended.

A board that is populated with components on both sides may require more than one reflow soldering cycle. In such a case, the process should ensure the module is only placed on the board submitted for a single final upright reflow cycle. A module placed on the underside of the board may detach during a reflow soldering cycle due to lack of adhesion.

The module can also tolerate an additional reflow cycle for rework purposes.

Wave soldering

Base boards with combined through-hole technology (THT) components and surface-mount technology (SMT) devices require wave soldering to solder the THT components. Only a single wave soldering process is encouraged for boards populated with modules.

Rework



CAUTION. Risk of device damage. Using a hot air gun is an uncontrolled process. It can lead to overheating and severely damage the module. Always avoid overheating the module.

After the module is removed from the oven, clean the pins before reapplying the solder paste, placing the module in the oven and proceeding with the reflow soldering of a new module.

-  Never attempt to alter the module itself, e.g. by replacing individual components. Such actions immediately void the warranty.

Conformal coating

Certain applications employ a conformal coating of the PCB using HumiSeal® or other related coating products. These materials affect the RF properties of the GNSS module and it is important to prevent them from flowing into the module. The RF shields do not provide 100% protection for the module from coating liquids with low viscosity. Apply the coating carefully.

-  Conformal coating of the module voids the warranty.

Casting

If casting is required, use viscose or another type of silicon pottant. The OEM is strongly advised to qualify that such processes are suitable for the module before implementing them in the production.

-  Casting voids the warranty.

Grounding metal covers

Attempts to improve grounding by soldering ground cables, wick or other forms of metal strips directly onto the EMI covers is done at customer's own risk. The numerous ground pins should be sufficient to provide optimum immunity to interferences and noise.

-  u-blox provides no warranty for damages to the module caused by soldering metal cables or any other forms of metal strips directly onto the EMI covers.

Use of ultrasonic processes

Some components on the module are sensitive to ultrasonic waves.

-  **CAUTION.** Risk of device damage. Use of any ultrasonic processes (cleaning, welding etc.) may cause damage to the receiver.
-  u-blox provides no warranty against damages to the module caused by ultrasonic processes.

Appendix

A Migration

u-blox is committed to ensuring that products in the same form factor are backwards compatible over several technology generations. This section describes important differences to consider when migrating from the u-blox 7/8/M8 to u-blox M10.

A.1 Hardware changes

Table 47 lists the key **hardware-related changes** between MAX-M10S and MAX-7/8/M8 modules.

Feature	Change	Action needed / Remarks
Absolute maximum ratings	The absolute maximum voltage for PIOs changed to $V_{IO} + 0.3\text{ V}$ in MAX-M10S.	A HW change is required if main supply voltage source is not within the specified range.
VCC, V_{IO}	The V_{IO} voltage range is selected with the V_{IO_SEL} pin. V_{IO} supply voltage cannot be higher than $VCC + 0.3\text{ V}$ in MAX-M10S.	For designs with 1.8 V supply at V_{IO} , switch off the V_{IO} supply 100 ms before VCC when transitioning to the hardware backup mode. Alternatively, put the receiver in the software standby mode by sending the UBX-RXM-PMREQ message before switching off V_{IO} and VCC. Refer to the minimum and maximum V_{IO} ramp requirements in the MAX-M10S data sheet [1].
VCC and V_{IO} supply range	MAX-M10S VCC: 1.76 V - 3.6 V MAX-M10S V_{IO} : 1.76 V - 1.98 V, 2.7 V - 3.6 V MAX-7C/8C/M8C: 1.65 V - 3.6 V	A HW change is required when migrating from a MAX-7C/8C/M8C design that uses less than 2.7 V VCC and V_{IO} supply because V_{IO_SEL} (pin 15) needs to be connected to GND in MAX-M10S and the voltage range for 1.8 V designs is different.
V_{BCKP} supply range	MAX-M10S: 1.65 V - 3.6 V MAX-7/8/M8: 1.4 V - 3.6 V	A HW change is required when migrating from a MAX-7/8/M8 design that uses less than 1.65 V V_{BCKP} supply.
Hardware backup current	The hardware backup current into V_{BCKP} in MAX-M10S is higher than in MAX-7Q/7W/8Q/M8Q/W modules. The hardware backup current in MAX-M10S is significantly lower than in MAX-7C/8C/M8C modules.	Check the backup battery capacity if it is still suitable for the target design requirements.
Software standby current	The software standby current into V_{IO} in MAX-M10S is higher than in MAX-7Q/8Q/M8Q/W modules. MAX-7W has higher software backup current than MAX-M10S. The software standby current in MAX-M10S is significantly lower than in MAX-7C/8C/M8C.	Check if the changes in current affects target design/application requirements.
Pin-out	1:1 pin-out mapping with MAX-7Q/7C/8Q/8C modules, but not with MAX-7W/M8W.	A HW change is required for MAX-7W/M8W designs that use V_{ANT} (pin 15) to supply an active antenna, which is V_{IO_SEL} in MAX-M10S.

Feature	Change	Action needed / Remarks
RF front-end	LNA-SAW MAX-M10S includes a Band 13 notch filter, LNA and a SAW filter on the RF path, allowing the use of passive antenna. MAX-M10S noise figure: 1.5 dB MAX-7/8/M8 noise figure: 3.5 dB	External BOM savings are possible if the same components have been added externally on the RF path of the MAX-7/8/M8 design. Use the internal LNA of the receiver in bypass mode when an active antenna is used. The internal LNA mode of the receiver can be configured to low gain and bypass modes because the MAX-M10S module has an LNA on the RF path. This allows flexible optimization of the receiver performance and power consumption with respect to the selected RF front-end/ antenna. The default internal LNA mode on MAX-M10S is set to low gain. An external SAW filter can be added in front of the RF_IN pin in MAX-M10S, allowing a SAW-LNA-SAW RF front-end circuit for improving out-of-band immunity against RF interference from other sources. This is especially useful when MAX-M10S is used in cellular applications. Check that the external RF path (antenna and filtering) is suitable for all of the GNSS signals in use.
Absolute max RF input power	MAX-M10S: 0 dBm MAX-M8 modules: +15 dBm MAX-7 modules: +13 dBm	A HW change: add external SAW filter in front of MAX-M10S for cellular applications.
SAFEBOOT_N, TIMEPULSE	The SAFEBOOT_N pin is internally connected to TIMEPULSE pin through a 1 kΩ series resistor .	Do not drive the TIMEPULSE pin low at startup because it will put the receiver in safeboot mode.
RESET_N	MAX-M10S: clears the RTC time and BBR contents (GNSS orbit data). MAX-7/8Q/M8: clears only the RTC time.	After resetting MAX-M10S with the RESET_N pin, the TTFF is similar to performing a cold start. After resetting the MAX-7/8Q/M8 modules, the TTFF is typically around 6 s in open-sky conditions.
LNA_EN	The LNA_EN pin is internally connected to the V_IO supply voltage through a transistor switch, which serves as a buffer.	The drive strength and voltage level of the LNA_EN signal is not identical to the PIOs.
Digital IO	External IO isolation is required. The output current of the digital IOs is 2 mA compared to 4 mA in MAX-7 and MAX-8Q/M8 modules, except for the TIMEPULSE pin, which still provides 4 mA drive strength.	Caution! Risk of receiver damage. If the IO pins are not isolated, the GNSS receiver may get permanently damaged when high current is drawn from the IO pins. Use external IO isolation to avoid receiver damage. Do not drive IOs when VCC and V_IO are not supplied. Removing the VCC or V_IO supply does not isolate the module IO pins from the the host device IO pins. Additional external pull-up resistors may be required to increase the drive strength of the digital IO output pin or adjust the load accordingly.

Table 47: MAX-M10S hardware features compared to MAX-7/8/M8 modules

Migrating from MAX-7W/M8W to MAX-M10S requires special care and may require some re-design because there are changes in the assignment of pins 13 and 15, as shown in [Figure 34](#). Consequently, pin 15 should be left open (i.e. not connected) when migrating from MAX-7W/M8W because there is no built-in antenna supervisor support in MAX-M10S. Therefore, in MAX-M10S, the active antenna supply and antenna supervisor circuitry (if used) need to be connected externally as shown in [Antenna supervisor designs](#).

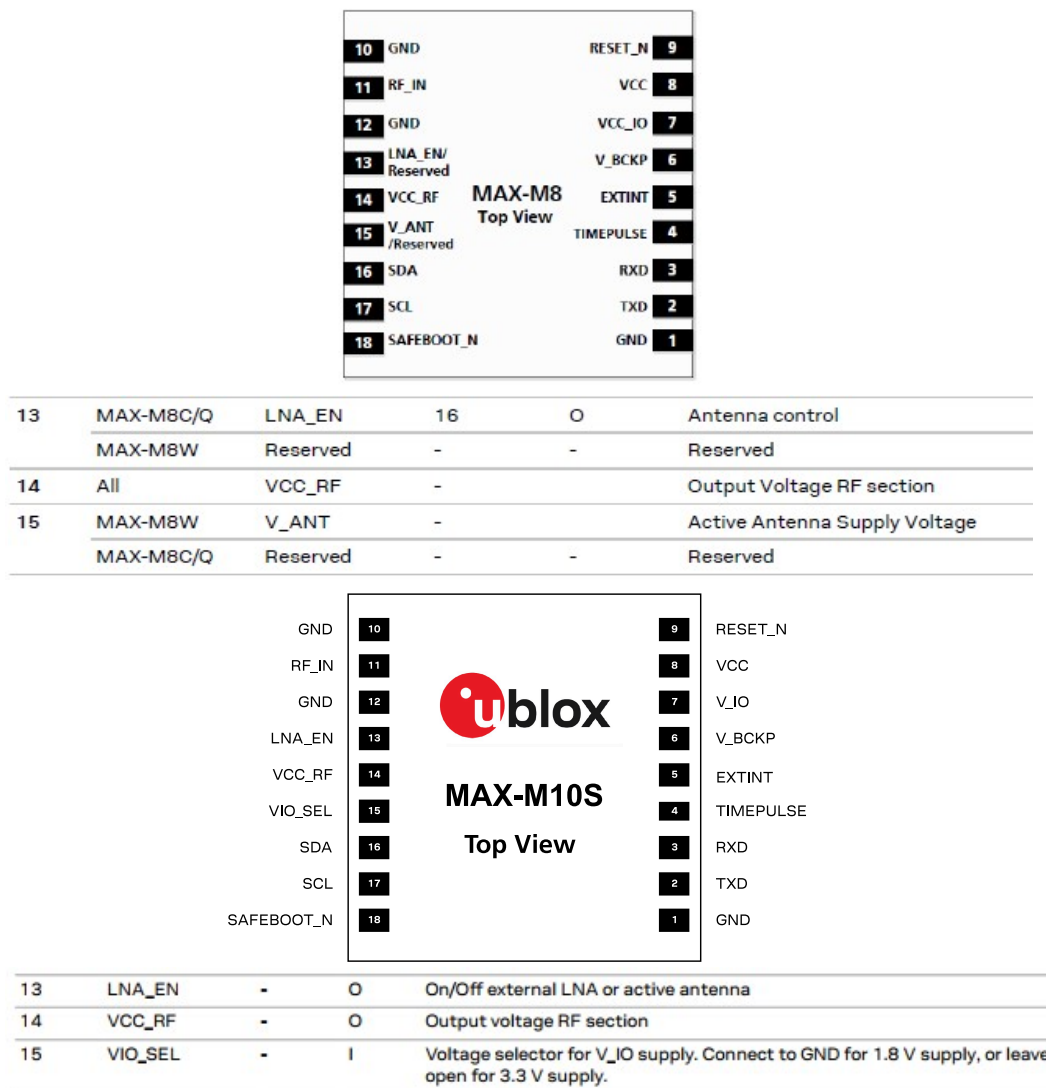


Figure 34: MAX-M8 vs. MAX-M10S comparison (pin 13 - 15)

Refer to the product Data sheets for details.

A.2 Firmware changes

Table 48 presents a summary of the key **firmware-related changes** between u-blox M10 and u-blox 7/8/M8, as well as required actions during migration.

Feature	Change	Action needed / Remarks
Signals		
Default GNSS configuration	MAX-M10S: GPS, Galileo, BeiDou B1I, QZSS and SBAS. MAX-M8Q/C/W: GPS, GLONASS, QZSS and SBAS. MAX-7Q/7C/7W/8Q/8C: GPS, QZSS and SBAS.	Code change (optional) HW change (optional): support for BeiDou B1I requires antenna selection or antenna frequency tuning, and any additional filtering in the previous MAX-7/8/M8 designs.

Feature	Change	Action needed / Remarks
BeiDou B1C	New signal. BeiDou satellite IDs up to 63 supported. Cannot be used simultaneously with BeiDou B1I. AssistNow and power save mode not supported.	Code change (optional)
BeiDou B1I	BeiDou satellite IDs up to 63 supported.	-
SBAS	New SBAS PRN selection: 123, 126-129, 131, 133, 136-138.	Code change (optional)
QZSS L1S	SLAS corrections are now applied for navigation.	Code change (optional)
QZSS IMES	Not supported.	Code change (optional)
Protocols		
NMEA	Supports NMEA V4.11, V4.10, V4.0, V2.3, and V2.1. NMEA V4.11 is enabled by default. In MAX-M10S, the NMEA GSV messages for zero signal C/N0 level are grouped separately as compared to u-blox 7/8/M8. If the signal's C/N0 level is 0, the FW treats it as an unknown signal for not being able to track it and sets the "signal ID" to zero. The firmware creates two separate GSV message groups. The first group contains GSV messages for the tracked signals with non-zero C/N0 and the second group contains GSV messages for the untracked signals with zero C/N0. In the untracked signals message group, the C/N0 field is blank (i.e. , ,). There is no configuration to revert this to the u-blox 7/8/M8 format.	Code change (optional)
RTCM	Not supported.	Code change
General		
Configuration concept	New configuration scheme using UBX-CFG-VALSET and UBX-CFG-VALGET messages.	Code change
Navigation update rate	Navigation update rate up to 18 Hz for single GNSS, 10 Hz for 3 GNSS, and 5 Hz for 4 GNSS constellations.	Code change (optional)
Super-S	New feature. Improves performance under weak signal conditions. Enabled by default in current firmware.	-
Power save mode (PSM)	PSM configuration options have changed: PSMCT: 4 Hz navigation rate not supported. PSMCT period is configured with CFG-RATE configuration group. BeiDou B1C not supported in power save modes.	Code change (new configuration messages)
Altitude limit	The maximum altitude limit increased to 80 000 m.	-
Features		
AssistNow	Simultaneous operation of AssistNow Online, Offline and Autonomous. New AssistNow Offline data download options are available. The time period can be limited to a resolution of days to reduce the data package size.	Code change (optional)
CloudLocate	New feature. Position calculation in the cloud provided by the u-blox CloudLocate service. Extends the life of energy-constrained IoT applications. Low payload messages supported.	Code change (optional)
Geofencing	Not supported.	Code change
Data logging	Not supported.	Code change
Galileo return link message	Galileo search and rescue (SAR) return link message UBX-RXM-RLM.	Code change (optional)
RF spectrum view	New message. UBX-MON-SPAN shows in-band RF spectrum around the GNSS band. This can be used to identify potential in-band RF interference sources in the design.	Code change (optional)
Location batching	New feature that can be used to reduce power consumption by saving navigation solutions on the receiver for up to 10 minutes (at 1 Hz) and polling the solutions afterwards to a host MCU.	Code change (optional)

Feature	Change	Action needed / Remarks
Protection level	New feature. Real-time position accuracy estimate with 95% confidence level with the message UBX-NAV-PL.	Code change (optional)
Security		
Message integrity	New feature. Authentication of data output based on private/public key pair.	Code change (optional)
Unique chip identifier	A unique chip identifier output in boot screen and in the UBX-SEC-UNIQID message.	Code change (optional)
Configuration lock	New security feature that is enabled with CFG-SEC-CFG_LOCK message for locking the receiver configuration.	Code change (optional)
Additional changes for u-blox-7 migration	Support for multiple GNSS and new signals: Galileo E1, BeiDou B1I, and GLONASS L1OF. Multiple GNSS assistance (MGA) with AssistNow services. Geofencing support. UBX message integrity mechanism added. Spoofing detection added. Power save modes support. Wrist dynamic model added. Odometer to measure traveled distance with support for different user profiles. Time pulse and time mark reference changed from GPS to UTC.	Code change (optional)

Table 48: MAX-M10S firmware features

For more information on the supported features and messages in the u-blox M10 receiver, refer to the SPG 5.10 Release note and Interface description [2], [3].

B Reference designs

The [External components](#) section provides the specification and recommendations for the external components that are shown in each reference design.

This section provides some reference designs for typical and antenna supervisor design cases.



Designs with 1.8 V main supply or with independent supply for VCC and V_IO must fulfill certain requirements when transitioning to the hardware backup mode. For more details, refer to [Supply design examples](#).

B.1 Typical design

Here are the key features for a MAX-M10S typical design:

- VCC and V_IO are connected together to a single supply. In designs with 3.3 V supply, the VIO_SEL pin must be left open, as shown in [Figure 35](#). In designs with 1.8 V supply, the VIO_SEL pin must be connected to GND, as shown in [Figure 36](#).
- V_BCKP supply is optional. If present, the hardware backup mode is supported. This mode maintains the time and GNSS orbit data in the battery-backed RAM memory if the main supply is switched off.
If there is no backup supply, time aiding with the UBX-MGA-INITIME_UTC message (optionally with a timing signal at the EXTINT pin) and the GNSS orbit data from the AssistNow services or stored on the host controller can be used to reduce the TTFF.
- A passive or active antenna can be used. An active antenna can be supplied either with the VCC_RF output from MAX-M10S, as shown in [Figure 37](#), or from an external supply, as shown in [Figure 38](#). Nevertheless, the internal LNA provides enough gain for passive antennas.

- MAX-M10S has an integrated RF circuit which includes band 13 notch filter followed by a low-noise amplifier (LNA) and a SAW filter, hence, no additional RF front-end component is needed. However, in cellular applications, an external SAW filter can be added in front of RF_IN as shown in [Figure 37](#), allowing a SAW-LNA-SAW RF front-end circuit for improved out-of-band immunity against RF interference from other sources.
- UART and I2C communication interfaces are available.
- For an absolute minimum design using UART, other PIOs (RESET_N, EXTINT, TIMEPULSE, SDA, SCL, SAFEBOOT_N) can be left open.

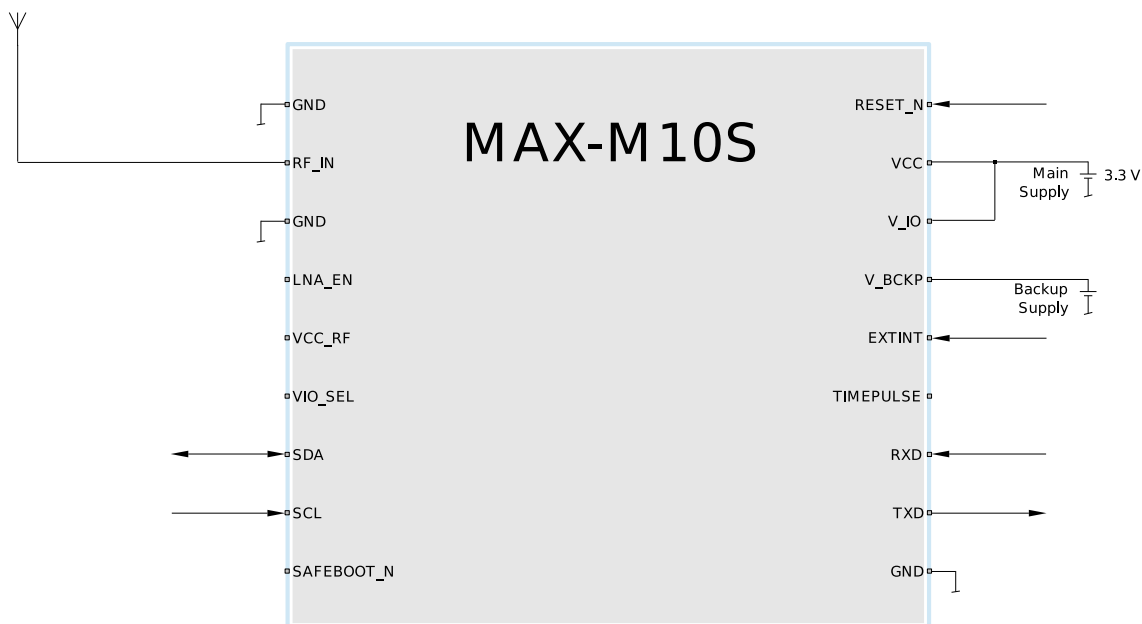


Figure 35: Typical 3.3 V design

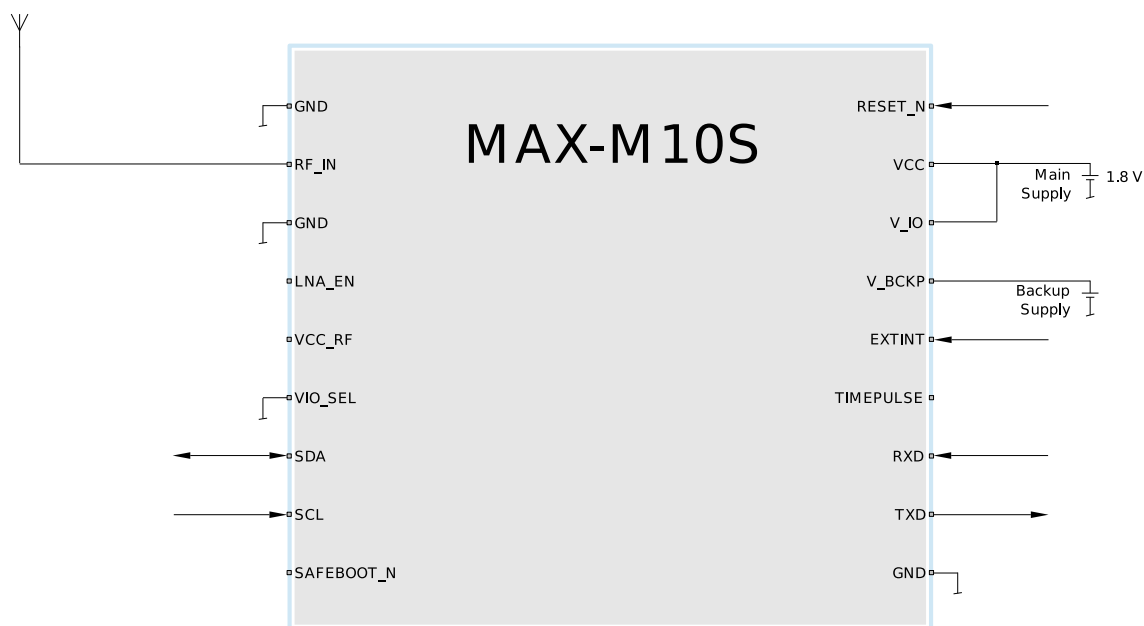


Figure 36: Typical 1.8 V design

B.2 Antenna supervisor designs

Figure 37 and Figure 38 show a reference design for a 2-pin and 3-pin antenna supervisor design respectively. Here are the key features:

- VCC and V_IO are connected together to a single supply.
- Supply at V_BCKP is optional. If present, the hardware backup mode is supported. This mode maintains the time and GNSS orbit data in the battery-backed RAM memory if the main supply is switched off.
If there is no backup supply, time aiding with the UBX-MGA-INITIME_UTC message (optionally with a timing signal at the EXTINT pin) and the GNSS orbit data from the AssistNow services or stored on the host controller can be used to reduce the TTFF.
- An external SAW filter can be placed on the RF path as shown in Figure 37, which allows an SAW-LNA-SAW RF front-end circuit for improving out-of-band immunity against RF interference from other sources. This is especially useful when MAX-M10S is used in cellular applications.
- An active antenna can be supplied with the VCC_RF output from MAX-M10S or from an external supply. VCC_RF is a filtered output voltage supply, which outputs VCC - 0.1 V. In addition, the active antenna supply can be turned on/off by the LNA_EN signal, which also controls the internal LNA of MAX-M10S.
- External open drain buffers and operational amplifiers are also needed depending on whether a 2-pin or 3-pin antenna supervisor design is used.
- UART and I2C communication interfaces are available. I2C PIOs (SDA and SCL) can be used in a 3-pin antenna supervisor design as shown in Figure 38. In this case, the I2C interface needs to be disabled before assigning the new function to the PIOs.



Disable the I2C interface with the CFG-I2C-ENABLED configuration key when I2C pins are used for antenna supervisor functions. Likewise, disable the UART interface (CFG-UART1-ENABLED) or TIMEPULSE (CFG-TP-TP1_ENA) when the pins are used for antenna supervisor functions.

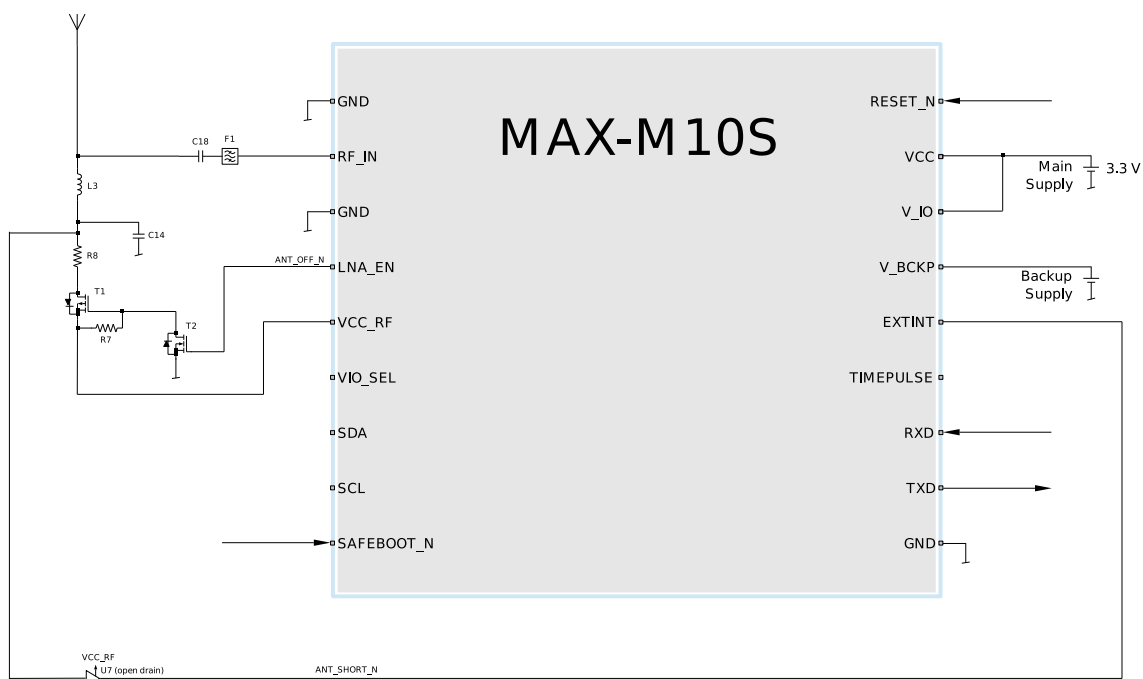


Figure 37: 2-pin antenna supervisor design

Table 49 lists the configuration settings required for the 2-pin antenna supervisor reference design given in Figure 37.

Configuration key	Value
CFG-HW-ANT_CFG_VOLTCTRL	1 (true), default (no configuration required)
CFG-HW-ANT_SUP_SWITCH_PIN	7, default (no configuration required)
CFG-HW-ANT_CFG_SHORTDET	1 (true)
CFG-HW-ANT_CFG_SHORTDET_POL	1 (true), default (no configuration required)
CFG-HW-ANT_SUP_SHORT_PIN	5
CFG-HW-ANT_CFG_PWRDOWN	1 (true)
CFG-HW-ANT_CFG_PWRDOWN_POL	0 (false), default (no configuration required)
CFG-HW-ANT_CFG_RECOVER	1 (true)

Table 49: Configuration for the 2-pin antenna supervisor design



Configuration key	Value
CFG-I2C-ENABLED	0 (false)
CFG-HW-ANT_CFG_VOLTCTRL	1 (true), default (no configuration required)
CFG-HW-ANT_SUP_SWITCH_PIN	7, default (no configuration required)
CFG-HW-ANT_CFG_SHORTDET	1 (true)
CFG-HW-ANT_CFG_SHORTDET_POL	1 (true), default (no configuration required)
CFG-HW-ANT_SUP_SHORT_PIN	3
CFG-HW-ANT_CFG_OPENDET	1 (true)
CFG-HW-ANT_CFG_OPENDET_POL	1 (true), default (no configuration required)
CFG-HW-ANT_SUP_OPEN_PIN	2
CFG-HW-ANT_CFG_PWRDOWN	1 (true)
CFG-HW-ANT_CFG_PWRDOWN_POL	0 (false), default (no configuration required)
CFG-HW-ANT_CFG_RECOVER	1 (true)

Table 50: Configuration for the 3-pin antenna supervisor design

This section lists the recommended values for the external components in the reference designs.

Examples of suitable GNSS antennas for u-blox M10 platform are listed in [Table 51](#).

Manufacturer	Order no.	Comments
Taoglas	CGGBP.18.4.A.02	GPS/Galileo/BeiDou/GLONASS 18 x 18 x 4 mm ³ passive

Manufacturer	Order no.	Comments
Taoglas	GP.1575.18.4.A.02	GPS/Galileo 18 x 18 x 4 mm ³ passive
Amotech	YDRA-A18-1575	GPS/Galileo 18x18x4 mm ³ passive
Amotech	AGA363913-S0-A1	GPS/Galileo/BeiDou/GLONASS 5 V / 14 mA active
INPAQ	PA1575MZ50J4G	GPS/Galileo 18.4 x 18.4 x 4 mm ³ passive
INPAQ	B3G02G-S3-XX-A	GPS/Galileo/BeiDou/GLONASS 2.7 to 3.9 V / 10 mA active

Table 51: L1 GNSS antennas

C.2 Standard capacitors

Table 52 presents the recommended capacitor values for MAX-M10S.

Name	Use	Type / Value
C14	RF Bias-T capacitor	10 nF, 10%, 16 V, X7R
C18	DC block	47 pF, 5%, 25 V, COG

Table 52: Standard capacitors

C.3 Standard resistors

Table 53 presents the recommended resistor values for MAX-M10S.

Name	Use	Type / Value
R5	Antenna supervisor voltage divider	560 Ω , 5%, 0.1 W
R6	Antenna supervisor voltage divider	100 k Ω , 5%, 0.1 W
R7	Pull-up resistor at antenna supervisor transistor	100 k Ω , 5%, 0.1 W
R8	Antenna supply current limiter/shunt resistor	10 Ω , 5%, 0.25 W

Table 53: Standard resistors

C.4 Inductors

Table 54 presents the recommended inductor values for MAX-M10S.

Name	Use	Type / Value	Recommended component
L3	RF Bias-T inductor	27 nH, 5%	Murata LQG15H, LQW15A series Johanson Technology L-07W series Any other inductor with impedance >500 Ω at GNSS frequency and current rating above 300 mA.

Table 54: Recommended inductors

C.5 Operational amplifier

Name	Manufacturer	Order no.
U6	Linear Technology	LT6000, LT6003

Table 55: Recommended parts list for the operational amplifier

C.6 Open drain buffers

Name	Manufacturer	Order no.
U7, U8	Fairchild	NC7WZ07P6X

Table 56: Recommended parts list for the open drain buffers

C.7 Switch transistors for antenna supervisor

Table 57 presents the recommended switch transistors for MAX-M10S.

Name	Manufacturer	Order no.	Comments
T1, T2	Vishay	Si1016X-T1-GE3	p-channel, n-channel MOSFET

Table 57: Recommended parts list for the antenna supervisor switch transistors

Related documents

- [1] MAX-M10S Data sheet, [UBX-20035208](#)
- [2] u-blox M10 SPG 5.10 Release notes, [UBX-22001426](#)
- [3] u-blox M10 SPG 5.10 Interface description, [UBX-21035062](#)
- [4] Product packaging reference guide, [UBX-14001652](#)
- [5] AssistNow Product change note, [UBXDOC-199822977-208025](#)
- [6] AssistNow service, <https://support.thingstream.io/>
- [7] Joint IPC/JEDEC standard, www.jedec.org



For regular updates to u-blox documentation and to receive product change notifications please register on our homepage <https://www.u-blox.com>.

Revision history

Revision	Date	Status / comments
R01	29-Apr-2021	Advance information
R02	08-Jul-2021	Added migration section.
R03	12-Jul-2022	New product type number MAX-M10S-00B-01 with ROM SPG 5.10 firmware. Product status is available in the data sheet [1]. Updates: pin assignment, receiver configuration, PIOs, power save modes, power supplies, RF front-end, migration, and reference designs sections. Added augmentation systems, navigation configuration, antenna supervisor description, reset, protection level, multiple GNSS assistance (MGA), RF interference, data batching, configuration lock, and CloudLocate sections.
R04	01-Jun-2023	Added sections • BeiDou B1I and B1C signals • High performance navigation update rate configuration • OTP memory configuration
R05	28-Apr-2026	Editorial changes throughout the document. Added sections: • Protection level: Validity requirements • AssistNow GNSS assistance: legacy AssistNow services Online and Offline have been replaced with the corresponding new services Live Orbits and Predictive Orbits. • Collecting debug logfiles: design-in guidance • External components: Antenna Updated sections: • Basic receiver configuration: Message output configuration • Basic receiver configuration: Antenna supervisor configuration • Navigation configuration: Super-Signal (Super-S) technology • Augmentation systems: SBAS • Communication interfaces and PIOs: I2C • Security • RF interference: Spectrum analyzer • Integration: Package footprint, copper and solder mask • Migration: added MAX-7 module Removed sections: • Multiple GNSS assistance (MGA) • Product handling: Packaging, MSL (moved to MAX-M10S Data sheet)

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